

**SEASONAL AND NON-SEASONAL ARMA MODELLING FOR
PM2.5 CONCENTRATION PREDICTION IN JOHOR**

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PM2.5 CONCENTRATION PREDICTION IN JOHOR**

by

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TABLE OF CONTENT

	Page
ACKNOWLEDGEMENT	i
TABLE OF CONTENT	ii
LIST OF TABLES	vi
LIST OF FIGURES	ix
LIST OF ABBREVIATIONS	xi
ABSTRACT	xii
ABSTRAK	xiii
CHAPTER 1 INTRODUCTION	1
1.1 Background of the Study	1
1.2 Problem Statement	3
1.3 Research Objectives	4
1.4 Significance of the Study	4
1.5 Scope of the Study	5
1.6 Organization of the Thesis	5
CHAPTER 2 LITERATURE REVIEW	7
2.1 Introduction	7
2.2 Time Series Applications in Air Quality Modelling	8
2.3 Malaysian Studies on PM2.5 Modelling	9
2.4 Research Gaps	10
2.5 Summary	11
CHAPTER 3 METHODOLOGY	15

3.1 Introduction	15
3.2 Data Sources.....	15
3.3 Data Preprocessing.....	16
3.4 Exploratory Data Analysis (EDA)	17
3.5 Stationary Time Series	18
3.5.1 Augmented Dickey-Fuller (ADF Testing for Stationarity)	19
3.5.2 Differencing.....	20
3.6 Time Series Model	21
3.6.1 Autoregressive (AR) Model	21
3.6.2 Moving Average (MA) Model	22
3.6.3 Autoregressive Moving Average (ARMA) Model.....	22
3.6.4 Autoregressive Integrated Moving Average (ARIMA) Model	22
3.6.5 Seasonal Autoregressive Moving Average (SARMA) Model	23
3.6.6 Seasonal Autoregressive Integrated Moving Average (SARIMA) Model....	23
3.7 Model Identification and Selection	24
3.7.1 Autocorrelation Function (ACF)	24
3.7.2 Partial Autocorrelation Function (PACF)	25
3.7.3 Model Selection using <code>auto.arima()</code>	26
3.7.4 Parameter Estimation	27
3.8 Diagnostics Checking.....	27
3.8.1 ACF and PACF Plots of Residuals.....	28

3.8.2 Test for Normality using Q-Q Plots	29
3.8.3 Ljung-Box Portmanteau Test	29
3.9 Forecasting	30
3.9.1 Forecast Accuracy Metric	30
CHAPTER 4 RESULTS AND DISCUSSION	32
4.1 Introduction	32
4.2 Exploratory Data Analysis	33
4.2.1 Summary Statistics	33
4.2.2 Time Series Plots	36
4.3 Stationarity and Differencing	40
4.4 Model Identification	41
4.4.1 ACF and PACF Plots	42
4.4.2 Model Selection Criteria	47
4.5 Model Estimation and Parameter Significance	53
4.6 Model Diagnostics	59
4.6.1 Residual Analysis	59
4.6.2 Forecast Accuracy	68
4.7 Out-of-Sample Forecasting	70
4.8 Comparison Across Stations	75
4.9 Discussion	78
4.9.1 Spatial Variation in PM2.5 Patterns	78

4.9.2 Pollution Behaviour in Rural vs Urban Areas	78
4.9.3 Meteorological and Cyclical Factors.....	78
4.9.4 Consistency with Previous Research.....	79
4.9.5 Practical Applications of Forecasting.....	79
4.10 Summary of the Chapter	80
CHAPTER 5 CONCLUSION	82
5.1 Concluding remarks	82
5.2 Limitations	83
5.3 Research Recommendations and Contributions.....	84
REFERENCES.....	87
APPENDICES	91

LIST OF TABLES

Table	Title	Page
2.1	Summary of Reviewed Literature	12
3.1	Air Quality Monitoring Station Locations in Johor	15
3.2	Theoretical characteristics of ACF and PACF of ARMA(p,q) model	20
4.1	Summary Statistics for PM2.5 Concentrations	30
4.2	Comparative analysis of the average daily PM2.5 concentrations exceeding selected threshold	32
4.3	Results of Augmented Dickey-Fuller (ADF) Test for PM2.5 Concentration Time Series	37
4.4	Variations of ARMA Models with Approximated AIC Values for Batu Pahat	44
4.5	Variations of SARMA Models with Approximated AIC Values for Batu Pahat	45
4.6	Variations of ARMA Models with Approximated AIC Values for Kluang	45
4.7	Variations of ARMA Models with Approximated AIC Values for Kota Tinggi	46
4.8	Variations of ARMA Models with Approximated AIC Values for Larkin	46
4.9	Variations of ARMA Models with Approximated AIC Values for Pasir Gudang	46
4.10	Variations of ARMA Models with Approximated AIC Values for Pengerang	46
4.11	Variations of ARMA Models with Approximated AIC Values for Segamat	47
4.12	Variations of SARMA Models with Approximated AIC Values for Segamat	47
4.13	Variations of ARMA Models with Approximated AIC Values for Tangkak	48
4.14	ARMA(1,2) Model Parameter Estimates for Batu Pahat	50
4.15	SARMA(1,0) (1,0) [7] Model Parameter Estimates for Batu Pahat	50

4.16	ARMA(1,2) Model Parameter Estimates for Kluang	50
4.17	ARMA(1,0) Model Parameter Estimates for Kota Tinggi	50
4.18	ARMA(1,0) Model Parameter Estimates for Larkin	51
4.19	ARMA(2,1) Model Parameter Estimates for Pasir Gudang	51
4.20	ARMA(2,2) Model Parameter Estimates for Pengerang	51
4.21	ARMA(4,3) Model Parameter Estimates for Segamat	51
4.22	SARMA(2,2)(0,1)[7] Model Parameter Estimates for Segamat	51
4.23	ARMA(2,2) Model Parameter Estimates for Tangkak	52
4.24	Ljung-Box Test Result for ARMA model Residuals of Batu Pahat	55
4.25	Ljung-Box Test Result for SARMA model Residuals of Batu Pahat	56
4.26	Ljung-Box Test Result for ARMA model Residuals of Kluang	56
4.27	Ljung-Box Test Result for ARMA model Residuals of Kota Tinggi	57
4.28	Ljung-Box Test Result for ARMA model Residuals of Larkin	58
4.29	Ljung-Box Test Result for ARMA model Residuals of Pasir Gudang	59
4.30	Ljung-Box Test Result for ARMA model Residuals of Pengerang	59
4.31	Ljung-Box Test Result for ARMA model Residuals of Segamat	60
4.32	Ljung-Box Test Result for SARMA model Residuals of Segamat	60
4.33	Ljung-Box Test Result for ARMA model Residuals of Tangkak	61
4.34	Forecast Accuracy of Best Models of all 8 Stations	63

4.35	14-Day PM2.5 Forecast for Batu Pahat Using SARMA Model	66
4.36	14-Day PM2.5 Forecast for Kluang Using ARMA Model	66
4.37	14-Day PM2.5 Forecast for Kota Tinggi Using ARMA Model	66
4.38	14-Day PM2.5 Forecast for Larkin Using ARMA Model	67
4.39	14-Day PM2.5 Forecast for Pasir Gudang Using ARMA Model	67
4.40	14-Day PM2.5 Forecast for Pengerang Using ARMA Model	67
4.41	14-Day PM2.5 Forecast for Segamat Using ARMA Model	68
4.42	14-Day PM2.5 Forecast for Tangkak Using ARMA Model	68
4.43	Performance Metrics of Best Models of all 8 Stations	70

LIST OF FIGURES

Figure	Title	Page
4.1	Time Series Plot of Batu Pahat	33
4.2	Time Series Plot of Kluang	33
4.3	Time Series Plot of Kota Tinggi	33
4.4	Time Series Plot of Larkin	33
4.5	Time Series Plot of Pasir Gudang	34
4.6	Time Series Plot of Pengerang	34
4.7	Time Series Plot of Segamat	34
4.8	Time Series Plot of Tangkak	34
4.9	ACF and PACF for ARMA of Batu Pahat	39
4.10	ACF and PACF for SARMA of Batu Pahat	39
4.11	ACF and PACF for ARMA of Kluang	39
4.12	ACF and PACF for ARMA of Kota Tinggi	40
4.13	ACF and PACF for ARMA of Larkin	40
4.14	ACF and PACF for ARMA of Pasir Gudang	40
4.15	ACF and PACF for ARMA of Pengerang	41
4.16	ACF and PACF for ARMA of Segamat	41
4.17	ACF and PACF for SARMA of Segamat	41
4.18	ACF and PACF for ARMA of Tangkak	41
4.19(a)	ACF and PACF for Residuals of ARMA(1,2) on PM2.5 of Batu Pahat	55
4.19(b)	Q-Q Plot of Residual for ARMA(1,2) of Batu Pahat	56
4.20(a)	ACF and PACF for Residuals of SARMA(1,0)(1,0) on PM2.5 of Batu Pahat	56
4.20(b)	Q-Q Plot of Residual for SARMA(1,0)(1,0) of Batu Pahat	56

4.21(a)	ACF and PACF for Residuals of ARMA(1,2) on PM2.5 of Kluang	57
4.21(b)	Q-Q Plot of Residual for ARMA(1,2) of Kluang	57
4.22(a)	ACF and PACF for Residuals of ARMA(1,0) on PM2.5 of Kota Tinggi	57
4.22(b)	Q-Q Plot of Residual for ARMA(1,0) of Kota Tinggi	58
4.23(a)	ACF and PACF for Residuals of ARMA(1,0) on PM2.5 of Larkin	58
4.23(b)	Q-Q Plot of Residual for ARMA(1,0) of Larkin	58
4.24(a)	ACF and PACF for Residuals of ARMA(2,1) on PM2.5 of Pasir Gudang	59
4.24(b)	Q-Q Plot of Residual for ARMA(2,1) of Pasir Gudang	59
4.25(a)	ACF and PACF for Residuals of ARMA(2,2) on PM2.5 of Pengerang	59
4.25(b)	Q-Q Plot of Residual for ARMA(2,2) of Pengerang	60
4.26(a)	ACF and PACF for Residuals of ARMA(4,3) on PM2.5 of Segamat	60
4.26(b)	Q-Q Plot of Residual for ARMA(4,3) of Segamat	60
4.27(a)	ACF and PACF for Residuals of SARMA(2,2)(0,1) on PM2.5 of Segamat	61
4.27(b)	Q-Q Plot of Residual for SARMA(2,2)(0,1) of Segamat	61
4.28(a)	ACF and PACF for Residuals of ARMA(2,2) on PM2.5 of Tangkak	61
4.28(b)	Q-Q Plot of Residual for ARMA(2,2) of Tangkak	62

LIST OF ABBREVIATIONS

ACF	Autocorrelation Function
ADF	Augmented Dickey-Fuller
AIC	Akaike Information Criterion
API	Air Quality Index
AR	Autoregressive
ARMA	Autoregressive Integrated Moving Average
DF	Degree of Freedom
DOE	Department Of Environment
LCL	Lower Confidence Level
MA	Moving Average
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
PACF	Partial Autocorrelation Function
PM2.5	Particulate Matter with Aerodynamic Diameter Less Than $2.5\ \mu m$
RMSE	Root Mean Square Error
TSP	Time Series Plot
UCL	Upper Confidence Level
WHO	World Health Organization

SEASONAL AND NON-SEASONAL ARMA MODELLING FOR PM2.5 CONCENTRATION PREDICTION IN JOHOR

ABSTRACT

This study aims to model PM2.5 concentrations in Johor, Malaysia. With rising concerns surrounding air quality and its detrimental effects on public health, accurate and timely modelling tools have become essential for environmental planning and risk mitigation. This research applies univariate time series models ARMA and its seasonal extension, SARMA across eight air quality monitoring stations in Johor to determine the most reliable predictive approach. Selection of the best model is based on the Akaike Information Criterion (AIC). While, model validation is carried out using residual diagnostics and error metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). A 14-day out-of-sample forecast is performed to assess predictive accuracy. Results shows that ARMA models are suitable for stations with minimal seasonal variation, while SARMA models exhibit improved accuracy in locations with observable seasonality. The findings suggest that forecasting performance varies according to local emission patterns. The study emphasizes the importance of selecting context-appropriate models to support air quality management strategies. It contributes to the advancement of time series forecasting methodologies and offers practical insights for environmental researchers and policymakers.

PEMODELAN ARMA BERMUSIM DAN TIDAK BERMUSIM UNTUK PERAMALAN KEPEKATAN PM_{2.5} DI JOHOR

ABSTRAK

Kajian ini bertujuan untuk memodel kepekatan PM_{2.5} di Johor, Malaysia. Dengan kebimbangan yang semakin meningkat terhadap kualiti udara dan kesannya yang buruk terhadap kesihatan awam, penggunaan alat pemodalan yang tepat dan berkesan adalah penting untuk perancangan alam sekitar dan pengurangan risiko. Kajian ini menggunakan pendekatan model siri masa univariat ARMA dan lanjutan bermusimnya, SARMA bagi lapan stesen pemantauan kualiti udara di Johor untuk menentukan model ramalan yang paling boleh dipercayai. Analisis melibatkan pra-pemprosesan data, semakan kekekalan siri menggunakan ujian Augmented Dickey-Fuller (ADF), serta pengecaman model menerusi graf Autokorelasi (ACF) dan Autokorelasi Separa (PACF). Pemilihan model terbaik dibuat berdasarkan Akaike Information Criterion (AIC). Manakala pengesahan model dijalankan melalui diagnostik sisaan dan metrik ralat seperti Ralat Kuasa Dua Purata (RMSE), Ralat Mutlak Purata (MAE) dan Ralat Peratusan Mutlak Purata (MAPE). Ramalan luar sampel selama 14 hari turut dijalankan untuk menilai ketepatan model. Hasil kajian menunjukkan bahawa model ARMA adalah sesuai untuk stesen yang tidak menunjukkan variasi bermusim yang ketara, manakala model SARMA memberi prestasi ramalan yang lebih baik di lokasi yang mempunyai pola bermusim. Penemuan ini menekankan pentingnya pemilihan model yang bersesuaian mengikut konteks dan corak data tempatan. Kajian ini menyumbang kepada kemajuan kaedah peramalan pencemaran udara dan menyediakan panduan praktikal kepada penyelidik serta pembuat dasar dalam pengurusan kualiti udara.

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Air pollution remains one of the most critical environmental and public health challenges faced globally. Rapid urbanization, industrial activities and increased vehicular emissions have significantly contributed to the deterioration of air quality in many parts of the world. According to the World Health Organization (WHO), exposure to polluted air, especially fine particulate matter, is responsible for millions of premature deaths each year and poses a significant threat to human health and the environment (World Health Organization, 2021). This issue is not only prevalent in highly industrialized countries but also affects developing nations, including Malaysia.

Among the various air pollutants, particulate matter with a diameter of 2.5 micrometres or less (PM_{2.5}) has garnered significant attention due to its harmful impacts. PM_{2.5} consists of microscopic particles small enough to penetrate deep into the respiratory system, reaching the lungs and even entering the bloodstream. Exposure to high levels of PM_{2.5} has been linked to numerous health problems, including respiratory diseases, cardiovascular complications and increased mortality rates (Brook et al., 2010; Lelieveld et al., 2015). From an environmental perspective, PM_{2.5} contributes to haze, reduces visibility and affects the overall quality of the ecosystem (Kumar et al., 2022).

In Malaysia, air pollution has been a growing concern, particularly with the recurring episodes of transboundary haze, often resulting from forest fires in neighbouring countries such as Indonesia (Gaveau et al., 2014). The state of Johor, located in the southern part of Peninsular Malaysia, is among the regions frequently

affected by such haze events. Besides transboundary factors, Johor has experienced rapid urban development and industrial growth, further contributing to deteriorating air quality (Department of Environment Malaysia, 2022). These factors have resulted in fluctuating PM_{2.5} levels, with occasional exceedances of safe limits, posing health risks to the population.

Given these challenges, forecasting air quality, particularly PM_{2.5} concentrations, has become increasingly important. Accurate air quality forecasts play a crucial role in supporting early warning systems, enabling authorities to implement timely mitigation measures and raising public awareness to reduce exposure to unhealthy air. Furthermore, reliable forecasting models are essential for guiding environmental policies aimed at improving air quality and safeguarding public health (Li et al., 2017).

One of the most widely used approaches for air quality forecasting involves time series models, such as the Autoregressive Integrated Moving Average (ARIMA) and its seasonal extension, Seasonal ARIMA (SARIMA). Additionally, Autoregressive Moving Average (ARMA) models used for stationary data serve as a foundational structure for ARIMA, excluding the differencing component. ARMA models are ideal when pollutant time series do not exhibit significant trends (Alexandra et al., 2019). For datasets with seasonal variation, Seasonal ARMA (SARMA) extends this logic by integrating seasonal autoregressive and moving average terms (Jadhav et al., 2023). While ARMA and SARMA are less commonly applied than ARIMA and SARIMA, they offer simplified structures and can be beneficial in modelling stable, seasonally patterned pollution behaviour especially for stations with consistent emissions and meteorological cycles (Alexandra et al., 2019). Together, these models form a robust framework for analysing and

forecasting air pollution. Their adaptability to different station characteristics urban, industrial or rural enhances their relevance for localized environmental planning in Johor and beyond. These models are well-suited for capturing historical patterns, trends and seasonal variations in air pollution data, making them effective tools for short-term air quality prediction. Their interpretability and statistical rigor make them valuable for both scientific research and practical environmental management (Box et al., 2016; Zhang et al., 2018).

1.2 Problem Statement

In the state of Johor, PM_{2.5} concentrations demonstrate significant temporal fluctuations, with certain periods exceeding the health-based thresholds suggested by the World Health Organization (WHO), thereby posing a tangible threat to public health (Hanafi et al., 2019). Numerous national-level studies have examined Malaysia's air pollution trends; however, there is a notable lack of localized forecasting research targeting Johor's PM_{2.5} levels, despite its susceptibility to transboundary haze and intense urban pressures (Sentian et al., 2019; Zaman et al., 2021). Moreover, existing forecasting efforts are often broad and sometimes inconsistent or delayed, limiting their utility for effective policy response, early warning implementation and public awareness campaigns.

Consequently, there is an urgent need for accurate, timely and interpretable forecasting models specifically tailored to Johor, leveraging historical PM_{2.5} data from regional monitoring stations. Such models would empower policymakers and health agencies to enact targeted mitigation strategies, improve proactive public advisories and strengthen the effectiveness of local air quality management interventions.

1.3 Research Objectives

General Objective:

To model PM_{2.5} concentrations in Johor using time series techniques for accurate air quality forecasting.

Specific Objectives:

1. To analyse the trend and seasonal components of PM_{2.5} concentrations in Johor.
2. To develop ARMA and SARMA models for selected PM_{2.5} monitoring stations in Johor.
3. To identify the best-performing model for each monitoring station based on forecasting accuracy.

1.4 Significance of the Study

This study offers significant practical and academic contributions in the field of air quality management and environmental forecasting. From a practical perspective, the findings of this research will assist the Department of Environment (DOE) Malaysia, policymakers and public health officials in making informed decisions to improve air quality in Johor. Accurate forecasting of PM_{2.5} concentrations can support the development of effective early warning systems, enable timely mitigation measures and enhance public awareness regarding air pollution risks. Academically, this study enriches the existing local forecasting literature on PM_{2.5}, particularly in Johor, where localized studies remain limited. The research also contributes towards achieving global Sustainable Development Goals (SDGs), specifically SDG 11, Sustainable Cities and Communities, by providing scientific evidence to support policies aimed at creating healthier and more sustainable urban environments. Furthermore, the use of interpretable statistical models, such as ARMA,

SARMA, ARIMA and SARIMA, promotes transparency and reliability in environmental planning, making the results accessible and understandable to both scientific and policy communities.

1.5 Scope of the Study

This study is geographically limited to the state of Johor, Malaysia, with a specific focus on all eight PM_{2.5} air quality monitoring stations distributed across the region. The inclusion of these stations ensures comprehensive coverage and provides a localized understanding of PM_{2.5} concentration trends within Johor. In terms of the temporal scope, the research utilizes historical daily PM_{2.5} concentration data collected over a one-year period, from January 1, 2021, to December 31, 2021, based on the data provided by the Department of Environment (DOE) Malaysia (Department of Environment Malaysia, 2022). Methodologically, the study is restricted to the use of time series forecasting techniques, specifically the Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) models. These models are widely recognized for their effectiveness in analysing time-dependent data and capturing seasonal patterns in air quality studies (Zhang et al., 2018). The research focuses on a univariate time series analysis, considering only PM_{2.5} concentrations as the primary variable of interest. Other meteorological factors, such as temperature, wind speed and humidity, are excluded from the scope to maintain a focused and interpretable modelling approach.

1.6 Organization of the Thesis

This thesis is structured into five chapters to provide a systematic presentation of the research. Chapter 1 introduces the background of the study, problem statement,

research objectives, significance, scope and organization of the thesis. Chapter 2 presents a review of relevant literature, including studies on air pollution, PM_{2.5}, forecasting techniques and the application of time series models in air quality forecasting. Chapter 3 describes the research methodology, detailing the data sources, data processing steps, model development, forecasting and evaluation methods used in this study. Chapter 4 provides the results and discussion, including the exploratory data analysis, model selection and performance evaluation, out of sample forecast and interpretation of findings. Finally, Chapter 5 concludes the research by summarizing the key findings, discussing limitations and providing recommendations for future research.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter aims to provide a comprehensive review of existing literature related to PM_{2.5} air pollution, with a specific focus on forecasting methods applied in both global and Malaysian contexts particularly in Johor. The purpose is to build a strong foundation for the current study by synthesising prior findings, identifying trends in research approaches and highlighting critical gaps that justify this investigation. The structure of the chapter begins with a general overview of PM_{2.5} and its health impacts, emphasising its significance as a pollutant of concern due to its link to cardiovascular and respiratory diseases (World Health Organization, 2021). It then narrows the focus to Malaysia and Johor, where recurring haze episodes, industrialisation and urbanisation contribute significantly to pollution patterns (Afzali, 2014; Ismail et al., 2016). Following this, the chapter delves into the importance of forecasting PM_{2.5} for early warning systems and public health preparedness, as supported by both local and international studies (Rahman et al., 2016; Lee & Tuen, 2012). A focused discussion on time series forecasting models is presented, particularly ARIMA and SARIMA, which have been widely used due to their simplicity, transparency and suitability for smaller, structured datasets (Li & Zhao, 2022; Abu Bakar et al., 2021). These models are especially relevant in regions like Johor, where high-frequency and long-term pollutant data are often limited. Comparative literature on machine learning approaches is also reviewed, revealing their strengths in accuracy but limitations in interpretability and data demands (Zhang et al., 2024). Additionally, global applications such as those by Kumar and Goyal, (2011) and Yadav and Sahu (2020) offer valuable insights into modelling strategies

under varying climatic and geographic contexts, reinforcing the relevance of time series methods for PM forecasting. Finally, this chapter outlines the main research gaps, specifically, the lack of high-resolution PM_{2.5} forecasting studies tailored to Johor's distinct urban and industrial zones providing strong justification for the present study.

2.2 Time Series Applications in Air Quality Modelling

Air quality forecasting has seen steady growth over the years as concerns about pollution and its health impacts continue to rise. Around the world and in Malaysia, researchers have increasingly focused on developing models that can not only track pollution trends but also predict future air quality levels. These studies are especially important for managing pollutants that pose serious health risks and for enabling proactive environmental planning. Among the various pollutants commonly studied, particulate matter such as PM_{2.5} and PM₁₀, as well as gaseous pollutants like NO₂, are the most frequently analysed due to their direct links to respiratory and cardiovascular diseases. In particular, PM_{2.5} is among the most concerning forms of air pollution due to its ability to penetrate deeply into the respiratory system and reach the bloodstream (World Health Organization, 2021). In forecasting applications, time series models such as ARIMA and SARIMA have been widely used. These models are known for their interpretability, efficiency and suitability for univariate time-dependent data. As explained in the literature, ARIMA is useful for identifying and forecasting general trends, while SARIMA extends this by incorporating seasonal effects, which are especially relevant in areas affected by monsoons, haze events and industrial cycles. Previous studies have validated the usefulness of these models in the Malaysian context. For instance, Lee and Tuen (2012) demonstrated the effectiveness of ARIMA in tracking air quality trends in Johor, while Ismail et al. (2016) highlighted

SARIMA's ability to handle seasonal fluctuations in southern Peninsular Malaysia. Globally, Jain et al. (2022) showed that SARIMA models performed well in forecasting PM_{2.5} in urban areas of Nepal, especially during seasonal transitions. Likewise, Abhiram (2018) applied ARIMA to predict PM₁₀ levels in Chennai, India and found the model effective in producing reliable short-term forecasts suitable for regulatory planning. These findings reinforce the role of traditional time series approaches as transparent and effective tools for air quality forecasting, particularly in regions like Johor where data availability may be limited and machine learning models may be less suitable due to their high computational costs and data requirements.

2.3 Malaysian Studies on PM_{2.5} Modelling

Numerous studies have investigated air pollution trends and forecasting models in Malaysia, reflecting the country's growing concern over environmental and public health impacts. These studies have explored various pollutants and modelling techniques across different regions, aiming to better understand pollution patterns and inform mitigation strategies. Several states and regions have been the focus of such research. For example, Lee and Tuen (2012) demonstrated the effectiveness of ARIMA in tracking air quality trends in Johor, while Ismail et al. (2016) highlighted SARIMA's ability to handle seasonal fluctuations in southern Peninsular Malaysia. Additionally, Afzali (2014) studied PM₁₀ levels in Pasir Gudang, establishing a connection between industrial emissions and meteorological conditions, insights that are equally relevant to PM_{2.5}. These studies primarily employed time series models such as ARIMA and SARIMA, which are well-regarded for their transparency, efficiency and suitability for univariate time-dependent data, especially in regions like Johor where data availability may be limited. Some studies have also incorporated comparisons with more complex models. Abu Bakar et al. (2021 compared ARIMA

with LSTM (Long Short-Term Memory networks) and found that while LSTM produced more accurate results, ARIMA was more accessible and easier to apply in data-scarce environments. Similarly, Hassan et al. (2019) applied ARIMA and Multiple Linear Regression (MLR) models to PM_{2.5} data in Klang Valley, demonstrating the practicality of traditional models in urban environments. Another study by Mohd Fadhil et al. (2020) explored pollutant trends across Penang and Kedah using simpler statistical models, revealing valuable insights but also highlighting the limitations of aggregated data approaches. These findings reinforce the practicality of traditional time series models for local forecasting tasks in Malaysia. Despite this growing body of work, there remains a gap in studies that focus specifically on high-frequency PM_{2.5} forecasting in Johor. Much of the existing literature has either concentrated on PM₁₀ or aggregated API data, which lacks the granularity needed to identify short-term pollution spikes. Furthermore, previous studies rarely account for the differences between industrial zones like Pasir Gudang and urban areas like Johor Bahru. These gaps highlight the need for more location-specific, high-resolution PM_{2.5} studies in Johor, using models tailored to its unique pollution dynamics.

2.4 Research Gaps

Despite the volume of research on air quality in Malaysia, there remains a gap in studies that focus specifically on high-frequency PM_{2.5} forecasting in Johor. Most existing literature has either concentrated on PM₁₀ or aggregated API data, which lacks the granularity needed to identify short-term pollution spikes. This limitation makes it difficult to provide timely and localised air quality warnings. Moreover, while some studies have employed SARIMA, there are few studies using seasonal models specifically for Malaysian PM_{2.5} data. Yet, seasonal effects such as haze events, monsoons and industrial cycles are highly relevant in Malaysia, particularly in

southern regions like Johor. The underutilisation of seasonal models represents a missed opportunity to capture these recurring patterns more accurately. Another notable gap is the insufficient exploration of model accuracy across different locations or stations. Previous studies rarely account for the differences between industrial zones like Pasir Gudang and urban areas like Johor Bahru, even though pollution sources and intensity can vary greatly. For instance, Hasim et al (2020) noted that most regional air quality studies tend to aggregate results without addressing spatial variability in emission sources. Similarly, Ahmad et al. (2021) pointed out that although SARIMA is well-suited for handling cyclical trends, its application remains sparse in Malaysia, especially for PM_{2.5} forecasting at station-level granularity. These issues highlight the need for more location-specific pollution monitoring and modelling that can offer tailored forecasts and improve decision-making for both policymakers and the public.

2.5 Summary

The reviewed literature highlights several key trends in air quality research, especially in the application of time series models like ARIMA and SARIMA for forecasting. These models are valued for their transparency, low data requirements and effectiveness in univariate forecasting settings (Lee & Tuen, 2012; Ismail et al., 2016). In Malaysia, numerous studies have explored air quality modelling using these traditional methods across different pollutants and regions. For instance, Hassan et al. (2019) analysed PM_{2.5} trends in the Klang Valley using ARIMA and MLR, while Mohd Fadhil et al. (2020) provided a statistical overview of PM_{2.5} and PM₁₀ in Northern Malaysia. Afzali (2014) focused on PM₁₀ in Pasir Gudang, highlighting industrial and meteorological influences that are also applicable to PM_{2.5}. Internationally, SARIMA and hybrid ARIMA models have been tested in urban environments such as Beijing (Li & Zhao, 2022) and Kathmandu (Jain et al., 2022),

showing that traditional models can be enhanced with seasonal or decomposed structures. Kumar and Goyal (2011) also demonstrated the use of principal component regression in Delhi, reflecting early efforts to combine statistical tools with pollution data. More recently, Yadav and Sahu (2020) validated hybrid ARIMA methods for daily AQI forecasting in India. Meanwhile, Abhiram (2018) employed ARIMA to track PM10 fluctuations in Chennai. Although machine learning models such as LSTM offer higher accuracy (Wong et al., 2021; Zhang et al., 2024), they demand substantial data and computational power. In data-scarce contexts like Johor, traditional models remain practical. Notably, Ahmad et al. (2021) reviewed seasonal forecasting in Southeast Asia and found limited studies addressing short-term, location-specific modelling. Likewise, Hasim et al. (2020) emphasized spatial differences in PM2.5 sources across Malaysian urban and semi-urban zones, yet many forecasting studies still generalise across regions. In Johor, gaps remain in high-frequency PM2.5 forecasting, particularly in distinguishing pollution dynamics between industrial zones like Pasir Gudang and urban centres like Johor Bahru. This study addresses those gaps by applying ARIMA and SARIMA to station-level data, aiming to generate interpretable, location-specific forecasts that can better guide environmental response and public health actions.

Table 2.1: Summary of Reviewed Literature

Study	Location	Model Used	Pollutant	Accuracy Metric	Key Findings
Lee & Tuen (2012)	Johor	ARIMA (Box-Jenkins)	API	Good fit	Demonstrated ARIMA's reliability for tracking pollution trends in Johor
Rahman et al. (2016)	Southern Malaysia	SARIMA	API	Not specified	SARIMA effectively captured seasonal variations

Afzali (2014)	Pasir Gudang	Trend Analysis	PM10	Not specified	Identified strong weather-pollution relationships in industrial settings
Abu Bakar et al. (2021)	Malaysia	ARIMA vs. LSTM	PM10	RMSE, MAE	ARIMA was efficient for smaller datasets; LSTM more accurate but data-heavy
Zhang et al. (2024)	Urban China	ARIMA, SARIMA, ML	PM2.5	Comparative review	ARIMA offers interpretability despite ML's superior accuracy
Li & Zhao (2022)	Beijing	Hybrid ARIMA	PM2.5	RMSE, MAE	Combining decomposition techniques with ARIMA improved forecast accuracy
Ismail et al. (2016)	Johor & Southern MY	SARIMA	API	Not specified	Highlighted SARIMA's strength for seasonal PM modelling
Wong et al. (2021)	Malaysia (Multiple)	LSTM, SVR, Random Forest	PM2.5	RMSE, MAE	ML models were more accurate but required extensive data and tuning
Kumar & Goyal (2011)	Delhi, India	Principal Component Regression	PM10	RMSE	Highlighted the importance of meteorological factors in pollution forecasting
Yadav & Sahu (2020)	India	ARIMA, Hybrid ARIMA	AQI	RMSE, MAE	Hybrid ARIMA models enhanced prediction accuracy over traditional ARIMA
Jain et al. (2022)	Kathmandu, Nepal	SARIMA	PM2.5	MAE	SARIMA captured seasonal PM2.5 patterns and offered actionable local forecasts

Hassan et al. (2019)	Klang Valley, Malaysia	ARIMA, MLR	PM2.5	Not specified	ARIMA showed reliable trend estimation; MLR incorporated external variables
Abhiram & Sivaramakrishnan (2018)	Chennai, India	ARIMA	PM10	Not specified	Validated ARIMA's utility in short-term PM10 forecasting in urban settings
Mohd Fadhil et al. (2020)	Northern Malaysia	Descriptive Statistics	PM2.5, PM10	Not specified	Statistical summary of pollution trends; emphasized industrial emission effects
Ahmed et al. (2021)	Southeast Asia	Literature Review	PM2.5	Not specified	Reviewed SARIMA's underuse for PM2.5 despite seasonal relevance in the region
Hasim et al. (2020)	Urban & Semi-urban MY	Spatiotemporal Modelling	PM2.5	Not specified	Emphasized regional variability; recommended station-level modelling

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter presents the methodological applied to analyse the PM2.5 concentration data collected from multiple monitoring stations across Johor. The primary objective of this chapter is to describe, justify and detail the processes undertaken to ensure rigorous and reliable time series modelling. Given the nature of air pollution data and its susceptibility to seasonal variations and random fluctuations, time series modelling techniques such as ARMA and SARMA were selected for this study. These models are well-suited for capturing underlying patterns, trends and seasonality in environmental datasets. This chapter provides an overview of the methodological steps, including data sourcing, preprocessing, exploratory analysis, stationarity testing, model development, diagnostics, forecasting and discussion of methodological limitations. This study uses R studio software to run the analysis.

3.2 Data Sources

The PM2.5 concentration data used in this study were obtained from the Department of Environment (DOE) Malaysia, an official government body responsible for monitoring air quality across the country. The dataset comprises daily average PM2.5 concentrations recorded at eight air quality monitoring stations located in Johor. These stations are situated in Batu Pahat, Kluang, Kota Tinggi, Larkin, Pasir Gudang, Pengerang, Segamat and Tangkak, providing broad geographical coverage that includes both urban and suburban areas. The data spans a full one-year period, from 1st January to 31st December 2021, offering a consistent and complete dataset for analysis. The daily frequency of the data enables the identification of short-term variations, seasonal patterns and long-term trends in PM2.5 concentrations across the

state. In this study, only PM_{2.5} data were considered, with no inclusion of meteorological variables such as temperature, wind speed or humidity.

Table 3.1: Air Quality Monitoring Station Locations in Johor

Station Name and Location	Latitude	Longitude	Zone	Classification
Segamat Klinik Kesihatan Bandar IOI, Segamat	02°29'38.09" N	102°51'45.69" E	South	Sub Urban
Batu Pahat MRSM Batu Pahat, Batu Pahat	01°55'09.56" N	102° 51' 59.82" E	South	Sub Urban
Kluang Kolej Kejururawatan Kluang, Kluang	02°02'16.37" N	103° 18' 43.42" E	South	Rural
Larkin Institut Perguruan Temenggong Ibrahim, Larkin	01°29'40.65" N	103° 44' 09.50" E	South	Urban
Pasir Gudang Sek. Men. Pasir Gudang 2, Pasir Gudang	01°28'12.43" N	103° 53' 36.44" E	South	Sub Urban
Pengerang Sek. Keb. Lepau, Pengerang	01° 23' 22.16" N	104° 08' 58.50" E	South	Industrial
Kota Tinggi Sek. Men. Agama Bandar Penawar, Kota Tinggi	01°33'50.60" N	104° 13' 31.10" E	South	Sub Urban
Tangkak Tapak Semaian Majlis Daerah Tangkak	2.294794	102.571595	Johor	Sub urban

3.3 Data Preprocessing

Before conducting the analysis, the dataset underwent a number of preparation steps to verify its completeness, accuracy and readiness for time series modelling. The datasets for all eight sites were thoroughly checked for missing values. The results verified that there were no missing values, which eliminated the requirement for data imputation or interpolation. Outliers in the data were identified by visual inspections of the time series plots and summary statistics. Although some high PM_{2.5} values

were found, particularly during recognized haze periods or pollution events, these results were deemed acceptable and included in the dataset to preserve the accuracy of real-world air quality conditions. No data treatments, such as log transformation, were used because preliminary visual inspections suggested that the raw data were adequate for modelling. Time formatting and alignment were conducted to ensure consistency across all stations, with dates uniformly formatted and datasets synchronised for accurate comparison and analysis.

3.4 Exploratory Data Analysis (EDA)

An exploratory data analysis was conducted to understand the distribution and temporal behaviour of PM_{2.5} concentrations across Johor. Descriptive statistics is a set of techniques used to summarize and convey the primary characteristics of a dataset. The objective of descriptive statistics is to concisely identify the monitoring data. Typically, these statistics provide information regarding a dataset's central tendency (minimum, mean, maximum), variability (standard deviation) and distribution (skewness and kurtosis). Descriptive statistics, including the number of observations (N), mean, median, standard deviation, minimum and maximum values, were calculated for each station to summarise air quality conditions. The standard deviation measures the average departure of each data point from the mean, whereas the coefficient of variation shows the degree of dispersion around the mean. Skewness analyses the asymmetry of a data distribution compared to its mean, whereas kurtosis evaluates its peak. In a normal distribution, the kurtosis is zero.

Time series plots were created for each station to help see patterns, trends and changes across time. These visualisations highlighted short-term variability, mid-year peaks that are most likely caused by seasonal haze occurrences and probable long-term stochastic tendencies.

3.5 Stationary Time Series

Time series data can be classified as either stationary or non-stationary time series. For time series analysis, data series must meet the stationarity assumption before fitting a model. To be considered stationary, it must meet three key characteristics. The series' mean should not change over time. Additionally, homoscedasticity requires that the series' variance be constant across time. If this criterion is not met, the series may show Autoregressive Conditional Heteroskedasticity (ARCH) effects. Finally, the covariance between terms should be based on the lag difference, not time. A series that fits all three characteristics is considered weakly stationary. If any of these conditions are not met, the series is considered non-stationary.

There are a few varieties of stationarity. The first form involves rigorous stationarity. Strict stationarity is mathematically defined as $\{y_1, y_2, \dots, y_n\}$ and $\{y_{1+h}, y_{2+h}, \dots, y_{n+h}\}$ having the same joint distributions for any integers h and $n > 0$. Second-order weak stationarity describes a stochastic process with constant mean, variance, autocovariance and autocorrelation throughout time. Assuming normality, this accurately describes the process.

For a time series $\{Y_t\}$ to be considered stationary, the following conditions must hold:

1. Constant Mean

$$\mu_t = E(Y_t) = \mu$$

The expected value (mean) is time-invariant.

2. Constant Variance

$$\sigma_t^2 = Var(Y_t) = \sigma^2$$

The dispersion of the data doesn't change over time.

3. Autocovariance Depends Only on Lag

$$\gamma(t, t + k) = Cov(Y_t, Y_{t+k}) = \gamma_k$$

The strength of the linear relationship between two observations depends only on the lag k , not the actual time points.

4. Autocorrelation Function (ACF)

$$\rho(t, t + k) = \frac{\gamma_k}{\gamma_0} = \rho_k$$

Normalized autocovariance; again, it should depend only on k .

Stationarity testing was a crucial step to determine the suitability of the data for ARMA and SARMA modelling. In time series analysis, stationarity ensures that the statistical properties of the series, such as mean and variance, remain constant over time. The Augmented Dickey-Fuller (ADF) test is one of the most widely used statistical tests to determine whether a time series is stationary or has a unit root (non-stationary).

3.5.1 Augmented Dickey-Fuller (ADF Testing for Stationarity)

The Augmented Dickey-Fuller (ADF) test is used to assess stationarity with the following hypotheses:

H_0 : The series has a unit root (non-stationary)

H_1 : The series is stationary

The ADF test equation is given by:

$$\Delta y_t = \alpha + \beta_t + \gamma y_{t-1} + \sum_{i=1}^p \theta_i \Delta y_{t-i} + \epsilon_t$$

Where Δy_t is the first difference of the series, γ is the key coefficient tested (if $\gamma = 0$, unit root exists) and p is the lag order to account for autocorrelation. The ADF test examines the presence of a unit root, with the null hypothesis indicating non-stationarity. If the p-value of the test is below 0.05, the null hypothesis is rejected, confirming stationarity.

3.5.2 Differencing

Differencing is a critical transformation technique applied to non-stationary time series data to achieve stationary in mean. This transformation eliminates trends by computing the differences between observation across time, by stabilizing the mean and enabling effective time series modelling.

If X_t denotes the value of a time series at time t , the first order difference is calculated as:

$$\Delta X_t = X_t - X_{t-1} = X_t - B X_t = (1 - B)X_t$$

Where ΔX_t is the differences value at time t . B is the backward shift operator such that $B X_t = X_{t-1}$.

Applying first-order differencing removes linear trend components. However, some series may still exhibit stochastic drift or curvature even after this transformation. In such cases, higher-order differencing is needed to achieve stationarity. The number of differencing operations required is denoted by d , referred to as the order of integration. For example, second-order differencing is defined as:

$$\Delta^2 X_t = \Delta(\Delta X_t) = (1 - B)^2 X_t$$

Once a series becomes stationary after differencing d times, the resulting transformed data can be used to estimate an ARIMA(p, d, q) model, where p is the autoregressive order, q is the moving average order, and d is the non-seasonal differencing order.

In addition to non-seasonal differencing, seasonal differencing is applied when the time series exhibits repeating seasonal patterns common in environmental or climatological data like PM2.5. Seasonal differencing at lag s is expressed as:

$$\Delta_s X_t = X_t - X_{t-s} = (1 - B^s)X_t$$

Where s is the seasonal period, $\Delta_s X_t$ represents the seasonally differenced series. To handle both trend and seasonal variation, combined differencing may be applied as:

$$\text{Differenced } X_t = (1 - B)^d \times (1 - B^s)^D X_t$$

Where d is the order of non-seasonal differencing, D is the order of seasonal differencing, s is the seasonal lag. This formulation prepares the series for SARIMA(p,d,q)(P,D,Q)[s] modelling, which captures both short-term and seasonal temporal dependencies.

3.6 Time Series Model

3.6.1 Autoregressive (AR) Model

The autoregressive model involves regressing a value from a time series against prior values from the same series. The fundamental autoregressive model, AR (1), has only one parameter, as shown below:

$$Y_t = \varphi^1 Y_{t-1} + \varepsilon_t$$

The general autoregressive model of order p can be mathematically expressed as General AR(p):

$$Y_t = \varphi^1 Y_{t-1} + \varphi^2 Y_{t-2} + \dots + \varphi_p Y_{t-p} + \varepsilon_t$$

where $\varphi^1, \varphi^2, \dots, \varphi_p$ are p unknown parameters. The AR (p) model is applied to a time series when the autocorrelation function (ACF) demonstrates tailing off exponentially quickly pattern and the Partial Autocorrelation Function (PACF) displays a cut off at lag p .

3.6.2 Moving Average (MA) Model

In contrast, the moving average model (MA(q)) connects current observations to historical errors. The MA model is expressed broadly as follows, with MA (1) being the most fundamental.

$$Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}$$

where $\theta_1, \theta_2, \dots, \theta_q$ are q unknown parameters. The MA (q) model is obtained when the autocorrelation function (ACF) shows a cut off at lag q and the Partial Autocorrelation Function (PACF) displays decay off pattern.

3.6.3 Autoregressive Moving Average (ARMA) Model

When building a model, it's often best to combine autoregressive and moving average methods. For stationary time series, an ARMA(p, q) model to the stationary series obtained by transforming $\{Y_t\}$ into $X_t = \Delta^d Y_t$. Under this setting, Equation above can be rewritten as an ARMA model:

$$\phi_p(B)X_t = \theta_q(B)\varepsilon_t.$$

where X_t is the stationary time series, $\phi_p(B)$ and $\theta_q(B)$ are polynomials in the backshift operator B , and ε_t represents white noise.

3.6.4 Autoregressive Integrated Moving Average (ARIMA) Model

However, many real-world time series, including environmental indicators like PM2.5 concentrations, are non-stationary and thus require transformation before applying ARMA. This leads to the use of the Autoregressive Integrated Moving Average (ARIMA) model, which introduces an additional component, differencing. The ARIMA(p, d, q) model incorporates the differencing operator of order d to remove trends and achieve stationarity. The model can be written as:

$$\phi_p(B)(1 - B)^d Y_t = \theta_q(B)\varepsilon_t$$

Where $\varphi_p(B)$ = autoregressive polynomial, $\theta_q(B)$ = moving average polynomial, B = backshift operator and ε_t = white noise error term

3.6.5 Seasonal Autoregressive Moving Average (SARMA) Model

The Seasonal Autoregressive Moving Average (SARMA) model extends the classical ARMA approach by incorporating seasonal dependencies into time series analysis. This model is suited for data that are already stationary but exhibit seasonal autocorrelation common in environmental and meteorological datasets. The SARMA model includes both non-seasonal ARMA terms and additional seasonal terms applied at lag multiples of the seasonal cycle. It is denoted as SARMA(p,q)(P,Q)[s]. where p and q are the non-seasonal AR and MA orders, P and Q are the seasonal AR and MA orders, and s is the length of the seasonal period. The general SARMA equation is given by:

$$\varphi_p(B^s) \varphi_p(B) X_t = \theta_Q(B^s) \theta_q(B) \varepsilon_t$$

where, X_t is a stationary series, $\varphi_p(B)$ and $\theta_q(B)$ are the non-seasonal AR and MA polynomials, $\varphi_P(B^s)$ and $\theta_Q(B^s)$ are the seasonal AR and MA polynomials with backshifts at lag s and ε_t is a white noise term. SARMA is particularly useful when the data is already differenced and stationarity has been achieved, yet seasonal structures still exist in the residual patterns.

3.6.6 Seasonal Autoregressive Integrated Moving Average (SARIMA) Model

When the time series data exhibit seasonal pattern, it's essential to model both non-seasonal and seasonal components. The SARIMA model extends ARMA by incorporating seasonal autoregressive and moving average terms along with seasonal differencing. The SARIMA(p,d,q)(P,D,Q) model is expressed as follows:

$$\Phi_p(B^s) \phi_p(B) (1 - B)^d (1 - B^s)^D Y_t = \theta_q(B^s) \theta_q(B) \varepsilon_t$$

Where, $\phi_p(B)$ = non-seasonal autoregressive (AR) polynomial, $\theta_q(B)$ = non-seasonal moving average (MA) polynomial, $\Phi_p(B^s)$ = seasonal autoregressive (SAR) polynomial, $\theta_q(B^s)$ = seasonal moving average (SMA) polynomial, B = backshift operator, s = seasonal period, d = order of non-seasonal differencing, D = order of seasonal differencing and ε_t = white noise error term.

3.7 Model Identification and Selection

The ARMA and SARMA models were selected for this study due to their robustness in capturing both non-seasonal and seasonal components of time series data. ARMA models incorporate three components the autoregressive (AR) terms, which capture the influence of past observation and moving average (MA) terms, which account for past forecast errors. In cases where seasonal patterns were evident, SARMA models were considered, extending the ARMA framework by including seasonal autoregressive and moving average terms, along with a defined seasonal period (s). Model identification was guided by the examination of Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots for each station's PM2.5 time series.

3.7.1 Autocorrelation Function (ACF)

The ACF plots the correlation coefficient against the lag, which is measured in terms of periods. A lag which denoted as k represents the interval after which a particular observation occurs in the time series.

For a stationary process $\{X_t\}$, with

$$E(X_t) = \mu,$$

$$\text{Var}(X_t) = E(X_t - \mu)^2 = \sigma^2,$$

the degree of dependency between two observations, k lags apart is determined by autocovariance, which is expressed as:

$$\gamma_k = \text{Cov}(X_t, X_{t+k}) = E(X_t - \mu)(X_{t+k} - \mu),$$

and the autocorrelation is computed as

$$\rho_k = \frac{\text{Cov}(X_t, X_{t+k})}{\sqrt{\text{Var}(X_t)}\sqrt{\text{Var}(X_{t+k})}} = \frac{\gamma_k}{\gamma_0},$$

where $\text{Var}(X_t) = \text{Var}(X_{t+k}) = \gamma_0$

Plotting the ACF helps see the correlation between time series observations over different lags, which is important for recognizing trends and developing appropriate models.

3.7.2 Partial Autocorrelation Function (PACF)

To consider the impact of all intermediate observations, the Partial Autocorrelation Function (PACF) is used. This function measures the dependence between X_t and X_{t+k} , while accounting for the effect of the intermediate observations, and is represented by ρ_{kk} :

$$\text{Corr}(X_t, X_{t+k} | X_{t+1}, \dots, X_{t+k-1}) = \rho_{kk},$$

This implies that when computing the partial dependence, we must consider the dependencies of all intermediate variables between them. These plots provided insights into the appropriate lag structures for the AR and MA terms, as well as the presence of seasonality.

The pattern of the ACF and PACF plots of the series are observed in order to determine the order of the ARMA and SARMA model. However, this order determination does not guarantee the model is the most appropriate model to fit the

series. Despite that, a good understanding on the patterns of common theoretical ACF and PACF plots can significantly increase the probability of discovering an appropriate model in a concise timeframe. Table 3.1 below has summarized the theoretical non-seasonal models derived from examining the correlograms.

Table 3.2: Theoretical characteristics of ACF and PACF of ARMA(p, q) model

Process	ACF	PACF
AR(p)	Tails off toward zero	Cuts off at lag p
MA (q)	Cuts of at lag q	Tails off to zero
ARMA(p, q)	Tails off	Tails off

Preliminary model specifications were determined based on the observed patterns and the *auto.arima* function in RStudio was employed to refine model selection by minimising the Akaike Information Criterion (AIC).

3.7.3 Model Selection using auto.arima()

For this study, model selection was conducted using the *auto.arima()* function from the forecast package in RStudio. This method automates the process of identifying the best-fitting ARMA or SARMA model by evaluating multiple parameter combinations and selecting the one that minimizes the Akaike Information Criterion (AIC). The formula for AIC in the context of ARMA models is as follows:

$$AIC(p, q) = \ln\left(\frac{\sum \varepsilon_t^2}{n}\right) + \frac{2}{n}(p + q)$$

Where $\sum \varepsilon_t^2$ is sum of squared residuals, n is number of observations and p, q is orders of the ARMA model. It is important to note that AIC values can be negative when the term $\ln\left(\frac{\sum \varepsilon_t^2}{n}\right)$ is smaller than $\frac{2(p+q)}{n}$. In such cases, a more negative AIC indicates a better-fitting model.

The `auto.arima()` function performs a stepwise search algorithm across possible values of the non-seasonal components (p,d,q) and, where applicable, the seasonal components (P,D,Q)[s] automatically accounts for stationarity and selects the model with the lowest AIC among the candidates. The selected model was then evaluated further using residual diagnostics and forecast accuracy metrics to confirm its adequacy for PM2.5 prediction.

3.7.4 Parameter Estimation

Model estimation was conducted using the Maximum Likelihood Estimation (MLE) approach, which is widely recognised for its efficiency and reliability in time series analysis. Maximum likelihood estimation (MLE) is similar to the least squares estimates that is obtained by minimizing $\sum_{t=1}^T \varepsilon_t^2$, the parameter estimates should be significant at the 5% level and meet the requirements of stationarity and invertibility. The RStudio software environment was used to implement ARMA and SARMA models, packages such as *forecast* and *tseries*. Parameter inclusion in the final models was based on statistical significance, with only parameters demonstrating p-values below 0.05 retained in the final model specifications.

3.8 Diagnostics Checking

Diagnostic checking on the model is essential in order to examine the goodness-of-fit of a proposed model. If the model is statistically inadequate, it is crucial to identify and implement modifications to improve the model's fit to the time series data. A key assumption in ARMA and SARMA modelling is that the error terms follow a white noise process, denoted as $\varepsilon_t \sim NID(0, \sigma_\varepsilon^2)$ meaning they are normally and independently distributed with zero mean and constant variance.

After fitting a time series model like ARMA or SARMA, it is vital to ensure that the model accurately captures the underlying data structure. This is often determined by looking at the model residuals, the difference between observed and fitted values. Proper diagnostic checks determine whether the residuals have the characteristics of white noise, such as zero mean, constant variance and no autocorrelation. Diagnostic methods used in this investigation were the autocorrelation function (ACF) and Partial Autocorrelation Function (PACF) plots, the quantile-quantile (Q-Q) plot and the Ljung-Box portmanteau test.

3.8.1 ACF and PACF Plots of Residuals

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of the residuals are useful visual tools for detecting the presence of autocorrelation that may remain after model fitting. These plots allow for examination of whether the model residuals behave like a white noise process. In the ACF plot, autocorrelation coefficients of residuals are plotted against different lags to reveal any cyclical or repetitive structure. Similarly, the PACF plot provides insight into the direct correlation between residuals at a given lag after accounting for the contributions of intermediate lags.

For a model to be considered well-fitted, the ACF and PACF plots of the residuals should show all spikes falling within the 95% confidence interval, which is represented by horizontal bands on the plot. When residual autocorrelation spikes lie outside this boundary, it suggests that the model has not fully captured the time-dependent structure of the data. In this study, the residual plots were evaluated for each fitted model to ensure there was no significant autocorrelation remaining. This step is essential in confirming that the model has adequately accounted for the underlying dependencies and that the forecasts generated are reliable.

3.8.2 Test for Normality using Q-Q Plots

A statistically adequate model has normally distributed residuals. The normality of residuals can be investigated by plotting a Q-Q normal plot of the residuals. If the plot shows most of the residuals lie on the reference line, this implies that the model captures the main patterns and sources of variation in the data series and that the errors are random and independent.

3.8.3 Ljung-Box Portmanteau Test

The Ljung-Box test was used to determine if the residuals of the fitted ARMA or SARMA models were independently distributed. This test is critical for determining whether the residuals match a white noise process, which is a necessary assumption for the validity of time series models. White noise residuals indicate that the model has caught all meaningful temporal patterns in the data, with no substantial autocorrelation remaining unexplained. The Ljung-Box test examines a set of autocorrelations at different delays and determines if they are all equal to zero. The hypotheses for the test are as follows:

Hypotheses:

H_0 : The residuals are independently distributed (no autocorrelation)

H_1 : The residuals are not independently distributed (autocorrelation is present)

Test Statistic:

$$Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}$$

Where Q is the Ljung-Box test statistic, ρ sample autocorrelation at lag k , n is the sample size and h is the number of lags considered. In this study, the Ljung-Box

test was performed on the residuals of each fitted model with an appropriate lag length. A model was regarded acceptable if the p-value of the test above the 0.05 significance level, suggesting that no substantial autocorrelation existed.

3.9 Forecasting

Short-term forecasts of 14 days for PM2.5 concentrations were generated using the ARMA and SARMA models, following model selection and validation. Forecasting is the process of predicting future values based on patterns detected in historical data. ARMA models were utilized for non-seasonal time series, whereas SARMA models were employed for data with strong seasonal characteristics. RStudio's forecast() function was used to provide predictions with 95% confidence intervals. These intervals represent a statistical range that captures the uncertainty surrounding each anticipated value.

3.9.1 Forecast Accuracy Metric

To evaluate the performance of the models', forecast accuracy was assessed using three standard metrics:

- 1) Root Mean Square Error(RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (Y_t - \hat{Y}_t)^2}$$

Indicates the average magnitude of squared errors More sensitive to larger errors.

- 2) Mean Absolute Error(MAE)

$$MAE = \frac{1}{n} \sum_{t=1}^n |Y_t - \hat{Y}_t|$$

Measures the average size of errors regardless of their direction.

- 3) Mean Absolute Percentage Errors(MAPE):

$$MAPE = \frac{100}{n} \sum_{t=1}^n \frac{|Y_t - \hat{Y}_t|}{Y_t}$$

Expresses error as a percentage making it interpretable across different scales.

Each of these metrics was used to compare model prediction against actual observed values. Lower value indicates better forecast performance and model accuracy.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results of the statistical analysis and forecasting of daily PM_{2.5} concentration levels in Johor, Malaysia, based on data collected from eight air quality monitoring stations for the year 2021. The purpose of this chapter is to provide a detailed explanation of the data characteristics, assess stationarity, identify the most suitable time series models, evaluate model adequacy and generate short-term forecasts for PM_{2.5} levels across the selected locations.

The analysis in this chapter follows the methodology outlined in Chapter 3, which involves applying the Autoregressive Moving Average (ARMA) and Seasonal ARMA(SARMA) models to the PM_{2.5} time series data. These models are widely recognized in air quality research for their ability to model temporal dependencies, account for seasonality and produce reliable short-term forecasts. ARMA models are particularly effective in capturing linear patterns and trends in time series data, while SARMA models extend this approach by incorporating seasonal components, making them suitable for datasets that exhibit periodic fluctuations, such as weekly or monthly cycles.

In this study, the seasonal period considered is seven days, to account for potential weekly variations in PM_{2.5} concentrations. The chapter begins with a descriptive analysis of the dataset, including summary statistics and time series plots for each station. This is followed by stationarity testing using the Augmented Dickey-Fuller (ADF) test, which is necessary to ensure the data meet the assumptions required for ARMA and SARMA modelling. Subsequent sections present the model identification process, including Autocorrelation Function (ACF) and Partial

Autocorrelation Function (PACF) plots, as well as model selection based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). The chapter concludes with model diagnostics, interpretation of the results and short-term forecasts for PM2.5 concentrations at each station.

4.2 Exploratory Data Analysis

An exploratory data analysis was conducted to provide an overview of the PM2.5 concentration levels across the eight selected air quality monitoring stations in Johor for the year 2021. This analysis helps to understand the general characteristics of the dataset, including the distribution, central tendency and variability of PM2.5 concentrations.

4.2.1 Summary Statistics

The PM2.5 data were obtained from the Department of Environment (DOE) Malaysia for eight monitoring stations located across Johor which are Larkin, Pasir Gudang, Kota Tinggi, Pengerang, Kluang, Segamat, Batu Pahat and Tangkak. Each station provided average daily PM2.5 concentration readings for the year 2021, measured in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$). A total of 365 observations were recorded for each station, corresponding to one reading per day, with no missing values throughout the dataset.

The summary statistics for each station including mean, median, standard deviation, minimum, maximum, range, skewness and kurtosis are presented in Table 4.1, offering a comprehensive overview of the distribution and variability of PM2.5 concentrations across the selected locations.

Table 4.1: Summary Statistics for PM2.5 Concentrations

Station	Mean	Median	Std. Dev.	Min	Max	Range	Skewness	Kurtosis
---------	------	--------	-----------	-----	-----	-------	----------	----------

Pasir Gudang	13.72	13.29	4.82	4.06	29.05	24.99	0.52	-0.02
Larkin	16.57	15.82	5.47	5.63	35.88	30.25	0.60	0.18
Pengerang	11.48	10.64	5.17	2.12	39.52	37.40	2.13	7.33
Kluang	10.60	9.71	4.30	2.14	24.35	22.21	0.74	0.25
Segamat	13.32	12.35	5.63	3.38	47.63	44.26	1.59	6.04
Kota Tinggi	10.14	9.56	3.96	2.38	24.81	22.43	0.71	0.30
Batu Pahat	12.69	11.40	5.85	3.01	40.35	37.34	1.47	3.54
Tangkak	13.87	12.60	5.82	5.76	50.49	44.73	2.26	8.73

The descriptive statistics reveal notable variations in PM_{2.5} concentration levels across the eight monitoring stations. Larkin recorded the highest mean PM_{2.5} concentration of 16.57 µg/m³, reflecting the influence of urbanisation, traffic congestion and industrial activities in the area. In contrast, Kota Tinggi reported the lowest mean PM_{2.5} level, at 10.14 µg/m³, indicating relatively better air quality conditions, likely due to its less urbanised environment. Other stations such as Pasir Gudang and Batu Pahat recorded moderate mean concentrations of 13.72 µg/m³ and 12.69 µg/m³ respectively, while areas such as Kluang, Segamat, Pengerang, and Tangkak reported mean PM_{2.5} levels within a similar range.

The standard deviation for each station was examined to assess the level of variability in PM_{2.5} concentrations throughout the year. Tangkak and Batu Pahat exhibited relatively high standard deviations of 5.82 µg/m³ and 5.85 µg/m³ respectively, indicating significant daily fluctuations in air quality in these locations.

Similarly, Segamat and Larkin also showed high variability, suggesting that these stations are subject to irregular pollution events that influence PM_{2.5} levels.

Across the stations, the minimum daily PM_{2.5} values ranged between 2.12 µg/m³ and 5.76 µg/m³, while the maximum values varied significantly, with Tangkak recording the highest daily PM_{2.5} concentrations at 50.49 µg/m³. Such extreme values may be linked to specific environmental events, such as haze episodes or periods of unfavourable weather that trap pollutants near the ground. Other stations, such as Segamat and Batu Pahat, also recorded high maximum PM_{2.5} levels of 47.63 µg/m³ and 40.35 µg/m³ respectively, highlighting the occurrence of occasional but severe air pollution incidents in these areas.

The following Table 4.2 presents a comparative analysis of the number of days in 2021 during which average daily PM_{2.5} concentrations exceeded key thresholds recorded at eight air quality monitoring stations in 2021.

Table 4.2: Comparative analysis of the average daily PM_{2.5} concentrations exceeding selected threshold

Station	Exceeds WHO Guideline (25 µg/m³)	Exceeds Malaysia Guideline (35 µg/m³)	Exceeds 20 µg/m³
Pasir Gudang	7	0	9
Larkin	29	1	92
Pengerang	8	4	16
Kluang	0	0	11
Segamat	8	3	41
Kota Tinggi	0	0	5
Batu Pahat	13	4	33
Tangkak	17	3	40

The table above highlights daily exceedance frequencies for three concentration benchmarks which are 20 µg/m³, the WHO guideline of 25 µg/m³, and the Malaysian Ambient Air Quality Standard of 35 µg/m³. Notably, Larkin recorded

the highest number of exceedance days, surpassing the WHO guideline 29 times and exceeding $20 \mu\text{g}/\text{m}^3$ on 92 days indicating prolonged exposure to moderate PM2.5 pollution. Segamat also showed relatively high exceedances, including 41 days above $20 \mu\text{g}/\text{m}^3$ and 8 days above the WHO guideline, while Pasir Gudang and Pengerang registered fewer, more intermittent violations. Stations such as Tangkak and Batu Pahat displayed moderate frequency exceedances, suggesting the presence of persistent background pollution rather than acute peaks. In contrast, Kluang and Kota Tinggi recorded consistently low pollution levels, with no exceedances above $25 \mu\text{g}/\text{m}^3$ or $35 \mu\text{g}/\text{m}^3$, and fewer than 12 days above the $20 \mu\text{g}/\text{m}^3$ benchmark reflecting relatively clean and stable air quality. Overall, the spatial distribution of exceedances highlights significant variability in PM2.5 exposure risk, emphasizing the need for targeted, location-specific air quality interventions in higher-risk areas like Larkin and Tangkak.

4.2.2 Time Series Plots

To visually explore the temporal behaviour of PM2.5 concentrations across the eight monitoring stations in Johor, time series plots were generated for each location. These plots which are Figure 4.1 to 4.8 illustrate the daily fluctuations in PM2.5 levels throughout the year 2021 and provide important insights into trends, peaks and potential seasonal patterns within the data.

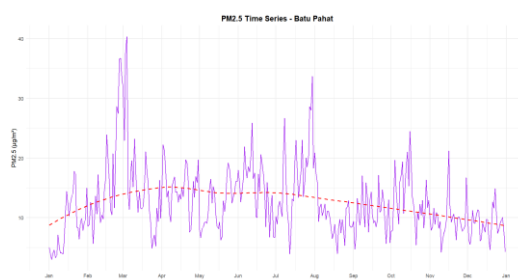


Figure 4.1 Time Series Plot of Batu Pahat

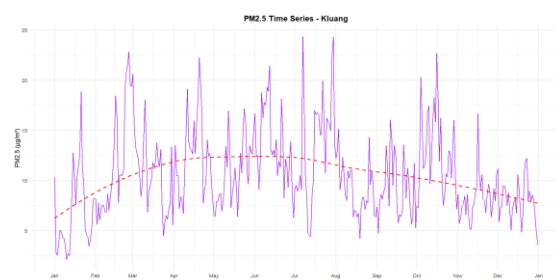


Figure 4.2 Time Series Plot of Kluang

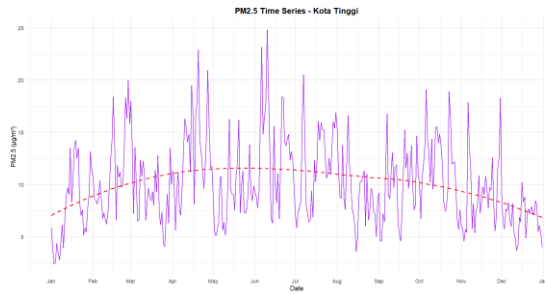


Figure 4.3 Time Series Plot of Kota Tinggi

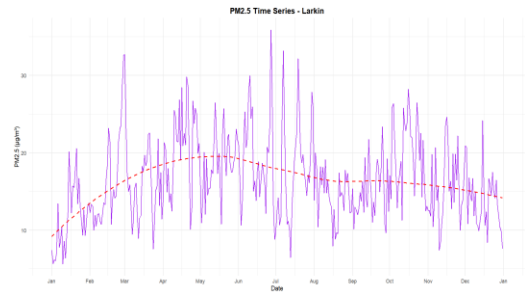


Figure 4.4 Time Series Plot of Larkin

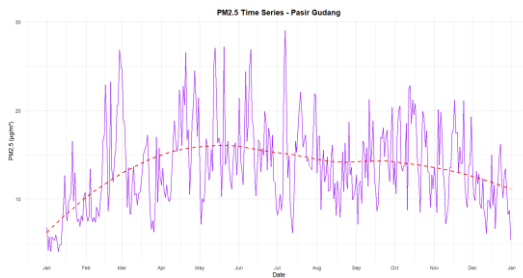


Figure 4.5 Time Series Plot of Pasir Gudang

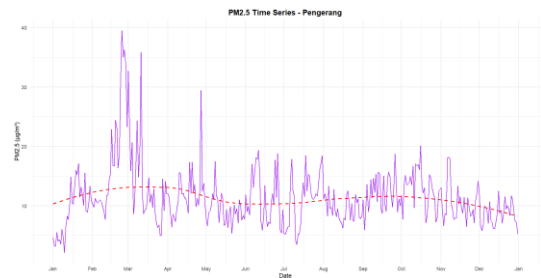


Figure 4.6 Time Series Plot of Pengerang

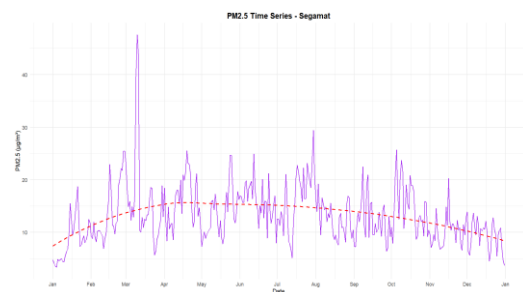


Figure 4.7 Time Series Plot of Segamat

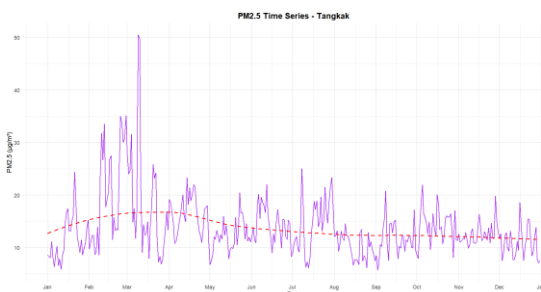


Figure 4.8 Time Series Plot of Tangkak

To investigate the annual behaviour of PM2.5 concentrations in Johor, time series plots were created for each of the eight monitoring stations over the entire year of 2021. These plots show the daily average PM2.5 concentrations for 365 days, from January 1st to December 31st, 2021 and provide information on short-term volatility, long-term trends and potential seasonal patterns.

The time series plot for Batu Pahat shows significant short-term changes, including dramatic peaks and troughs in PM2.5 concentrations. A growing trend is seen from early January to early March, followed by a slow fall until early May. The series then increases again, peaking about mid-May, before commencing a long-term decreasing trend that lasts until the end of the year, with only rare variations. The absence of a clear deterministic trend and the continuous variability suggest the presence of a long-term stochastic trend, where PM2.5 levels vary over time without stabilising around a consistent mean.

Similarly, the time series plot for Kluang shows persistent short-term swings, with frequent increases and decreases in concentration levels, indicating transient pollution episodes. The first few months of the year exhibit modest PM2.5 levels, then steadily rise through April and May, ending in a significant mid-year surge possibly caused by seasonal pollution events or haze. This is followed by a slow but uneven drop punctuated by occasional surges, indicating a stochastic trend influenced by both seasonal patterns and random environmental variables.

The time series for Kota Tinggi shows continuous short-term changes throughout the year, with slight upward movement at the start of the year, followed by minor oscillations. A considerable peak appears in the middle months, likely due to seasonal haze incidents. The series then falls until December, broken by sudden, brief spikes. The overall pattern lacks a deterministic trend, with stochastic pattern dominate the changes, indicating that seasonal and random causes influence PM2.5 levels.

The time series plot for Larkin indicates regular fluctuations in PM2.5 levels, with a progressive increase from March to April, culminating around mid-year. This height is most likely caused by seasonal variables like dry weather or haze. Following

the peak, the series gradually declines but remains uneven, with occasional abrupt rises. The irregular changes and the lack of a strong, predictable trend suggest a stochastic trend influenced by both environment and human influences.

Pasir Gudang's PM_{2.5} concentrations also fluctuate in the short term, with dramatic peaks and dips throughout the year. The early months are rather stable, followed by a gradual increase through March and April, culminating in a significant mid-year high. This heightened phase could be attributed to seasonal variables like smog or low rainfall. The latter half of the year exhibits an overall decreasing trend, although sporadic upward and downward movements remain, indicating a stochastic pattern with no clear long-term mean.

The Pengerang plot shows frequent short-term variations, including rapid climbs and sudden drops, indicating the changeable character of local air quality. A noticeable peak appears at week 10, indicating a brief but significant pollution incident, followed by an uneven decline with sporadic spikes. The lack of a recurrent pattern, as well as the overall unpredictability, indicate a stochastic trend in PM_{2.5} levels, which is driven by seasonal fluctuations, industrial activity and unpredictable environmental factors.

Segamat has considerable short-term fluctuation, with distinct peaks indicating episodic contamination occurrences. The early months are rather consistent, followed by a gradual increase that culminates in a conspicuous mid-year high, most likely due to the dry season or haze. Following this peak, the data indicates a slow fall throughout the following months, punctuated by brief bursts. The absence of a clear, repeated pattern, as well as the constant variations, point to a stochastic trend moulded by both predictable seasonal impacts and random external factors.

Finally, Tangkak has high fluctuation in PM2.5 levels, with values varying throughout the year. Around Day 100, PM2.5 levels surge to over 50 $\mu\text{g}/\text{m}^3$, indicating a localized pollution incident. Following this, the series slowly decreases off, stabilising in lower concentration levels with sporadic swings. The lack of a regular periodic pattern and the occurrence of abrupt shifts imply a stochastic tendency driven by seasonal fluctuation and external variables such as industrial pollutants, traffic and weather changes.

In summary, time series plots for every station show a mix of short-term swings, mid-year peaks likely caused by periodic haze or dry weather and long-term stochastic tendencies. These patterns demonstrate the complex, dynamic nature of PM2.5 behaviour in Johor, necessitating the use of advanced time series models to accurately capture and forecast air quality variations.

4.3 Stationarity and Differencing

To formally assess that stationarity exists the Augmented Dickey-Fuller (ADF) test was performed on the daily PM2.5 concentration series for each of Johor's eight monitoring stations. The ADF test is one of the most often used statistical tests for detecting the presence of a unit root, indicating non-stationarity.

The ADF test results for all eight stations are shown in Table 4.3. As indicated in the table, the p-values for all stations were less than the 0.05 significance level, suggesting that the PM2.5 time series for each site is stationary and does not require differencing.

Table 4.3: Results of Augmented Dickey-Fuller (ADF) Test for PM2.5 Concentration Time Series

Station	ADF Test Statistic	p-value	Stationarity Conclusion
Pasir Gudang	-4.51	0.01	Stationary
Larkin	-5.23	0.01	Stationary
Pengerang	-4.88	0.01	Stationary
Kluang	-4.39	0.01	Stationary
Segamat	-5.72	0.01	Stationary
Kota Tinggi	-4.12	0.02	Stationary
Batu Pahat	-5.01	0.01	Stationary
Tangkak	-4.95	0.01	Stationary

The ADF test results confirm that all eight PM2.5 concentrations are stationary at the 5% significance level. This implies that differencing is not required for any of the datasets in this study. The stationarity of the series simplifies the modelling process of the subsequent ARMA and SARMA model development. Visual inspection of the time series plots further supports the statistical test results. After accounting for seasonal patterns and short-term fluctuations, the series demonstrated consistent mean and variance over time, reinforcing the conclusion that the datasets are stationary.

With confirmation of stationarity for all eight stations, the study proceeds to the model identification stage, where suitable ARMA and SARMA models will be specified for each time series. The next section discusses this process in detail, including the use of Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots, as well as model selection based on statistical criteria.

4.4 Model Identification

This section outlines the process of selecting the most appropriate models for PM2.5 concentrations at each monitoring station in Johor. The selection was based on both graphical analysis and statistical model evaluation to ensure the chosen models

adequately captured the underlying patterns, trends and potential seasonal behaviours in the time series data. Two primary approaches were applied which are visual guide by Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots for model selection, followed by formal model comparison using the Akaike Information Criterion (AIC).

4.4.1 ACF and PACF Plots

The ACF and PACF plots serve as essential tools in the preliminary identification of appropriate ARMA or SARMA models. These plots provide insight into the correlation structure of the PM2.5 time series data, highlighting patterns of autocorrelation, seasonality and the potential order of autoregressive and moving average terms. In this study, the ACF and PACF plots were generated for each of the eight monitoring stations in Johor. For stations exhibiting seasonal patterns, both ARMA and SARMA models were considered, while non-seasonal stations were evaluated using ARMA models alone. Figures 4.9 to 4.18, present the ACF and PACF plots for all stations, serving as a visual guide for model specification.

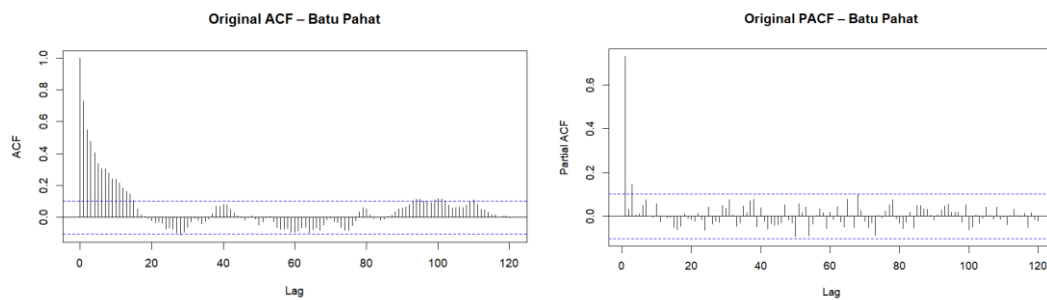


Figure 4.9: ACF and PACF for ARMA of Batu Pahat

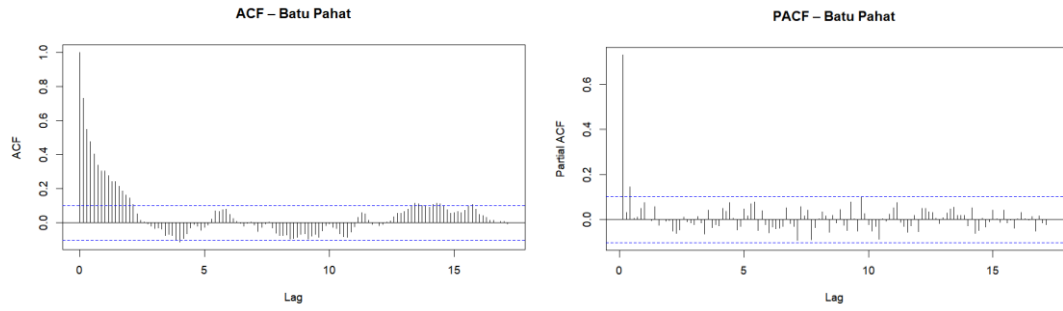


Figure 4.10: ACF and PACF for SARMA of Batu Pahat

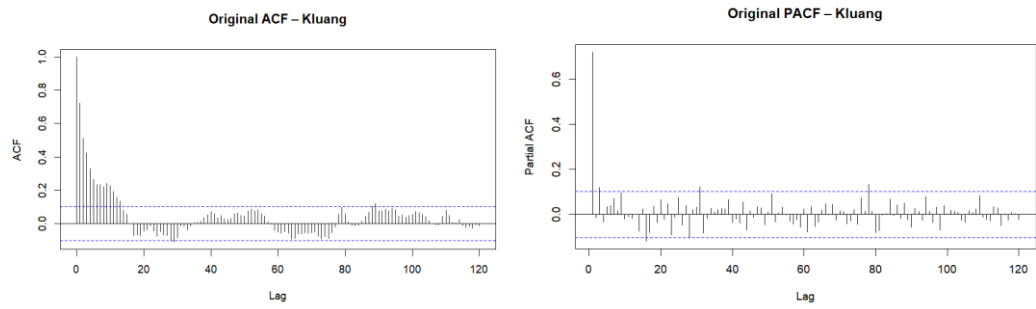


Figure 4.11: ACF and PACF for ARMA of Kluang

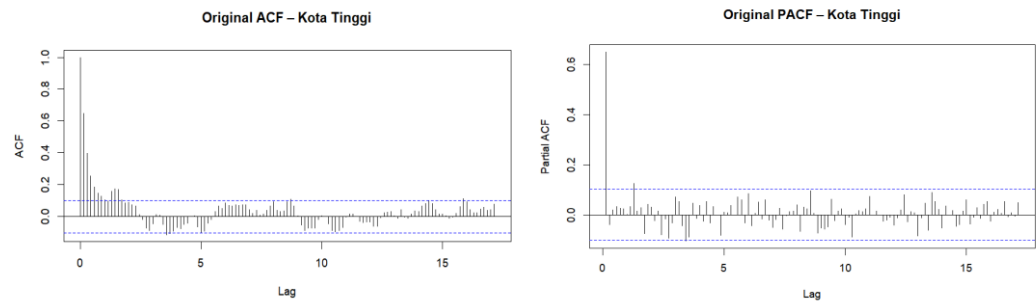


Figure 4.12: ACF and PACF for ARMA of Kota Tinggi

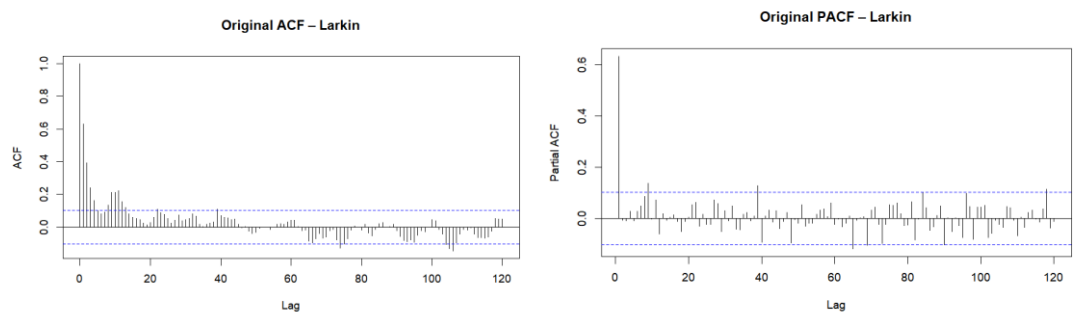


Figure 4.13: ACF and PACF for ARMA of Larkin

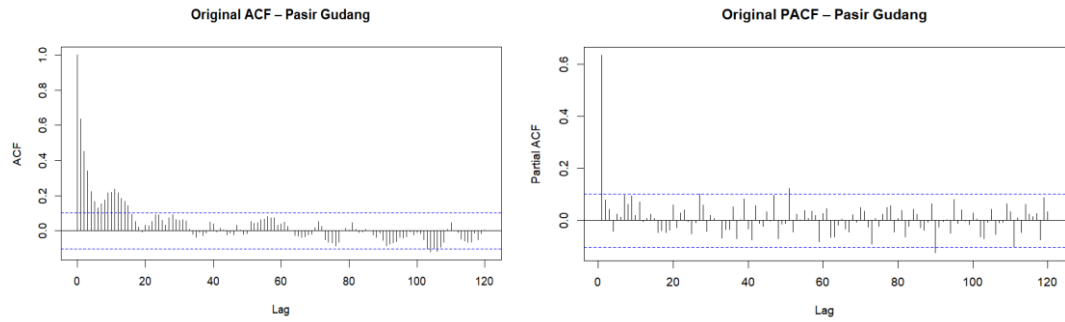


Figure 4.14: ACF and PACF for ARMA of Pasir Gudang

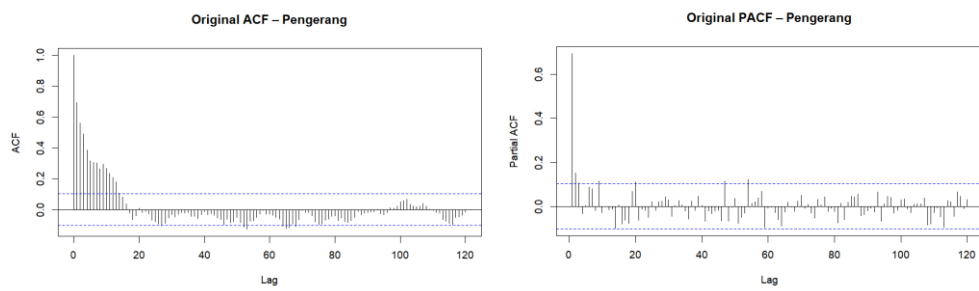


Figure 4.15: ACF and PACF for ARMA of Pengerang

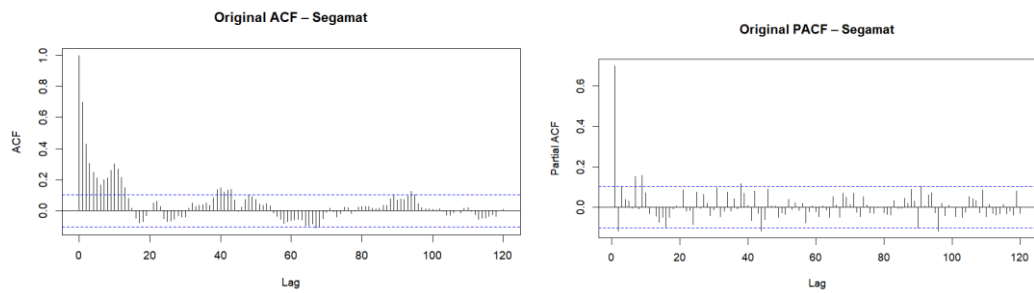


Figure 4.16: ACF and PACF for ARMA of Segamat

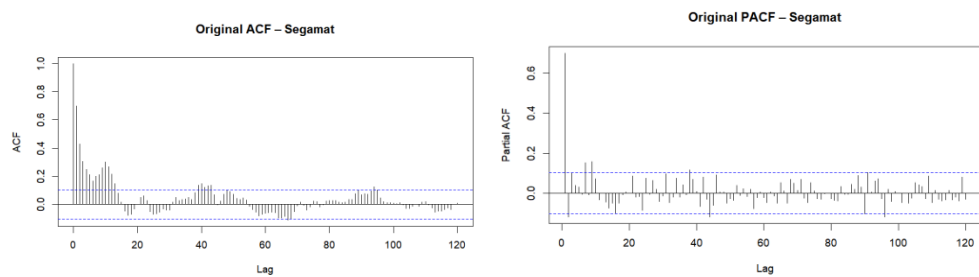


Figure 4.17: ACF and PACF for SARMA of Segamat

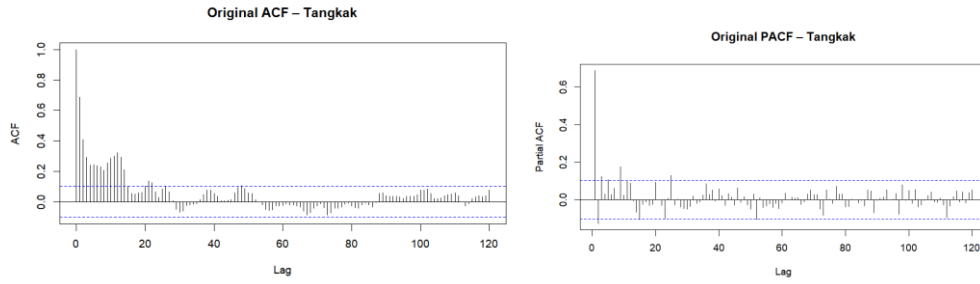


Figure 4.18: ACF and PACF for ARMA of Tangkak

Firstly, we begin by focusing on the station at Batu Pahat, where the ACF and PACF plots of the original series revealed strong autocorrelation at several early lags and a scattered pattern of partial correlations. This indicated that a simple AR or MA model would not suffice. Instead, a combination of both autoregressive and moving average terms was required, leading to the selection of an ARMA(1, 2) model, as visualised in Figure 4.9. Further seasonal analysis displayed a noticeable peak at lag 7 in the ACF plot, confirming the presence of weekly seasonality. This justified the use of a SARMA(1,0)(1,0)[7] model, which incorporated both non-seasonal and seasonal autoregressive terms to better capture the recurring weekly patterns in PM2.5 fluctuations, as shown in Figure 4.10.

For Kluang station, the ACF plot exhibited a gradual decaying pattern, consistent with short-term memory, as presented in Figure 4.11. The PACF plot demonstrated clear spikes at lags 1 and 3, suggesting the presence of autoregressive behaviour, albeit without a distinct cut-off or tail-off structure, complicating precise model identification. Given the absence of strong seasonal components and the presence of moderate autocorrelation, an ARMA(1,2) model was deemed suitable. Based on Figure 4.12, the ACF plot for Kota Tinggi station did not exhibit a smooth decay but rather displayed fluctuations across the early lags. The PACF plot revealed spikes at lags 1 and 9, indicating a mix of short-term memory and irregular patterns.

Due to the lack of strong or consistent autocorrelation structures and the relatively stable nature of the data, a simple ARMA(1,0) model was selected.

Similarly, based on Figure 4.13 at Larkin station, the autocorrelation patterns resembled those of Kota Tinggi. The ACF plot displayed several moderate spikes with no smooth tail-off and the PACF plot had an initial spike at lag 1 followed by random fluctuations. Given the absence of a clear structure, an ARMA(1,0) model was also applied at this station.

For Pasir Gudang, the ACF plot showed a strong spike at lag 1 with a gradual tailing-off pattern thereafter based on Figure 4.14, indicating autoregressive behaviour. The PACF plot exhibited a sharp drop after lag 1, suggesting that most of the predictive power resided within the first lag. Based on these characteristics, along with residual diagnostics, an ARMA(2,1) model was selected. No evidence of seasonality was observed.

Based on Figure 4.15, the ACF plot of Pengerang demonstrated a typical tailing-off pattern while the PACF plot displayed significant spikes at lags 1, 2, and 8. These patterns indicated the presence of both autoregressive and moving average components, with no seasonal signature present. Consequently, an ARMA(2,2) model was chosen.

The Segamat station's ARMA ACF plot in Figure 4.16 revealed moderate autocorrelation initially, followed by consistent decay. The PACF plot showed pronounced spikes at lags 1, 2 and 3, with smaller increases at lags 7 and 8, indicating a complex structure not adequately captured by simple AR or MA models. An ARMA(4,3) model was initially selected based on strong in-sample performance. However, based on Figure 4.17, seasonal analysis revealed a repeating pattern in the

ACF around lag 7, confirming weekly seasonality. This warranted the use of a SARMA(2,2)(0,1)[7] model to account for recurring weekly cycles.

Finally, at Tangkak, the ACF plot in Figure 4.18 demonstrated a tailing-off trend after strong initial spikes, implying that current PM_{2.5} values were influenced by previous observations. The PACF plot exhibited notable spikes at lags 1, 2 and 3, with less consistent behaviour beyond these lags. Considering the moderate memory structure and absence of clear seasonal patterns, an ARMA(2,2) model was deemed most appropriate.

In conclusion, the analysis of ACF and PACF plots across all stations revealed differing patterns of autocorrelation and partial autocorrelation, reflecting the diverse air quality behaviours at each site. Simpler models sufficed for relatively stable locations such as Kota Tinggi and Larkin, while the presence of seasonality and complex structures at stations like Batu Pahat and Segamat necessitated the use of SARMA models. Meanwhile, moderate volatility at stations such as Pengerang and Pasir Gudang justified more complex ARMA structures, underscoring the importance of adopting a tailored modelling approach for each monitoring station.

4.4.2 Model Selection Criteria

To determine the best-fitting model for each station, several ARMA and SARMA model variations were developed and compared based on their Akaike Information Criterion (AIC) values. The AIC is a widely accepted statistical measure that balances model goodness-of-fit with complexity, where lower AIC values indicate a more suitable model. For each station, multiple models were estimated with varying orders of autoregressive and moving average terms, along with seasonal components where applicable. Tables 4.4 to 4.13 summarise the model variations tested and their

corresponding AIC values, providing a clear basis for the final model selection at each monitoring station.

Table 4.4: Variations of ARMA Models with Approximated AIC Values for Batu Pahat

Model	Mean Type	AIC
ARMA(1,2)	Non-zero mean	2043.33
ARMA(2,2)	Non-zero mean	2045.35
ARMA(1,3)	Non-zero mean	2045.37
ARMA(1,0)	Non-zero mean	2048.66
ARMA(1,1)	Non-zero mean	2050.01
ARMA(2,3)	Non-zero mean	2047.10
ARMA(2,1)	Non-zero mean	2051.92
ARMA(0,2)	Non-zero mean	2099.38
ARMA(0,3)	Non-zero mean	2081.15
ARMA(0,1)	Non-zero mean	2140.94
ARMA(0,0)	Non-zero mean	2328.78
ARMA(0,0)	Zero mean	2962.80
ARMA(1,2)	Zero mean	Infinite

Table 4.5: Variations of SARMA Models with Approximated AIC Values for Batu Pahat

Model Structure	AIC Value
ARMA(2,2)(1,1)[7]	2048.53
ARMA(0,0) with non-zero mean	2328.78
ARMA(1,0)(1,0)[7]	2047.50
ARMA(0,1)(0,1)[7]	2130.63
ARMA(0,0) with zero mean	2962.80
ARMA(1,0)	2048.66
ARMA(1,0)(2,0)[7]	2048.32
ARMA(1,0)(1,1)[7]	2048.98
ARMA(1,0)(0,1)[7]	2047.88
ARMA(1,0)(2,1)[7]	2050.28
ARMA(0,0)(1,0)[7]	2294.03
ARMA(2,0)(1,0)[7]	2049.35
ARMA(1,1)(1,0)[7]	2049.23
ARMA(0,1)(1,0)[7]	2127.92
ARMA(2,1)(1,0)[7]	2051.02
ARMA(1,0)(1,0)[7] with zero mean	2089.92

Table 4.6: Variations of ARMA Models with Approximated AIC Values Kluang

Model	Mean Type	AIC
ARMA(1,2)	Non-zero mean	1832.98
ARMA(2,2)	Non-zero mean	1834.74
ARMA(1,3)	Non-zero mean	1834.96
ARMA(1,0)	Non-zero mean	1835.07
ARMA(2,3)	Non-zero mean	1836.55
ARMA(2,1)	Non-zero mean	1836.86
ARMA(1,1)	Non-zero mean	1837.02
ARMA(0,2)	Non-zero mean	1876.60
ARMA(0,3)	Non-zero mean	1855.14
ARMA(0,1)	Non-zero mean	1911.24
ARMA(0,0)	Non-zero mean	2103.11
ARMA(0,0)	Zero mean	2816.57
ARMA(1,2)	Zero mean	Infinite

Table 4.7: Variations of ARMA Models with Approximated AIC Values for Kota Tinggi

Model	Mean Type	AIC
ARMA(1,0)	Non-zero mean	1843.11
ARMA(1,1)	Non-zero mean	1844.67
ARMA(2,0)	Non-zero mean	1844.70
ARMA(2,1)	Non-zero mean	1846.04
ARMA(2,2)	Non-zero mean	1846.34
ARMA(0,1)	Non-zero mean	1897.14
ARMA(1,0)	Zero mean	1902.50
ARMA(0,0)	Non-zero mean	2043.32
ARMA(0,0)	Zero mean	2780.76

Table 4.8: Variations of ARMA Models with Approximated AIC Values for Larkin

Model	Mean Type	AIC
ARMA(1,0)	Non-zero mean	2092.24
ARMA(2,1)	Non-zero mean	2092.62
ARMA(2,2)	Non-zero mean	2094.14
ARMA(1,1)	Non-zero mean	2094.28
ARMA(2,0,0)	Non-zero mean	2094.28
ARMA(0,1)	Non-zero mean	2147.24
ARMA(1,0)	Zero mean	2156.38
ARMA(0,0)	Non-zero mean	2279.64
ARMA(0,0)	Zero mean	3125.19

Table 4.9: Variations of ARMA Models with Approximated AIC Values for Pasir Gudang

Model	Mean Type	AIC
ARMA(2,1)	Non-zero mean	1993.21
ARMA(1,1)	Non-zero mean	1995.75
ARMA(2,0)	Non-zero mean	1996.08
ARMA(1,0)	Non-zero mean	1996.66
ARMA(1,2)	Non-zero mean	1997.56
ARMA(3,1)	Non-zero mean	1997.61
ARMA(3,0)	Non-zero mean	1997.31
ARMA(2,2)	Non-zero mean	1997.77
ARMA(0,1)	Non-zero mean	2057.32
ARMA(0,0)	Non-zero mean	2186.65
ARMA(0,0)	Zero mean	2991.91
ARMA(2,1)	Zero mean	Infinite
ARMA(3,2)	Non-zero mean	Infinite

Table 4.10: Variations of ARMA Models with Approximated AIC Values for Pengerang

Model	Mean Type	AIC
ARMA(2,2)	Non-zero mean	1986.78
ARMA(1,1)	Non-zero mean	1987.00
ARMA(1,2)	Non-zero mean	1987.99
ARMA(2,1)	Non-zero mean	1987.89
ARMA(3,3)	Non-zero mean	1987.74
ARMA(2,3)	Non-zero mean	1988.83
ARMA(3,1)	Non-zero mean	1988.85
ARMA(1,3)	Non-zero mean	1989.71
ARMA(3,2)	Non-zero mean	1989.75
ARMA(1,0)	Non-zero mean	1996.73
ARMA(0,1)	Non-zero mean	2088.47
ARMA(0,0)	Non-zero mean	2237.64
ARMA(0,0)	Zero mean	2886.81
ARMA(2,2)	Zero mean	2006.19

Table 4.11: Variations of ARMA Models with Approximated AIC Values for Segamat

Model	Mean Type	AIC
ARMA(4,3)	Non-zero mean	2037.21
ARMA(3,4)	Non-zero mean	2041.54
ARMA(4,4)	Non-zero mean	2042.68
ARMA(4,2)	Non-zero mean	2044.46

ARMA(3,1)	Non-zero mean	2044.68
ARMA(3,3)	Non-zero mean	2045.43
ARMA(2,1)	Non-zero mean	2046.47
ARMA(3,2)	Non-zero mean	2046.66
ARMA(4,1)	Non-zero mean	2046.70
ARMA(5,2)	Non-zero mean	2046.15
ARMA(2,2)	Non-zero mean	2047.66
ARMA(1,2)	Non-zero mean	2047.96
ARMA(3,0)	Non-zero mean	2047.91
ARMA(5,3)	Non-zero mean	2047.82
ARMA(1,1)	Non-zero mean	2048.44
ARMA(2,0)	Non-zero mean	2049.94
ARMA(4,0)	Non-zero mean	2049.33
ARMA(1,0)	Non-zero mean	2052.71
ARMA(0,1)	Non-zero mean	2106.06
ARMA(0,0)	Non-zero mean	2300.85
ARMA(0,0)	Zero mean	2987.99
ARMA(4,3)	Zero mean	Infinite

Table 4.12: Variations of SARMA Models with Approximated AIC Values for Segamat

Model Structure	AIC
ARMA(2,2)(0,1)[7]	2044.50
ARMA(3,1)(0,1)[7]	2044.57
ARMA(2,2)(1,0)[7]	2044.57
ARMA(1,1)(0,1)[7]	2045.05
ARMA(1,3)(0,1)[7]	2045.85
ARMA(3,2)(0,1)[7]	2046.38
ARMA(2,1)(0,1)[7]	2046.58
ARMA(2,3)(0,1)[7]	2046.46
ARMA(2,2)(0,2)[7]	2046.53
ARMA(1,2)(0,1)[7]	2046.34
ARMA(2,2)(1,1)[7]	2048.20
ARMA(2,2)(0,0)[7]	2047.66
ARMA(1,0)(1,0)[7]	2052.49
ARMA(0,1)(0,1)[7]	2096.12
ARMA(0,0) with non-zero mean	2300.85
ARMA(0,0) with zero mean	2987.99

Table 4.13: Variations of ARMA Models with Approximated AIC Values for Tangkak

Model	Mean Type	AIC
ARMA(2,2)	Non-zero mean	2083.35
ARMA(1,3)	Non-zero mean	2084.44
ARMA(2,3)	Non-zero mean	2085.43
ARMA(1,2)	Non-zero mean	2087.06
ARMA(2,1)	Non-zero mean	2089.22
ARMA(3,3)	Non-zero mean	2085.22
ARMA(3,2)	Non-zero mean	2093.10
ARMA(3,1)	Non-zero mean	2091.19
ARMA(1,1)	Non-zero mean	2091.59
ARMA(1,0)	Non-zero mean	2098.01
ARMA(0,1)	Non-zero mean	2137.10
ARMA(0,0)	Non-zero mean	2331.48
ARMA(0,0)	Zero mean	3025.01
ARMA(2,2)	Zero mean	Infinite

First, Batu Pahat station was modelled using both ARMA and SARMA structures. Among the tested models, the ARMA(1,2) with non-zero mean emerged as the best model with the lowest AIC of 2043.33 as illustrated in Table 4.4. Although the seasonal counterpart, SARMA(1,0)(1,0)[7] resulted in a slightly higher AIC of 2047.50 as illustrated in Table 4.5, it was still considered beneficial for capturing weekly seasonality. Notably, although the SARMA model had slightly greater complexity, it offered a clearer addressed seasonal autocorrelation that the ARMA could not.

Moreover, Kluang station's model selection pointed to ARMA(1,2) model with non-zero mean, which recorded the lowest AIC at 1832.98, outperforming several other configurations as indicated in Table 4.6. Furthermore, Kota Tinggi station followed a similar trend, where ARMA(1,0) was chosen based on the lowest AIC of 1843.11 according to Table 4.7 and effectively modelling the location's relatively stable air quality trends.

Additionally, Larkin station followed suit, the selection process also favoured simplicity. The ARMA(1,0) model recorded the lowest AIC value of 2092.24 as reflected in the Table 4.8, confirming that simple models were enough for stations with low autocorrelation complexity. Followed by, for Pasir Gudang, among various alternatives, the ARMA(2,1) model delivered the lowest AIC at 1993.21 as can be seen in Table 4.9. This suggested a slightly more complex structure was warranted to capture the variability in PM_{2.5} concentrations, likely influenced by its semi-industrial characteristics.

Besides that, Pengerang was best represented by ARMA(2,2) with non-zero mean, achieving the lowest AIC of 1986.78 as presented in Table 4.10. The model outperformed higher-order models that added unnecessary complexity. Subsequently, in Segamat, which showed hints of both autoregressive and seasonal behaviour, the ARMA(4,3) model with non-zero mean had the lowest AIC of 2037.21 among ARMA structures as demonstrated in Table 4.11. Whereas, the SARMA(2,2)(0,1)[7] model delivered the lowest AIC of 2044.50, suggesting its suitability for capturing weekly patterns as demonstrated in Table 4.12. While ARMA outperformed SARMA slightly in AIC, SARMA's inclusion of a seasonal term allowed for better interpretability, especially given the subtle weekly seasonality observed.

Lastly, in Tangkak, the ARMA(2,2) model stood out with the lowest AIC of 2083.35 as Table 4.13 suggests. Though alternative models with slightly higher AR or MA orders were tested, they failed to outperform ARMA(2,2).

4.5 Model Estimation and Parameter Significance

The Table 4.14 to 4.23 present the model estimation results and parameter significance for the best ARMA models applied to all eight stations. Additionally, for

the two stations exhibiting seasonal patterns that are Batu Pahat and Segamat, best SARMA model estimations and parameter significance are also reported.

Table 4.14: ARMA(1,2) Model Parameter Estimates for Batu Pahat

Parameter	Estimate	Standard Error	Z-value
AR(1)	0.88***	0.04	20.20
MA(1)	-0.18**	0.07	-2.54
MA(2)	-0.21***	0.07	-3.06
Intercept	12.44***	1.06	11.75

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.15: SARMA(1,0) (1,0) [7] Model Parameter Estimates for Batu Pahat

Parameter	Estimate	Standard Error	Z-value
AR(1)	0.72 ***	0.04	19.44
SAR(1)	0.05 **	0.05	1.79
Intercept	12.53 ***	0.82	15.35

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.16: ARMA(1,2) Model Parameter Estimates for Kluang

Parameter	Estimate	Standard Error	Z-value
AR(1)	0.84 ***	0.06	14.30
MA(1)	-0.10	0.08	-1.15
MA(2)	-0.20 *	0.08	-2.44
Intercept	10.50 ***	0.66	15.94

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.17: ARMA(1,0) Model Parameter Estimates for Kota Tinggi

Parameter	Estimate	Standard Error	Z-value
AR(1)	0.65 ***	0.04	16.49
Intercept	10.09 ***	0.45	22.36

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.18: ARMA(1,0) Model Parameter Estimates for Larkin

Parameter	Estimate	Standard Error	Z-value
AR(1)	0.64 ***	0.04	15.81
Intercept	16.49 ***	0.61	27.03

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.19: ARMA(2,1) Model Parameter Estimates for Pasir Gudang

Parameter	Estimate	Standard Error	Z-value
AR(1)	1.51 ***	0.07	20.62
AR(2)	-0.52 ***	0.07	-8.01
MA(1)	-0.92 ***	0.05	-18.73
Intercept	12.96 ***	1.48	8.74

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.20: ARMA(2,2) Model Parameter Estimates for Pengerang

Parameter	Estimate	Standard Error	Z-value
AR(1)	-0.03	0.10	-0.35
AR(2)	0.74 ***	0.07	10.04
MA(1)	0.61 ***	0.11	5.54
MA(2)	-0.30 ***	0.07	-4.20
Intercept	11.31 ***	0.84	13.40

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.21: ARMA(4,3) Model Parameter Estimates for Segamat

Parameter	Estimate	Standard Error	Z-value
AR(1)	1.62 ***	0.09	17.50
AR(2)	-0.15	0.19	-0.78
AR(3)	-1.10 ***	0.17	-6.37
AR(4)	0.56 ***	0.07	8.01
MA(1)	-0.85 ***	0.08	-10.41
MA(2)	-0.69 ***	0.10	-7.27

MA(3)	0.83 ***	0.07	11.79
Intercept	13.11 ***	0.85	15.36

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.22: SARMA(2,2)(0,1)[7] Model Parameter Estimates for Segamat

Parameter	Estimate	Standard Error	Z-value
AR(1)	1.36 ***	0.14	9.43
AR(2)	-0.39 ***	0.12	-3.32
MA(1)	-0.56 ***	0.14	-3.96
MA(2)	-0.27 ***	0.07	-3.89
SMA(1)	0.09 **	0.06	1.57
Intercept	12.92 ***	1.12	11.53

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

Table 4.23 ARMA(2,2) Model Parameter Estimates for Tangkak

Parameter	Estimate	Standard Error	Z-value
AR(1)	1.25 ***	0.11	10.56
AR(2)	-0.28 **	0.11	-2.66
MA(1)	-0.48 ***	0.12	-4.15
MA(2)	-0.33 ***	0.07	-4.46
Intercept	13.62 ***	1.18	11.57

Note: The level of significance *** = 1% level, ** = 5% level, * = 10% level

At Batu Pahat stations, both the ARMA(1,2) and SARMA(1,0)(1,0)[7] models were identified to be statistically sound. In the ARMA model, all estimated parameters which are AR(1), MA(1), MA(2) and the intercept are statistically significant, with AR(1) showing a notably strong positive relationship to the previous day's value as illustrated in Table 4.14. The SARMA model also yielded a highly significant non-seasonal AR(1) term, whereas the seasonal AR(1) component was only marginally significant as illustrated in Table 4.15. Despite this, the seasonal term enhanced the model's ability to account for weekly variations in pollution.

Moreover, in Kluang station, the ARMA(1,2) model highlighted a strongly significant AR(1) coefficient, confirming short-term autocorrelation. However, while the MA(2) term was marginally significant, the MA(1) term lacked statistical significance, implying it contributed little to the model's accuracy. Still, the model's intercept was highly significant, anchoring the model's reliability in forecasting around a stable mean. This is indicated in Table 4.16

Furthermore, in Kota Tinggi station, a much simpler ARMA(1,0) model proved effective. Both the AR(1) term and the intercept were statistically significant at the 1% level according to Table 4.17. This simplicity indicates that the PM2.5 series in Kota Tinggi is best described by short-term dependence without the need for more complex dynamics.

Additionally, in Larkin station, the ARMA(1,0) model also worked well, both the AR(1) coefficient and intercept reaching high statistical significance. The AR(1) value indicates a strong relationship between the current and previous day's PM2.5 levels, reflecting a steady and predictable trend in the data as reflected in Table 4.18.

Followed by, in Pasir Gudang station, required a more complex ARMA(2,1) model. All three coefficients which are AR(1), AR(2) and MA(1) are proven to be highly significant. The large AR(1) coefficient suggests a very strong influence from the previous day, while the AR(2) and MA(1) terms accounted for delayed and residual-based fluctuations. The intercept was also significant, indicating a steady mean level can be seen from Table 4.19.

Besides that, in Pengerang, the ARMA(2,2) model included both significant and non-significant terms. While AR(1) was not statistically meaningful, AR(2),

MA(1) and MA(2) were all highly significant as presented in Table 4.20. This suggests delayed lag and error structures were more influential than immediate past values. The intercept was again significant, reinforcing the stability of the average concentration level.

Subsequently, Segamat's ARMA(4,3) model was complex but informative. As observed, significant coefficient are AR(1), AR(3), AR(4), all MA terms (MA(1), MA(2) and MA(3)) and the intercept, whereas only AR(2) was seen to lack significance as demonstrated in Table 4.21. This indicates that PM_{2.5} levels in Segamat are influenced by multiple lags and require a nuanced model to capture these dependencies. In contrast, the SARMA model for Segamat also showed excellent parameter significance, with all non-seasonal and seasonal components (including SMA(1)) contributing meaningfully to the model as demonstrated in Table 4.22, supporting the model's capacity to account for both short-term variation and weekly repetition.

Finally, Tangkak's ARMA(2,2) model featured four statistically significant parameters. AR(1) emerged particularly dominant, while AR(2) played a moderating role as Table 4.23 suggests. Both MA terms (MA(1) and MA(2)) were negative and significant, reflecting that previous forecast errors were appropriately adjusted in the model. The intercept remained highly significant, indicating a consistent central tendency.

In essence, most stations showed high statistical significance across key model parameters, particularly for the AR(1) and intercept terms. Simpler models sufficed for stable places like Kota Tinggi and Larkin, whereas more complex structures were required for variable stations such as Pasir Gudang, Segamat and Tangkak. The

SARMA models used in Batu Pahat and Segamat further demonstrated the importance of accounting for seasonality when weekly cycles are present. Together, these findings validate the appropriateness and statistical strength of the selected models for each station's data.

4.6 Model Diagnostics

Comprehensive model diagnostics were performed for each of the eight stations, focusing primarily on the evaluation of the residuals and the assessment of forecast accuracy.

4.6.1 Residual Analysis

Comprehensive residual diagnostics were carried out for the selected ARMA models across all eight stations and for the SARMA models at Batu Pahat and Segamat, where seasonality was observed. Table 4.24 to 4.33 present the results of the Ljung-Box test applied to the residuals of ARMA models for all eight stations. A similar diagnostic was performed on the SARMA model residuals for the two stations exhibiting seasonal patterns. In addition, the accompanying Figure 4.19 to 4.28 display the ACF and PACF plots, as well as Q-Q plots for residuals from both ARMA and SARMA models, encompassing all eight stations and the two stations respectively.

4.6.1.1 Residual Analysis for ARMA(1,2) for station Batu Pahat

Table 4.24 Ljung-Box Test Result for ARMA model Residuals of Batu Pahat

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
4.91	9	0.84	12

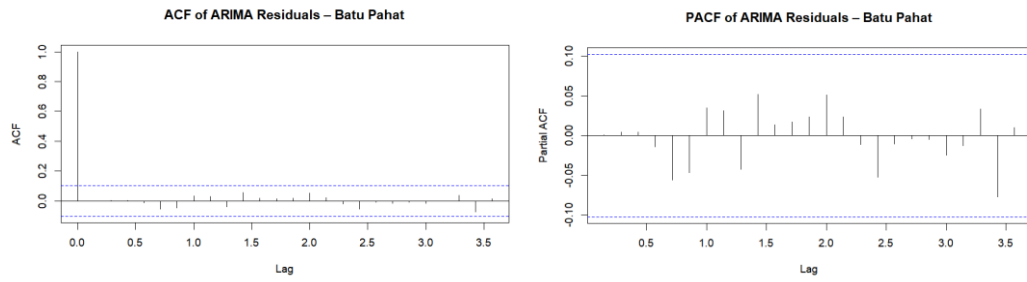


Figure 4.19(a): ACF and PACF for Residuals of ARMA(1,2) on PM2.5 of Batu Pahat

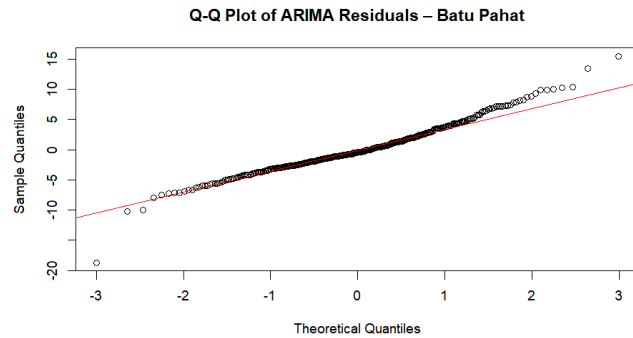


Figure 4.19(b): Q-Q Plot of Residual for ARMA(1,2) of Batu Pahat

4.6.1.2 Residual Analysis for SARMA(1,0)(1,0) for station Batu Pahat

Table 4.25 Ljung-Box Test Result for SARMA model Residuals of Batu Pahat

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
9.69	10	0.47	12

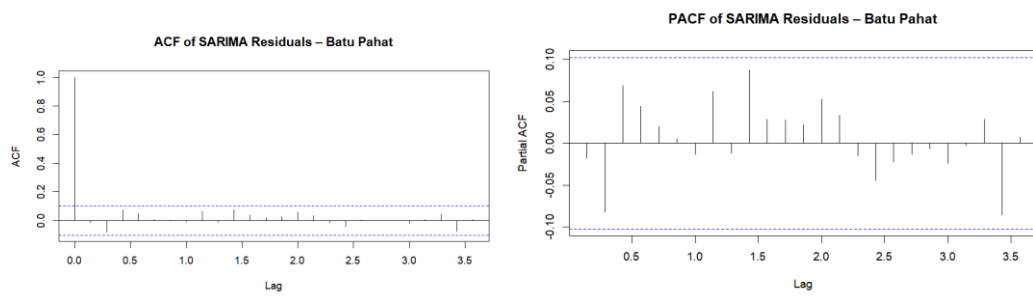


Figure 4.20(a): ACF and PACF for Residuals of SARMA(1,0)(1,0) on PM2.5 of Batu Pahat

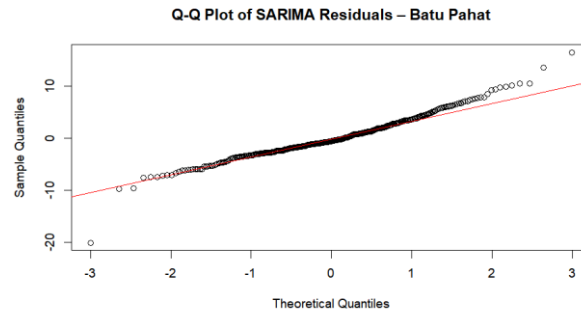


Figure 4.20(b): Q-Q Plot of Residual for SARMA(1,0)(1,0) of Batu Pahat

4.6.1.3 Residual Analysis for ARMA(1,2) for station Kluang

Table 4.26 Ljung-Box Test Result for ARMA model Residuals of Kluang

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
6.93	9	0.64	12

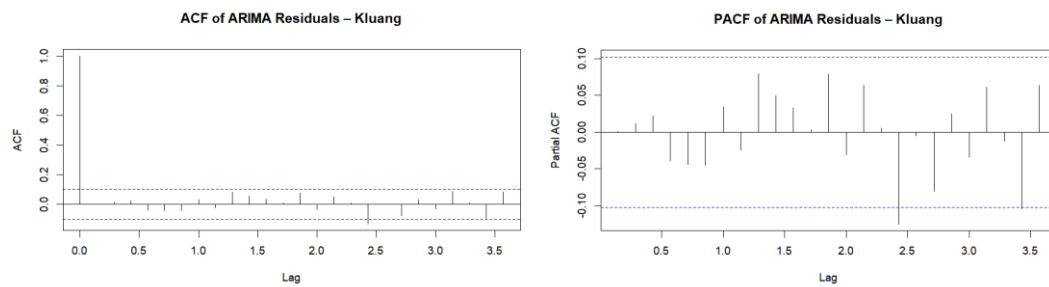


Figure 4.21(a): ACF and PACF for Residuals of ARMA(1,2) on PM2.5 of Kluang

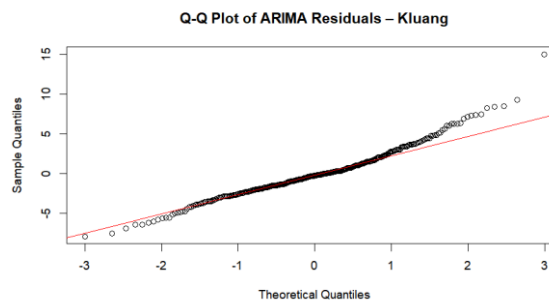


Figure 4.21(b): Q-Q Plot of Residual for ARMA(1,2) of Kluang

4.6.1.4 Residual Analysis for ARMA(1,0) for station Kota Tinggi

Table 4.27 Ljung-Box Test Result for ARMA model Residuals of Kota Tinggi

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
11.25	11	0.42	12

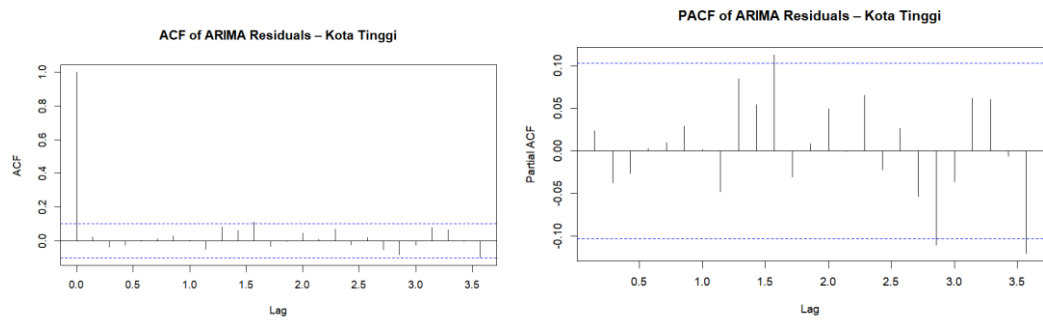


Figure 4.22(a): ACF and PACF for Residuals of ARMA(1,0) on PM2.5 of Kota Tinggi

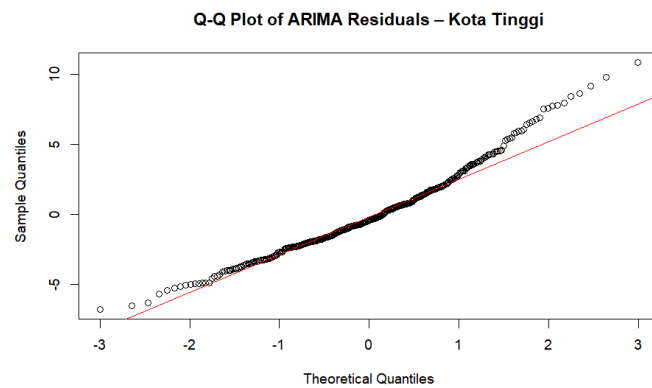


Figure 4.22(b): Q-Q Plot of Residual for ARMA(1,0) of Kota Tinggi

4.6.1.5 Residual Analysis for ARMA(1,0) for station Larkin

Table 4.28: Ljung-Box Test Result for ARMA model Residuals of Larkin

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
13.68	11	0.25	12

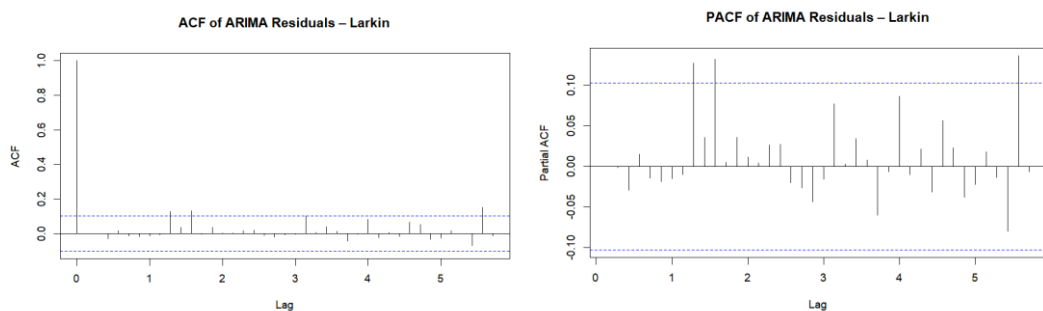


Figure 4.23(a): ACF and PACF for Residuals of ARMA(1,0) on PM2.5 of Larkin

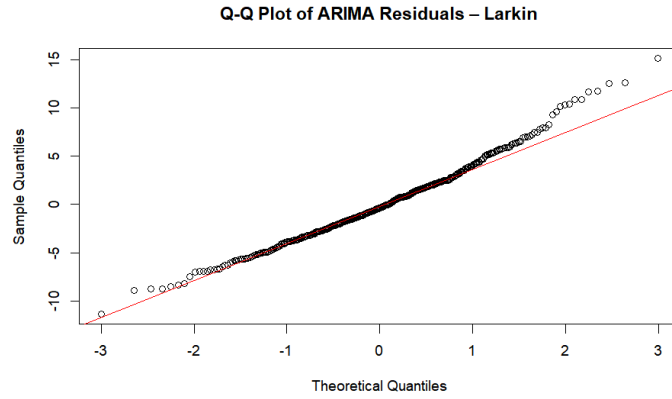


Figure 4.23(b): Q-Q Plot of Residual for ARMA(1,0) of Larkin

4.6.1.6 Residual Analysis for ARMA(2,1) for station Pasir Gudang

Table 4.29 Ljung-Box Test Result for ARMA model Residuals of Pasir Gudang

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
8.78	9	0.46	12

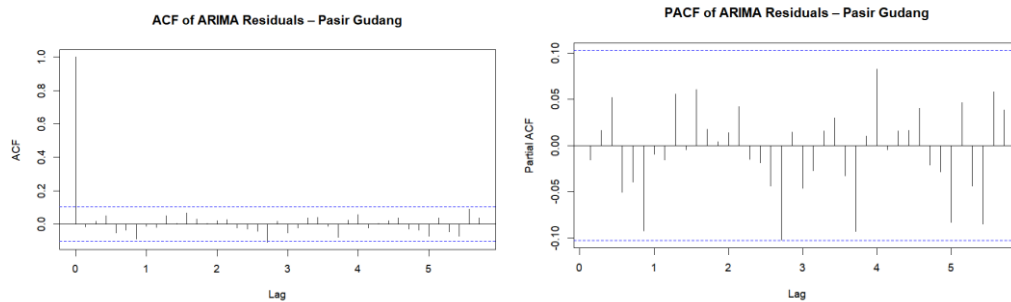


Figure 4.24(a): ACF and PACF for Residuals of ARMA(2,1) on PM2.5 of Pasir Gudang

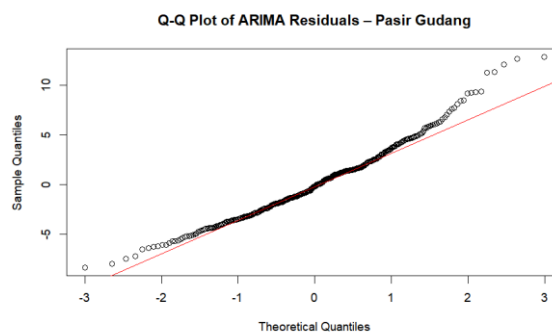


Figure 4.24(b): Q-Q Plot of Residual for ARMA(2,1) of Pasir Gudang

4.6.1.7 Residual Analysis for ARMA(2,2) for station Pengerang

Table 4.30: Ljung-Box Test Result for ARMA model Residuals of Pengerang

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
11.92	8	0.15	12

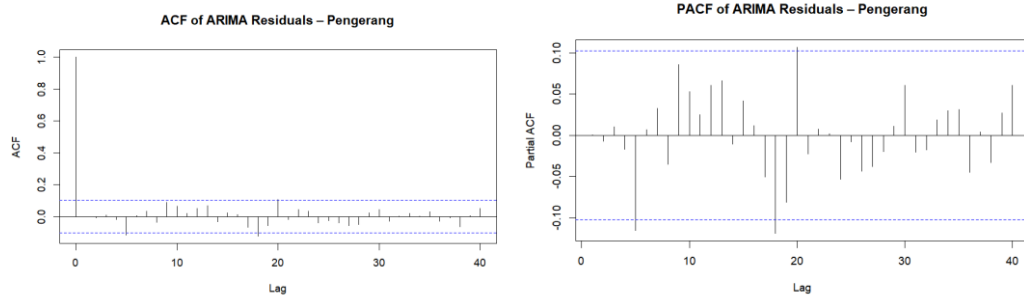


Figure 4.25(a): ACF and PACF for Residuals of ARMA(2,2) on PM2.5 of Pengerang

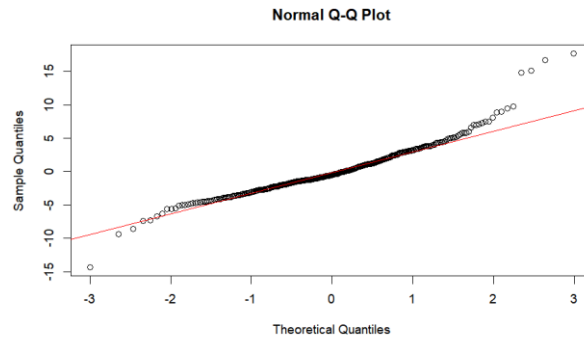


Figure 4.25(c): Q-Q Plot of Residual for ARMA(2,2) of Pengerang

4.6.1.8 Residual Analysis for ARMA(4,3) for station Segamat

Table 4.31 Ljung-Box Test Result for ARMA model Residuals of Segamat

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
11.10	5	0.06	12

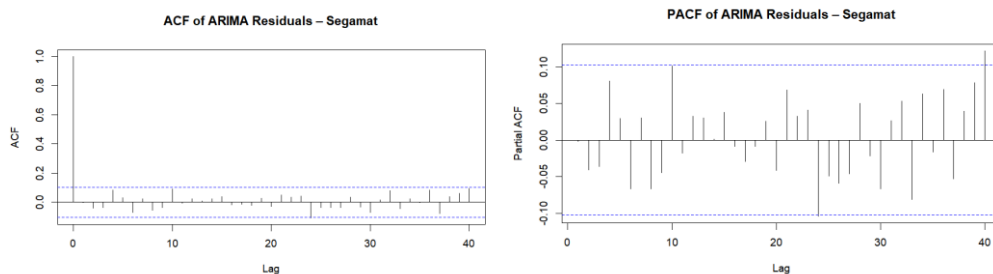


Figure 4.26(a): ACF and PACF for Residuals of ARMA(4,3) on PM2.5 of Segamat

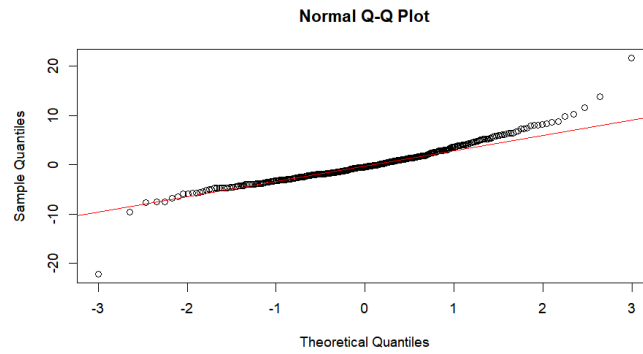


Figure 4.26(b): Q-Q Plot of Residual for ARMA(4,3) of Segamat

4.6.1.9 Residual Analysis for SARMA(2,2)(0,1) for station Segamat

Table 4.32 Ljung-Box Test Result for SARMA model Residuals of Segamat

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
18.99	7	0.08	12

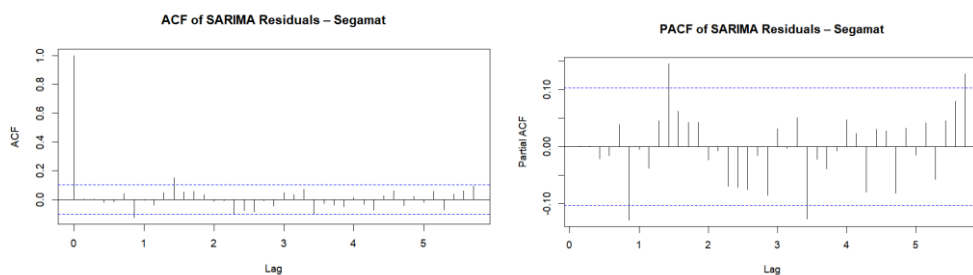


Figure 4.27(a): ACF and PACF for Residuals of SARMA(2,2)(0,1) on PM2.5 of Segamat

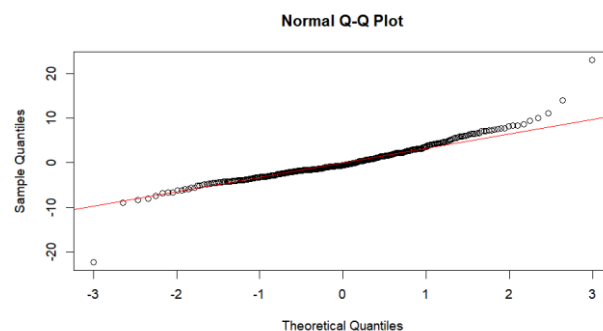


Figure 4.27(b): Q-Q Plot of Residual for SARMA(2,2)(0,1) of Segamat

4.6.1.10 Residual Analysis for ARMA(2,2) for station Tangkak

Table 4.33 Ljung-Box Test Result for ARMA model Residuals of Tangkak

Q-Statistic (Q*)	Degrees of Freedom (df)	p-value	Lags Tested
15.19	8	0.06	12

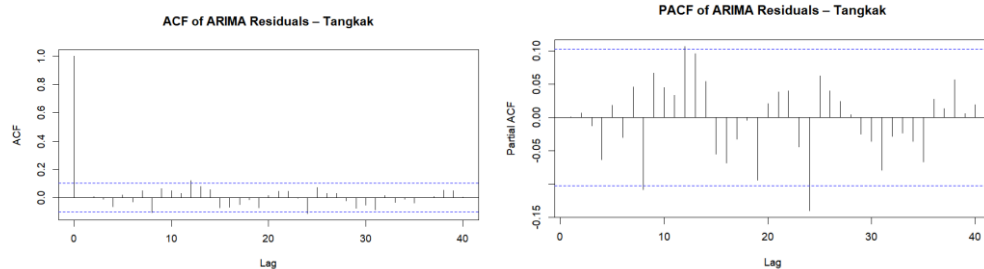


Figure 4.28(a): ACF and PACF for Residuals of ARMA(2,2) on PM2.5 of Tangkak

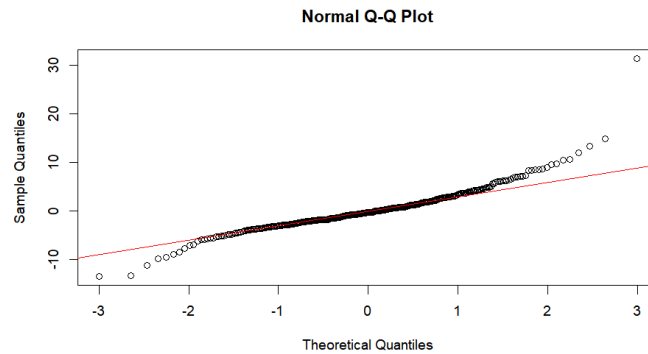


Figure 4.28(b): Q-Q Plot of Residual for ARMA(2,2) of Tangkak

For all the monitored stations, the residual ACF and PACF plots indicated that almost all spikes fell within the 95% confidence intervals, indicating and absence of significant autocorrelation. This result confirmed that both ARMA and SARMA models successfully captured the time-dependent structure in the PM2.5 data. In particular, stations like Batu Pahat, Kluang, Kota Tinggi and Larkin demonstrated clean ACF and PACF patterns, reinforcing the validity of their simpler model of ARMA(1,0) and ARMA(1,2) structures as shown in Figure 4.19(a), Figure 4.21 (a), Figure 4.22(a) and Figure 4.23(a) respectively. Similarly, more complex models used in Pasir Gudang, Pengerang, Tangkak and even more complex structure which is

Segamat as shown in Figure 4.26 (a), Figure 4.24 (a), Figure 4.25 (a) and Figure 4.28 (a) also displayed residuals resembling white noise, supporting the adequacy of these models even in more volatile environments.

The Ljung-Box test results supported this visual evidence. All eight stations reported p-values greater than the 0.05 significance level, suggesting residual independence up to lag 12. For example, Batu Pahat's ARMA and SARMA models yielded p-values of 0.84 and 0.47, as illustrated in Table 4.24 and Table 4.25 respectively meaning both comfortably exceeding the threshold. Segamat and Tangkak approached the threshold with p-values 0.06 and 0.06 respectively as presented in Table 4.31 and Table 4.33 respectively, but still met the diagnostic criteria.

Q-Q plots offered additional confirmation. Q-Q plots across all stations, confirmed the normality assumption. Even in stations with more variability, such as Pengerang and Segamat, the Q-Q plots did not show major departures from normality as visualised in Figure 4.23(b) and Figure 4.24(b).

To summarise the diagnostic findings confirmed that the fitted models produced residuals that were approximately normal, uncorrelated and statistically independent. This affirmed both the soundness and forecasting reliability of the selected ARMA and SARMA models. In simpler regions like Kota Tinggi and Larkin, basic ARMA models sufficed, while in more complex or seasonally-influenced stations like Batu Pahat and Segamat, SARMA provided the necessary flexibility. Even in cases with near-threshold Ljung-Box values, such as Segamat and Tangkak, the residuals remained within acceptable statistical bounds.

4.6.2 Forecast Accuracy

Table 4.34 below provides a summary of the forecast accuracy metrics for the best-performing models across all eight stations. The selected evaluation measures include RMSE, MAE and MAPE.

Table 4.34 Forecast Accuracy of Best Models of all 8 Stations

Stations	Best Models	RMSE	MAE	MAPE(%)
Batu Pahat	ARMA(1,2)	4.04	3.09	27.50
	SARMA(1,0)(1,0)[7]	3.95	2.99	26.80
Kluang	ARMA(1,2)	2.94	2.17	23.45
Kota Tinggi	ARMA(1,0)	2.99	2.33	25.80
Larkin	ARMA(1,0)	4.21	3.31	22.13
Pasir Gudang	ARMA(2,1)	3.66	2.88	22.81
Pengerang	ARMA(2,2)	3.61	2.66	25.19
Segamat	ARMA(4,3)	3.84	2.83	23.41
	SARMA(2,2)(0,1)[7]	3.90	2.89	23.72
Tangkak	ARMA(2,2)	4.09	2.85	21.36

For stations with more stable PM_{2.5} trends, such as Kota Tinggi and Larkin simple models like ARMA(1,0) worked very well. Kota Tinggi, for instance, recorded a low RMSE of 2.99 and MAE of 2.33, with a MAPE of 25.80%, which is a fairly reliable level of accuracy as detailed in Table 4.34 for Kota Tinggi Station. Larkin, despite recording a slightly higher RMSE of 4.21, actually had the second lowest MAPE among all stations at just 22.13%, indicating excellent performance in terms of percentage-based prediction accuracy as can be seen from Table 4.34 for Larkin Station. These results highlight that in stable environments, simpler models can yield efficient and just as accurate.

In contrast, more variable stations like Segamat and Pengerang required more complex models to handle the fluctuation in PM_{2.5} levels. Segamat's ARMA(4,3) model attained an RMSE of 3.84, an MAE of 2.83 and a MAPE of 23.41% as Table 4.34 suggests for Segamat Station. Pengerang's ARMA(2,2) followed closely, with an

RMSE of 3.61 and MAE of 2.66 as reflected in Table 4.34 for Pengerang Station. Despite the increased model complexity, both produced satisfactory results in their respective dynamic environments.

Tangkak, another station with higher variability, had the second highest RMSE (4.09) among all stations, indicating larger forecast deviations. However, it still achieved the lowest MAPE (21.36%), indicating that relative accuracy remained high even with larger absolute errors as presented in Table 4.34 for Tangkak Station.

Only Segamat and Batu Pahat were evaluated and fitted with both ARMA and SARMA models to address weekly seasonality. For Segamat, ARMA(4,3) marginally outperformed its seasonal counterpart SARMA(2,2)(0,1)[7]. The ARMA model had a lower RMSE (3.84 vs. 3.90), lower MAE (2.83 vs. 2.89) and a slightly better MAPE (23.41% vs. 23.72%) as demonstrated in Table 4.34 for Segamat Station and Batu Pahat Station respectively. This suggests that the ARMA model was flexible enough to account for seasonal effects without needing a more complex seasonal structure, therefore, ARMA was chosen.

In Batu Pahat, both the ARMA(1,2) and SARMA(1,0)(1,0)[7] models produced very similar results. However, the SARMA model achieved slightly lower RMSE (4.04 vs 3.95), MAE (3.09 vs 2.99) and MAPE (27.50% vs 26.80%) scores, as shown in Table 4.34 for Batu Pahat Station. This suggests that SARMA performed marginally better, likely due to its ability to account for weekly seasonality. Nonetheless, the performance gains were modest, so the simpler ARMA model could still be considered sufficient when ease of implementation is prioritised over capturing seasonal patterns explicitly. Yet, capturing seasonal patterns is prioritised, hence, SARMA model is preferred.

Overall, Kluang's ARMA(1,2) was the top performer with the lowest RMSE (2.94) and lowest MAE (2.17) across all stations as shown in Table 4.34 for Kluang Station. While Tangkak also stood out for its relative accuracy with the lowest MAPE (21.36%) as presented in Table 4.34 for Tangkak Station. These outcomes show that model effectiveness is highly dependent on local environmental variability. Simpler models sufficed for stable areas, whereas slightly more complex models were needed to handle cyclical or unstable pollution patterns.

4.7 Out-of-Sample Forecasting

The tables below present the out-of-sample forecasting results, specifically detailing the 14-day PM2.5 predictions for all eight stations using either ARMA or SARMA models.

Table 4.35: 14-Day PM2.5 Forecast for Batu Pahat Using SARMA Model

Day	Forecast	Lower Limit of 95%	Upper Limit of 95%
1	6.18	-1.59	13.95
2	7.87	-1.72	17.46
3	9.13	-1.28	19.55
4	10.05	-0.77	20.87
5	10.70	-0.32	21.72
6	10.91	-0.21	22.04
7	10.91	-0.27	22.09
8	11.32	0.04	22.61
9	11.65	0.31	23.00
10	11.89	0.52	23.27
11	12.07	0.68	23.46
12	12.20	0.80	23.59
13	12.26	0.86	23.66
14	12.29	0.89	23.69

Table 4.36: 14-Day PM2.5 Forecast for Kluang Using ARMA Model

Day	Forecast	Lower Limit of 95%	Upper Limit of 95%
1	5.49	-0.33	11.31
2	6.89	-0.30	14.07
3	7.89	0.0	15.70
4	8.63	0.51	16.74
5	9.16	0.88	17.43
6	9.54	1.19	17.89
7	9.82	1.42	18.21
8	10.02	1.60	18.44
9	10.17	1.74	18.60
10	10.27	1.84	18.71
11	10.35	1.91	18.79
12	10.40	1.96	18.84
13	10.44	2.00	18.89
14	10.47	2.03	18.91

Table 4.37: 14-Day PM2.5 Forecast for Kota Tinggi Using ARMA Model

Day	Forecast	Lower Limit of 95%	Upper Limit of 95%
1	6.07	0.19	11.96
2	7.46	0.43	14.50
3	8.37	0.90	15.84
4	8.96	1.31	16.62
5	9.35	1.63	17.08
6	9.61	1.85	17.37
7	9.77	2.00	17.55
8	9.88	2.10	17.66
9	9.95	2.17	17.74
10	10.00	2.22	17.78
11	10.03	2.25	17.81
12	10.05	2.27	17.83
13	10.07	2.28	17.85
14	10.07	2.29	17.86

Table 4.38: 14-Day PM2.5 Forecast for Larkin Using ARMA Model

Day	Forecast	Lower Limit of 95%	Upper Limit of 95%
1	10.80	2.52	19.08
2	12.85	3.02	22.68
3	14.16	3.76	24.56
4	15.00	4.37	25.62

5	15.53	4.82	26.25
6	15.88	5.12	26.63
7	16.09	5.33	26.86
8	16.24	5.46	27.01
9	16.33	5.55	27.10
10	16.38	5.61	27.16
11	16.42	5.64	27.20
12	16.44	5.67	27.22
13	16.46	5.68	27.24
14	16.47	5.69	27.24

Table 4.39: 14-Day PM2.5 Forecast for Pasir Gudang Using ARMA Model

Day	Forecast	Lower Limit of 95%	Upper Limit of 95%
1	8.05	0.84	15.25
2	9.49	1.4	17.84
3	10.28	1.52	19.03
4	10.72	1.79	19.65
5	10.97	1.94	19.99
6	11.12	2.04	20.20
7	11.22	2.09	20.35
8	11.29	2.12	20.45
9	11.34	2.15	20.54
10	11.39	2.16	20.61
11	11.43	2.18	20.68
12	11.46	2.19	20.74
13	11.50	2.20	20.80
14	11.53	2.21	20.85

Table 4.40: 14-Day PM2.5 Forecast for Pengerang Using ARMA Model

Day	Forecast	Lower Limit of 95%	Upper Limit of 95%
1	6.91	-0.22	14.04
2	7.94	-0.31	16.18
3	8.16	-0.63	16.94
4	8.91	-0.36	18.18
5	9.05	-0.47	18.56
6	9.61	-0.15	19.36
7	9.69	-0.18	19.56
8	10.10	0.11	20.09
9	10.15	0.10	20.19
10	10.45	0.34	20.56
11	10.47	0.34	20.61

12	10.70	0.53	20.86
13	10.71	0.53	20.89
14	10.88	0.68	21.07

Table 4.41: 14-Day PM2.5 Forecast for Segamat Using ARMA Model

Day	Forecast	Lower Limit of 95%	Upper Limit of 95%
1	6.04	-1.56	13.64
2	9.13	-0.45	18.72
3	10.01	-0.05	20.06
4	11.18	0.91	21.45
5	10.86	0.57	21.16
6	10.94	0.61	21.27
7	10.31	-0.04	20.66
8	10.29	-0.14	20.71
9	10.073	-0.45	20.60
10	10.47	-0.23	21.16
11	10.81	-0.05	21.67
12	11.53	0.51	22.55
13	12.09	0.98	23.21
14	12.75	1.57	23.92

Table 4.42: 14-Day PM2.5 Forecast for Tangkak Using ARMA Model

Day	Forecast	Lower Limit of 95%	Upper Limit of 95%
1	14.17	6.09	22.25
2	12.76	2.58	22.94
3	12.39	1.84	22.94
4	12.33	1.64	23.02
5	12.35	1.58	23.13
6	12.40	1.56	23.24
7	12.45	1.55	23.35
8	12.50	1.55	23.46
9	12.55	1.55	23.56
10	12.60	1.56	23.65
11	12.65	1.57	23.74
12	12.69	1.57	23.82
13	12.74	1.58	23.89
14	12.78	1.60	23.96

Among all stations, Batu Pahat was modelled using SARMA and displayed steadily increasing PM_{2.5} concentrations over the 14-day forecast period. Batu Pahat's forecasts gradually rose from 6.18 µg/m³ to 12.29 µg/m³ over 14 days, with tight 95% confidence intervals in later days, reflecting a consistent yet predictable increase in pollution levels as was shown in Table 4.35. In contrast, Segamat was modelled using ARMA, and its forecasts showed a similar upward trend from 6.04 µg/m³ to 12.75 µg/m³ over the 14 days, but with wider confidence intervals in the earlier days, for example, at day 1 the lower bound was at -1.56 µg/m³, narrowing slightly over time. This suggest the ARMA model captured the general upward pattern effectively, though with greater initial uncertainty due to Segamat's more volatile pollution behaviour, as reflected in the Table 4.41.

In contrast, ARMA-based forecasts varied more based on local station characteristics. Urban stations like Kota Tinggi and Kluang, forecasts were relatively smooth, increasing from around 5–6 µg/m³ and rising to approximately 10–11 µg/m³ by day 14. The narrowing confidence intervals in these stations pointed to low volatility and well-behaved pollution levels with minimal fluctuation as presented in Table 4.37 and Table 4.36 respectively.

However, stations with more volatile emission subject to greater variability, such as Larkin and Pasir Gudang showed sharper forecast rises, reaching up to 16.47 µg/m³ and 11.53 µg/m³ respectively by day 14. The wider initial intervals for these stations reflected more uncertainty in short-term emissions, likely tied to unpredictable industrial activities as illustrated in Table 4.38 and Table 4.39 respectively. Similarly, Pengerang's forecast remained within 6.91 to 10.88 µg/m³, suggesting intermittent fluctuations likely due to localised activities according to Table 4.40.

Tangkak recorded the highest starting value among all stations at $14.17 \mu\text{g}/\text{m}^3$, with forecasts remaining elevated across the 14 days. Despite its rural nature, the high PM_{2.5} forecast could hint at localised emission sources or weather-driven retention of pollutants. Notably, Tangkak's confidence intervals remained relatively wide, implying ongoing volatility in its pollution trends as Table 4.42 suggests.

Overall, the SARMA model at Batu Pahat provided stable and clear forecasts, particularly suited to stations with identifiable seasonal cycles. Meanwhile, ARMA models, including at Segamat, delivered reliable short-term forecasts in station where seasonal effects were less pronounced, capturing upward trends while allowing for flexibility in handling stations with more variable pollution patterns. These out-of-sample forecasts offer valuable early warnings, enabling local authorities to better anticipate and respond to air quality risks across diverse settings.

4.8 Comparison Across Stations

Table 4.43 provides a compilation of performance metrics for the best-fitting models identified at each of the eight stations. These metrics include the AIC, BIC, RMSE, MAE and MAPE.

Table 4.43: Performance Metrics of Best Models of all 8 Stations

Station	Best Model	AIC	BIC	RMSE	MAE	MAPE(%)
Batu Pahat	ARMA(1,2)	2049.75	2062.66	4.04	3.09	27.50
	SARMA(1,0)(1,0)[7]	2047.39	2062.99	3.95	2.99	26.80
Kluang	ARMA(1,2)	1832.82	1852.32	2.94	2.17	23.45
Kota Tinggi	ARMA(1,0)	1843.05	1854.75	2.99	2.33	25.80
Larkin	ARMA(1,0)	2092.17	2103.87	4.21	3.31	22.13
Pasir Gudang	ARMA(2,1)	1993.05	2012.55	3.66	2.88	22.81
Pengerang	ARMA(2,2)	1986.55	2009.95	3.61	2.66	25.19
Segamat	ARMA(4,3)	2036.70	2071.80	3.84	2.83	23.41
	SARMA(2,2)(0,1)[7]	2044.18	2071.48	3.90	2.89	23.72
Tangkak	ARMA(2,2)	2083.12	2106.53	4.09	2.85	21.36

Kluang emerged as one of the strongest performers in terms of model accuracy. Its ARMA(1,2) model recorded the lowest RMSE (2.94) and MAE (2.17), along with

a low AIC (1832.82), as shown in Table 4.43 for Kluang Station, highlighting its effectiveness in capturing the station's stable PM_{2.5} trends. Similarly, Kota Tinggi and Pasir Gudang also showed strong results as illustrated in Table 4.43 for Kota Tinggi Station and Pasir Gudang Station respectively. Both used simpler ARMA structures and recorded low RMSE values (2.99 and 3.66 respectively), with moderate MAPE percentages which are 25.80% and 22.81% respectively, indicating consistent air quality that did not necessitate complex models.

In Tangkak, the ARMA(2,2) model achieved the lowest MAPE (21.36%), reflecting high relative forecasting accuracy. Even though, the RMSE was slightly higher at 4.09 as demonstrated in Table 4.43 for Tangkak Station, the model still performed well in proportional error terms. On the other hand, Larkin showed slightly higher forecast error, with an RMSE of 4.21 but relatively low MAPE of 22.13%, suggesting the model maintained general forecasting consistency despite some variation in individual values as demonstrated in Table 4.43 for Larkin Station.

Segamat, by contrast, was one of the more challenging stations to model due to its highly volatile PM_{2.5} levels. The ARMA(4,3) model outperformed the SARMA(2,2)(0,1)[7] model across all error metrics, achieving lower RMSE (3.84 vs. 3.90), MAE (2.83 vs. 2.89), and MAPE (23.41% vs. 23.72%), as shown in Table 4.43. These results indicate that while SARMA was designed to capture weekly seasonal patterns, in practice the ARMA(4,3) provided better predictive accuracy for Segamat. Therefore, ARMA was selected as the preferred model for this station, balancing complexity with forecast performance.

Batu Pahat also exhibited evidence of weekly seasonality. In this case, the SARMA(1,0)(1,0)[7] model performed slightly better than the ARMA(1,2), as

reflected in the performance metrics in Table 4.43. Specifically, SARMA achieved lower RMSE (3.95 vs. 4.04), MAE (2.99 vs. 3.09) and MAPE (26.80% vs. 27.50%), indicating more accurate forecasts. This suggests that incorporating seasonal components marginally improved the model's ability to capture the weekly cycles present in Batu Pahat's PM2.5 data, making SARMA the preferable choice for this station.

When comparing all stations overall, the results showed that different model complexities were needed for different environments. Kluang, Kota Tinggi and Pasir Gudang benefited from simpler ARMA models due to their stable air quality trends. In contrast, more volatile stations like Segamat required more complex ARMA to effectively capture the fluctuations. Tangkak, while more variable in value range, still delivered highly accurate predictions on a percentage basis. Batu Pahat showed consistent weekly seasonality, the SARMA model showed lower forecast accuracy and lower AIC value, which justified the use of SARMA models for the location.

In summary, the best-fitting models differed across stations, reflecting the diverse behaviour of PM2.5 levels in Johor. Simpler ARMA models worked well in stable urban areas, while more advanced or seasonal models were necessary in locations with fluctuating or cyclical pollution patterns. These findings highlight the importance of tailoring time series models to the unique environmental dynamics of each station.

4.9 Discussion

4.9.1 Spatial Variation in PM2.5 Patterns

The findings from this study reveal that PM2.5 patterns vary significantly across the different locations in Johor, and these variations can be explained by the environmental context surrounding each station. Industrial and urban stations such as Pasir Gudang, Pengerang and Larkin exhibited more complex pollution dynamics, necessitating ARMA models with multiple autoregressive and moving average terms. In Pasir Gudang, the ARMA(2,1) model was particularly well-suited, with all coefficients statistically significant. This indicates that pollution levels in such regions are not only consistently high but also influenced by past values and disturbances, likely due to steady industrial emissions and vehicular traffic.

4.9.2 Pollution Behaviour in Rural vs Urban Areas

Simpler pollution dynamics were observed in rural or semi-urban areas like Segamat and Tangkak. Their models reflected moderate autocorrelation. This makes sense, given that rural areas typically experience less daily industrial activity but may still be affected by factors such as agricultural burning, weather changes or transboundary haze. In urban commercial zones like Kluang and Larkin, daily human and vehicle movement appears to drive PM2.5 levels. These stations had moderately high values, indicating that while pollution is present, the behaviour is not overly erratic or seasonal, making them well-suited for simpler ARMA structures.

4.9.3 Meteorological and Cyclical Factors

Meteorological factors also appear to play an important role. For example, Batu Pahat displayed weekly cyclical patterns in their autocorrelation plots which likely is driven by repeating weather conditions or operational schedules of nearby industrial

sources. The incorporation of seasonal terms in SARMA model helped account for these periodic fluctuations.

4.9.4 Consistency with Previous Research

When comparing these findings to existing research, there's strong alignment. Previous studies, such as those by Lee and Tuen (2012) and Ismail et al. (2016), also found ARMA and SARMA models to be reliable for air quality prediction in Malaysia. This study builds on those efforts by delivering station-specific insights. Other research, such as Afzali (2014), emphasized industrial zones like Pasir Gudang as pollution hotspots, a conclusion strongly supported by your model's results for that location. Additionally, Zhang et al. (2024) emphasised that despite the rise of AI-based forecasting models, traditional statistical models like ARMA and SARMA remain relevant due to their interpretability and ease of deployment, particularly in data-sparse environments.

4.9.5 Practical Applications of Forecasting

The practical applications of the models are also noteworthy. Being able to forecast PM2.5 levels up to 14 days in advance means local authorities and communities can take preventive action rather than just reacting to air pollution events. In industrial areas, forecasted high pollution periods could lead to stricter enforcement of emissions controls or temporary shutdowns. In busy urban areas, the data could support traffic planning or the issuance of public health advisories. Moreover, the fact that different stations required different models confirms the need for localised air quality policies. A blanket approach won't work because the pollution sources and behaviour vary too much between rural, urban and industrial zones. This study provides strong evidence that area-specific modelling and forecasting should inform

decision-making processes, especially for a state like Johor with its diverse environmental and economic landscape.

Lastly, this research shows that affordable and explainable forecasting methods can be effective even without complex AI models. This is important for smaller municipalities or government agencies that may not have the resources for high-end data science but still want to act on air quality data in meaningful ways.

4.10 Summary of the Chapter

This chapter provided the findings and discussion of a time series analysis of PM2.5 concentration data collected from eight monitoring sites in Johor for the year 2021. The investigation began with a complete descriptive evaluation, which indicated spatial differences in PM2.5 levels, with urban and industrial regions such as Larkin and Pasir Gudang having higher average concentrations than more suburban places like Kota Tinggi.

Stationarity testing with the Augmented Dickey-Fuller (ADF) test revealed that all eight PM2.5 time series were stationary at the 5% threshold. As a result, no differencing was necessary for model construction, simplifying the modelling process while maintaining the validity of future ARMA and SARMA model implementations.

Following stationarity confirmation, model identification and selection were performed using Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots, as well as the `auto.arima` function in RStudio. The Akaike Information Criterion (AIC) used to choose models that struck the best balance between model simplicity and quality of fit. The best-fitting models determined for each station were as follows: The top models for Batu Pahat were

SARMA(1,0)(1,0)[7], Kluang with ARMA(1,2), and Kota Tinggi with ARMA(1,0). Similarly, ARMA(1,0) was chosen for Larkin, ARMA(2,1) for Pasir Gudang and ARMA(2,2) for Pengerang, with acceptable models for Segamat and Tangkak identified using the same statistical criteria.

The diagnostic examination of these models revealed that the residuals behaved like white noise and the Ljung-Box test validated the suitability of the models chosen for each station. The forecast accuracy was also evaluated using conventional performance indicators such as the Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE). The models produced reasonable forecast accuracy, with SARMA models outperforming others in capturing seasonal trends, as seen in Batu Pahat.

Overall, the findings in this chapter support the appropriateness of the ARMA and SARMA models for modelling and forecasting PM_{2.5} concentrations in Johor. The findings provide light on air quality dynamics in various parts of the state, laying a foundation for informed decision-making and air quality management initiatives. The next chapter gives the conclusions and suggestions based on these findings.

CHAPTER 5

CONCLUSION

5.1 Concluding remarks

This study aimed to model and forecast daily PM2.5 concentrations across eight different districts in Johor using time series techniques, specifically ARMA and SARMA models. The selected locations, including Batu Pahat, Kluang, Kota Tinggi, Larkin, Pasir Gudang, Pengerang, Segamat, and Tangkak, represented a diverse mix of urban, semi-urban and industrial settings. Throughout the process, the data is carefully analysed using a range of tools such as AIC, BIC, RMSE, MAE, MAPE, as well as residual diagnostics like ACF and PACF plots, Q-Q plots and the Ljung-Box test to ensure the adequacy of model fit and the model's reliability in the sense of statistical soundness.

Between the two models, SARMA stood out for stations that showed clear seasonal patterns therefore proved to be superior and effective. Locations such as Batu Pahat and Segamat especially support the claim on the SARMA models. By including a weekly seasonal component, SARMA was better able to account for recurring PM2.5 variations, which likely influenced by weekly patterns in human activity or weather. This resulted in better performance, shown through lower error rates and stronger model fit statistics.

In contrast, for districts such as Kluang, Kota Tinggi, Larkin, Pasir Gudang, Pengerang and Tangkak, where seasonality was not obvious, ARMA models were well-suited as they seem to have performed exceptionally well. Even with a simpler structure, these models passed all diagnostic checks. Their ability to capture the essential trends and noise in the data such as short-term dependencies shows that

ARMA can be a highly effective tool for forecasting air quality, particularly in urban areas where pollution levels may fluctuate less predictably.

In short, SARMA models are the better choice when the data shows regular patterns over time, while ARMA models are ideal for more stable or less seasonal data patterns. The results of this study show the critical importance of choosing the right model for each location. By doing so, we can improve the accuracy of PM_{2.5} forecasts and provide meaningful support for proactive planning and air quality management that ultimately contributes to improved public health outcomes.

5.2 Limitations

While this study has demonstrated ARMA and SARMA models can be effective tools for forecasting PM_{2.5} concentrations in Johor, there are still a few limitations to acknowledge.

Firstly, the models employed in this study were univariate, meaning they relied solely on past PM_{2.5} values to make future predictions. This approach excludes important factors like weather (temperature, humidity, wind speed) and external influences like transboundary haze from neighbouring countries. The absence of these inputs may lower the models' ability to anticipate abrupt changes or extreme pollution episodes.

Secondly, the analysis was according to the daily average PM_{2.5} values, even though they are suitable for identifying general trends, yet they may fail to capture short-term fluctuations or intra-day pollution peaks. Using hourly data in future studies could offer a more detailed view and improve the model's ability to provide early warnings during pollution events.

Another challenge posed is data availability and completeness. Not all monitoring stations had complete or consistent records. In some cases, missing values had to be filled in, which can affect the overall accuracy of the models. Also, the presence and strength of seasonal patterns varied across locations, making it more difficult to use SARMA models consistently

Finally, while ARMA and SARMA are practical and easy to interpret, they may not provide the highest possible forecasting accuracy. Advanced techniques like LSTM neural networks or hybrid models could yield improved results, especially with more comprehensive datasets. These newer models, however, come with their own challenges like being harder to explain and needing more computing power.

5.3 Research Recommendations and Contributions

Looking ahead, there are several ways this research could be expanded and improved. A primary recommendation is to extend the current univariate time series approach by incorporating additional exogenous variables. For instance, variables such as temperature, rainfall, wind speed and humidity play a significant role in air pollution levels. Including these variables would enable the development of multivariate time series models, which could yield more accurate and responsive forecasts, especially during irregular pollution events like haze episodes or industrial emission surges.

In addition, future studies should consider utilising higher-resolution data, like hourly PM_{2.5} measurements. This would help capture sudden pollution spikes, which are often missed in daily averages. With more granular data, researchers can better fine-tune the model's ability to provide timely and targeted alerts to both the public and policymakers.

Another recommendation is the application of advanced machine learning models, such as Long Short-Term Memory (LSTM) networks, Random Forests or hybrid combinations like ARMA-LSTM. These models are known for their ability to handle complex, nonlinear patterns in data. Although these methods require more computational resources and complex tuning, they have the potential to improve forecasting performance in high-variability environments like urban and industrial areas.

Despite its univariate limitations, this study offers meaningfully to regional air quality forecasting. First, it provides location-specific PM_{2.5} forecasts for eight different stations in Johor, covering both urban centres and industrial zones. This adds a layer of practical insight for local policymakers who need tailored information for different types of areas. Second, the study clearly confirms the utility of SARMA models in locations with identifiable seasonal cycles, while reconfirming the effectiveness of simpler ARMA models in stations without strong seasonality and is more stable.

Additionally, the research highlights the importance of comprehensive model diagnostics. Evaluating residuals for white noise and autocorrelation, to ensure the chosen models are statistically sound and reliable for forecasting. These checks are often overlooked but are critical for ensuring statistical robustness and reliability in forecasting applications.

Finally, this project illustrates how simple, interpretable forecasting tools like ARMA and SARMA can still play a powerful role in environmental management. Their ease of implementation and transparency make them highly suitable for

informing policy decisions and advancing air quality management, ultimately helping to protect public health in a fast-developing and pollution-prone region like Johor.

REFERENCES

- Abhiram, V., & Sivaramakrishnan, B. (2018). Time series analysis and forecasting of air pollutant PM10 in Chennai using ARIMA model. *International Journal of Civil Engineering and Technology*, 9(8), 1146–1154.
- Abu Bakar, S. A., Ismail, S. N., & Yusoff, M. A. M. (2021). Comparative performance of ARIMA and LSTM for air pollution forecasting in Malaysia. *Journal of Environmental Sciences*, 110, 12–25.
- Afzali, M. (2014). Air quality trend analysis and weather impact assessment in Pasir Gudang industrial area, Malaysia. *Atmospheric Pollution Research*, 5(2), 316–326.
- Ahmad, S. M., Jamil, N. I., & Mohd Noor, N. S. (2021). A review on seasonal air quality forecasting models in Southeast Asia. *International Journal of Environmental Research and Public Health*, 18(6), 3141. <https://doi.org/10.3390/ijerph18063141>
- Alexandra, C., Tantri, T., & Gunawan, F. E. (2019). Analyzing the performance criteria of ARMA model for air quality forecasting in Jakarta. *Journal of Environmental Management and Tourism*, 10(7), 1591–1600. <https://journals.aserspublishing.eu/jemt/article/view/4285>
- Box, G. E., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2016). *Time series analysis: Forecasting and control* (5th ed.). John Wiley & Sons. https://students.aiu.edu/submissions/profiles/resources/onlineBook/G6j8h8_Time_Series_Analysis_Forecasting5.pdf
- Brook, R. D., Rajagopalan, S., Pope, C. A., Brook, J. R., Bhatnagar, A., Diez-Roux, A. V., ... Kaufman, J. D. (2010). Particulate matter air pollution and cardiovascular disease: An update to the scientific statement from the American Heart Association. *Circulation*, 121(21), 2331–2378. <https://doi.org/10.1161/CIR.0b013e3181dbeece1>
- Department of Environment Malaysia. (2022). *Malaysia environmental quality report 2021*. Ministry of Environment and Water.
- Gaveau, D. L., Salim, M. A., Hergoualc’h, K., Locatelli, B., Sloan, S., Wooster, M., Molidena, E. (2014). Major atmospheric emissions from peat fires in Southeast Asia

during non-drought years: Evidence from the 2013 Sumatran fires. *Scientific Reports*, 4, 6112. <https://doi.org/10.1038/srep06112>

Hanafi, N. D. N., Yusof, N. F. F. M., Sahani, M., & Abdullah, N. A. (2019). Overview of health impacts due to haze pollution in Johor, Malaysia. *EnvironmentAsia*, 12(2), 80–92. <https://doi.org/10.14456/ea.2019.21>

Hasim, M. S., Latif, M. T., & Tang, R. (2020). Spatiotemporal variability and source apportionment of PM_{2.5} in urban and semi-urban areas of Malaysia. *Atmospheric Research*, 240, 104915. <https://doi.org/10.1016/j.atmosres.2020.104915>

Hassan, M. N., Mohd, N. S., & Yusof, N. F. M. (2019). Time series analysis of PM_{2.5} concentrations in Klang Valley using ARIMA and Multiple Linear Regression models. *Journal of Sustainability Science and Management*, 14(4), 45–55.

IQAir. (2025). Johor air quality index (AQI) and PM_{2.5} air pollution. Retrieved from <https://www.iqair.com/malaysia/johor>

Jadhav, A., Gaikwad, H., & Harihar, S. (2023). Time series analysis of AQI using ARMA and SARIMA models. *GitHub Repository*. <https://link.springer.com/article/10.1007/s00477-009-0361-8>

Jain, R., Bajracharya, R., & Sharma, S. (2022). Forecasting PM_{2.5} concentrations using SARIMA model in Kathmandu Valley, Nepal. *Environmental Monitoring and Assessment*, 194(9), 712. <https://doi.org/10.1007/s10661-022-10317-1>

Kumar, P., Hama, S., Omidvarborna, H., Sharma, A., Abhijith, K. V., Tiwari, A., ... Morawska, L. (2022). Temporary reduction in air pollution due to anthropogenic activity lockdown during COVID-19 pandemic. *Sustainable Cities and Society*, 65, 102629. <https://doi.org/10.1016/j.scs.2020.102629>

Kumar, U., & Goyal, P. (2011). Forecasting of air quality in Delhi using principal component regression technique. *Atmospheric Pollution Research*, 2(4), 436–444.

Lee, H. S., & Tuen, A. A. (2012). Modelling of air quality trends using ARIMA: A case study in Johor. *Environmental Monitoring and Assessment*, 184(10), 6239–6249.

- Lelieveld, J., Evans, J. S., Fnais, M., Giannadaki, D., & Pozzer, A. (2015). The contribution of outdoor air pollution sources to premature mortality on a global scale. *Nature*, 525(7569), 367–371. <https://doi.org/10.1038/nature15371>
- Li, J., & Zhao, Y. (2022). Hybrid ARIMA model for PM_{2.5} prediction in Beijing: A decomposition-based approach. *Environmental Modelling & Software*, 148, 105272.
- Li, X., Peng, L., Hu, Y., Shao, J., & Chi, T. (2017). Deep learning architecture for air quality predictions. *Environmental Science and Pollution Research*, 24(3), 22453–22464. <https://doi.org/10.1007/s11356-017-9867-8>
- Mohd Fadhil, M. H., Zulkipli, N. A., & Nasir, N. M. (2020). A statistical overview of PM_{2.5} and PM₁₀ trends in Northern Malaysia. *Malaysian Journal of Analytical Sciences*, 24(2), 300–312.
- Rahman, M. A., Latif, M. T., & Ismail, S. N. (2016). SARIMA-based seasonal air quality forecast in Malaysia. *Air Quality, Atmosphere & Health*, 9(3), 223–234.
- Sentian, J., Yusof, N. F. F. M., & Norazian, H. (2019). Urban air quality temporal variation and its relationship with traffic density and meteorological factors in Johor Bahru, Malaysia. *EnvironmentAsia*, 12(1), 136–148. <https://doi.org/10.14456/ea.2019.19>
- Wong, D. C., Tan, Y. C., & Abdullah, M. A. (2021). Machine learning applications in PM_{2.5} prediction across Malaysia: A comparison between LSTM, SVR, and Random Forest. *Atmospheric Pollution Research*, 12(5), 101059.
- World Health Organization. (2021). *WHO global air quality guidelines: Particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide*. World Health Organization. Retrieved from <https://www.who.int/publications/i/item/9789240034228>
- World Health Organization. (2021). Air pollution. Retrieved from <https://www.who.int/health-topics/air-pollution>
- Yadav, S. K., & Sahu, S. K. (2020). Forecasting of daily air quality index in India using ARIMA and hybrid ARIMA models. *Atmospheric Environment*, 232, 117474.

Zaman, H. M. F., Yusof, N. F. F. M., & Tahir, N. M. (2021). Assessment of PM2.5 and PM10 concentrations in major cities in Peninsular Malaysia using statistical and GIS techniques. *Malaysian Journal of Science*, 40(2), 95–106. <https://doi.org/10.22452/mjs.vol40no2.9>

Zhang, Y., Bocquet, M., Mallet, V., Seigneur, C., & Baklanov, A. (2018). Real-time air quality forecasting, part I: History, techniques, and current status. *Atmospheric Environment*, 60, 632–655. <https://doi.org/10.1016/j.atmosenv.2012.06.031>

Zhang, Y., Liu, C., & Wang, T. (2024). Comparative performance of ARIMA, SARIMA, and machine learning models for PM2.5 forecasting in urban China. *Environmental Research*, 232, 116158.

APPENDICES

APPENDIX A: DAILY PM2.5 DATA SAMPLE OF ALL 8 STATIONS

Standard Query (DAILY)															
STATION ID	LOCATIO N	DATE TIME	PM ₁₀ AVG (µg/m ³)	PM _{2.5} AVG (µg/m ³)	SO ₂ AVG (ppm)	NO ₂ AVG (ppm)	NO AVG (ppm)	NOX AVG (ppm)	O ₃ AVG (ppm)	CO AVG (ppm)	WIND DIRECTIO N AVG(°)	WIND SPEED AVG(m/s)	RELATIVE HUMIDIT Y AVG(%)	SOLAR RADIATI ON AVG(W/ m ²)	Ambient Temperat ure AVG(°c)
CA31J	Batu Pahat, JOHOR	1/1/2021	10.479	5.003	0.0015	0.004	0.0028	0.0068	0.0193	0.545	308.181	1.1429	94.9323	25.0472	23.4103
CA31J	Batu Pahat, JOHOR	2/1/2021	7.924	3.78	0.0016	0.0041	0.0016	0.0057	0.0191	0.489	282.87	1.0497	97.3818	2.1438	22.7048
CA31J	Batu Pahat, JOHOR	3/1/2021	7.615	3.008	0.0016	0.0025	0.0029	0.0055	0.0137	0.424	293.2111	1.0651	95.3861	23.7079	22.3953
CA31J	Batu Pahat, JOHOR	4/1/2021	10.884	4.097	0.0017	0.003	0.0039	0.0069	0.0135	0.469	274.5906	1.1212	89.0597	121.8196	24.0695
CA31J	Batu Pahat, JOHOR	5/1/2021	11.223	4.575	0.0017	0.0037	0.0029	0.0066	0.0181	0.458	299.1529	1.4206	83.4838	187.4475	26.286
CA31J	Batu Pahat, JOHOR	6/1/2021	7.587	3.274	0.0016	0.0022	0.0011	0.0034	0.0143	0.386	311.8894	1.991	89.4523	161.457	24.9354
CA31J	Batu Pahat, JOHOR	7/1/2021	8.27	3.697	0.0016	0.0027	0.0054	0.0081	0.0101	0.441	212.1989	1.3136	92.0119	44.3013	23.7745
Standard Query (DAILY)															
STATION ID	LOCATIO N	DATE TIME	PM ₁₀ AVG (µg/m ³)	PM _{2.5} AVG (µg/m ³)	SO ₂ AVG (ppm)	NO ₂ AVG (ppm)	NO AVG (ppm)	NOX AVG (ppm)	O ₃ AVG (ppm)	CO AVG (ppm)	WIND DIRECTIO N AVG(°)	WIND SPEED AVG(m/s)	RELATIVE HUMIDIT Y AVG(%)	SOLAR RADIATI ON AVG(W/ m ²)	Ambient Temperat ure AVG(°c)
CA32J	Kluang, JOHOR	1/1/2021	19.177	10.324	NA	NA	NA	NA	NA	NA	156.583	1.4388	96.3579	22.4205	23.562
CA32J	Kluang, JOHOR	2/1/2021	5.756	2.822	NA	NA	NA	NA	NA	NA	116.686	1.3798	97.9985	0.0009	22.3845
CA32J	Kluang, JOHOR	3/1/2021	6.304	2.539	NA	NA	NA	NA	NA	NA	136.2438	0.5387	98	5.7135	21.7146
CA32J	Kluang, JOHOR	4/1/2021	8.425	3.776	NA	NA	NA	NA	NA	NA	119.5784	0.3809	92.5758	77.994	23.5099
CA32J	Kluang, JOHOR	5/1/2021	11.24	5.058	NA	NA	NA	NA	NA	NA	69.9738	0.8829	85.585	154.21	26.0288
CA32J	Kluang, JOHOR	6/1/2021	10.136	4.879	NA	NA	NA	NA	NA	NA	78.0968	0.5107	90.411	85.4938	25.2447
CA32J	Kluang, JOHOR	7/1/2021	10.184	4.3	NA	NA	NA	NA	NA	NA	155.9325	0.6619	97.1728	21.169	23.4168
CA32J	Kluang, JOHOR	8/1/2021	13.713	4.271	NA	NA	NA	NA	NA	NA	112.2235	1.009	90.7552	106.522	25.6305
CA32J	Kluang, JOHOR	9/1/2021	11.503	3.493	NA	NA	NA	NA	NA	NA	48.5323	0.9088	91.1597	89.592	25.1405
CA32J	Kluang, JOHOR	10/1/2021	5.358	2.136	NA	NA	NA	NA	NA	NA	111.7297	1.645	97.3899	18.6635	23.4784
Standard Query (DAILY)															
STATION ID	LOCATIO N	DATE TIME	PM ₁₀ AVG (µg/m ³)	PM _{2.5} AVG (µg/m ³)	SO ₂ AVG (ppm)	NO ₂ AVG (ppm)	NO AVG (ppm)	NOX AVG (ppm)	O ₃ AVG (ppm)	CO AVG (ppm)	WIND DIRECTIO N AVG(°)	WIND SPEED AVG(m/s)	RELATIVE HUMIDIT Y AVG(%)	SOLAR RADIATI ON AVG(W/ m ²)	Ambient Temperat ure AVG(°c)
CA36J	Kota Tinggi, JOHOR	1/1/2021	15.218	5.845	0.001	0.0008	0.0002	0.001	0.0356	0.564	36.1983	2.8418	89.9473	75.1405	26.3654
CA36J	Kota Tinggi, JOHOR	2/1/2021	7.847	3.706	0.001	0.0014	0.0003	0.0016	0.0264	0.495	221.0732	2.7348	96.1254	1.2358	23.5919
CA36J	Kota Tinggi, JOHOR	3/1/2021	5.32	2.38	0.001	0.0023	0.0004	0.0027	0.0141	0.49	313.0904	1.052	97.6084	19.1295	23.4628
CA36J	Kota Tinggi, JOHOR	4/1/2021	5.209	2.513	0.0009	0.0015	0.0003	0.0018	0.0168	0.398	301.4208	1.4354	91.4329	81.4378	24.337
CA36J	Kota Tinggi, JOHOR	5/1/2021	7.832	4.336	0.001	0.001	0.0003	0.0013	0.0177	0.322	186.5019	1.7088	90.7597	141.1379	25.5701
CA36J	Kota Tinggi, JOHOR	6/1/2021	7.555	3.457	0.0011	0.0011	0.0002	0.0014	0.014	0.303	219.2461	1.379	89.8731	117.5625	25.6843

Standard Query (DAILY)															
STATION ID	LOCATIO N	DATE TIME	PM ₁₀ AVG (µg/m ³)	PM _{2.5} AVG (µg/m ³)	SO ₂ AVG (ppm)	NO ₂ AVG (ppm)	NO AVG (ppm)	NOX AVG (ppm)	O ₃ AVG (ppm)	CO AVG (ppm)	WIND DIRECTIO N AVG(°)	WIND SPEED AVG(m/s)	RELATIVE HUMIDIT Y AVG(%)	SOLAR RADIATI ON AVG(W/m ²)	Ambient Temperat ure AVG(°c)
CA33J	Larkin, JOHOR	1/1/2021	12.557	7.401	0.0019	0.0096	0.0015	0.011	0.0165	0.319	260.268	0.9868	96.2709	17.4856	23.8888
CA33J	Larkin, JOHOR	2/1/2021	8.104	5.674	0.002	0.0083	0.0015	0.0097	0.0157	0.223	263.8388	1.2267	98.5472	0.071	22.8269
CA33J	Larkin, JOHOR	3/1/2021	10.138	6.173	0.002	0.0066	0.0023	0.009	0.007	0.217	287.9434	0.6088	94.5931	36.2662	23.9474
CA33J	Larkin, JOHOR	4/1/2021	10.492	6.041	0.0019	0.0063	0.0024	0.0087	0.0109	0.196	301.1253	0.7982	88.3823	91.734	25.3364
CA33J	Larkin, JOHOR	5/1/2021	12.798	7.16	0.0021	0.0076	0.0026	0.0102	0.0131	0.197	322.2432	0.8675	83.0415	136.8455	27.4656
CA33J	Larkin, JOHOR	6/1/2021	20.128	13.45	0.0022	0.006	0.0021	0.0082	0.011	0.162	318.9873	0.8793	84.1065	139.8051	26.998
CA33J	Larkin, JOHOR	7/1/2021	12.916	7.777	0.0021	0.0094	0.0074	0.0169	0.0045	0.22	311.1267	0.5415	93.1117	28.7589	24.4046
CA33J	Larkin, JOHOR	8/1/2021	17.541	8.478	0.0019	0.0096	0.0046	0.0142	0.0105	0.263	167.5435	0.6855	93.0934	79.163	25.5312
CA33J	Larkin, JOHOR	9/1/2021	20.017	10.425	0.0019	0.0083	0.0024	0.0107	0.0138	0.22	270.9769	0.5119	93.7542	59.9692	25.2685
CA33J	Larkin, JOHOR	10/1/2021	8.723	5.63	0.0019	0.0077	0.0008	0.0085	0.0179	0.221	321.8806	1.2158	98.0439	3.3578	23.8686
Standard Query (DAILY)															
STATION ID	LOCATIO N	DATE TIME	PM ₁₀ AVG (µg/m ³)	PM _{2.5} AVG (µg/m ³)	SO ₂ AVG (ppm)	NO ₂ AVG (ppm)	NO AVG (ppm)	NOX AVG (ppm)	O ₃ AVG (ppm)	CO AVG (ppm)	WIND DIRECTIO N AVG(°)	WIND SPEED AVG(m/s)	RELATIVE HUMIDIT Y AVG(%)	SOLAR RADIATI ON AVG(W/m ²)	Ambient Temperat ure AVG(°c)
CA34J	Pasir Gudang, JOHOR	1/1/2021	11.928	6.825	0.0013	0.0113	0.0047	0.016	0.0163	0.732	239.0443	2.1285	93.5193	15.186	25.9777
CA34J	Pasir Gudang, JOHOR	2/1/2021	6.878	4.187	0.0013	0.0123	0.0029	0.0152	0.0123	0.574	254.2738	2.9405	95.2195	0.002	24.1317
CA34J	Pasir Gudang, JOHOR	3/1/2021	9.137	5.749	0.0013	0.0099	0.0043	0.0142	0.0051	0.565	308.5435	1.0057	92.8389	25.8195	25.5955
CA34J	Pasir Gudang, JOHOR	4/1/2021	7.192	4.097	0.0012	0.0057	0.0034	0.009	0.0078	0.495	301.332	0.9451	87.041	70.2542	26.9868
CA34J	Pasir Gudang, JOHOR	5/1/2021	10.106	5.715	0.0013	0.0052	0.0037	0.0088	0.0102	0.534	196.2409	1.135	82.361	123.4372	28.9476
CA34J	Pasir Gudang, JOHOR	6/1/2021	9.918	5.513	0.0012	0.0045	0.0023	0.0069	0.0084	0.441	312.061	1.3829	82.6518	121.5531	28.9003
CA34J	Pasir Gudang, JOHOR	7/1/2021	8.534	5.331	0.0012	0.0062	0.0037	0.01	0.0048	0.491	281.6972	1.2061	91.592	19.4176	26.1933
Standard Query (DAILY)															
STATION ID	LOCATIO N	DATE TIME	PM ₁₀ AVG (µg/m ³)	PM _{2.5} AVG (µg/m ³)	SO ₂ AVG (ppm)	NO ₂ AVG (ppm)	NO AVG (ppm)	NOX AVG (ppm)	O ₃ AVG (ppm)	CO AVG (ppm)	WIND DIRECTIO N AVG(°)	WIND SPEED AVG(m/s)	RELATIVE HUMIDIT Y AVG(%)	SOLAR RADIATI ON AVG(W/m ²)	Ambient Temperat ure AVG(°c)
CA35J	Pengeran g, JOHOR	1/1/2021	11.401	4.647	0.0019	0.0019	0.0009	0.0028	NA	NA	86.5415	1.4308	95.1095	56.8636	23.482
CA35J	Pengeran g, JOHOR	2/1/2021	6.026	3.23	0.0019	0.0029	0.0008	0.0037	NA	NA	224.7161	1.5721	98.1855	1.6319	21.5159
CA35J	Pengeran g, JOHOR	3/1/2021	6.334	3.179	0.0019	0.0018	0.0013	0.0031	NA	NA	148.0357	0.5019	96.9419	17.516	22.3078
CA35J	Pengeran g, JOHOR	4/1/2021	9.011	5.545	0.0014	0.0013	0.0012	0.0024	NA	NA	30.8356	0.6298	91.7574	89.8562	23.3618
CA35J	Pengeran g, JOHOR	5/1/2021	8.564	4.015	0.0008	0.0013	0.0013	0.0026	NA	NA	50.7788	0.8725	91.4495	161.9683	24.2006
CA35J	Pengeran g, JOHOR	6/1/2021	9.575	4.308	0.0007	0.0011	0.001	0.0021	NA	NA	43.8445	0.6753	88.7324	130.9338	24.8979
CA35J	Pengeran g, JOHOR	7/1/2021	7.36	3.575	0.0008	0.0012	0.0011	0.0022	NA	NA	126.004	0.6692	96.1538	21.5833	22.2958
CA35J	Pengeran g, JOHOR	8/1/2021	13.391	6.047	0.0007	0.0009	0.0009	0.0018	NA	NA	51.5523	0.9005	95.1457	82.4599	23.7844
CA35J	Pengeran g, JOHOR	9/1/2021	13.638	4.635	0.0008	0.0009	0.0007	0.0016	NA	NA	32.7356	1.1135	93.6892	90.7605	24.0784

Standard Query (DAILY)															
STATION ID	LOCATIO N	DATE TIME	PM ₁₀ AVG (µg/m ³)	PM _{2.5} AVG (µg/m ³)	SO ₂ AVG (ppm)	NO ₂ AVG (ppm)	NO AVG (ppm)	NOX AVG (ppm)	O ₃ AVG (ppm)	CO AVG (ppm)	WIND DIRECTIO N AVG(°)	WIND SPEED AVG(m/s)	RELATIVE HUMIDIT Y AVG(%)	SOLAR RADIATI ON AVG(W/m ²)	Ambient Temperat ure AVG(°c)
CA29J	Segamat, JOHOR	1/1/2021	8.346	4.745	0.0022	0.0045	0.0014	0.0059	0.0238	0.716	41.5219	1.5358	97.0785	27.9217	23.5019
CA29J	Segamat, JOHOR	2/1/2021	7.706	4.097	0.0022	0.0043	0.0021	0.0064	0.0149	0.744	40.0617	0.9662	98.058	6.2403	23.224
CA29J	Segamat, JOHOR	3/1/2021	5.271	3.595	0.0022	0.0055	0.0045	0.01	0.0113	0.748	44.9579	1.4895	99.816	6.5631	21.7787
CA29J	Segamat, JOHOR	4/1/2021	6.094	3.375	0.0027	0.004	0.0029	0.0068	0.0099	0.738	44.1907	1.1675	93.4962	77.65	23.8463
CA29J	Segamat, JOHOR	5/1/2021	9.587	4.866	0.0029	0.0036	0.0036	0.0072	0.0108	0.711	55.1263	1.4173	85.575	145.516	26.4219
CA29J	Segamat, JOHOR	6/1/2021	7.789	4.582	0.0024	0.0042	0.0046	0.0088	0.0086	0.715	65.9005	0.8039	93.3817	65.7846	24.681
CA29J	Segamat, JOHOR	7/1/2021	8.688	4.781	0.002	0.0048	0.0063	0.0111	0.0058	0.743	82.0516	0.9496	96.5125	25.8559	23.6785
CA29J	Segamat, JOHOR	8/1/2021	11.6	5.038	0.0023	0.0038	0.0026	0.0064	0.0136	0.676	74.096	1.4457	90.3815	105.7086	25.5048
CA29J	Segamat, JOHOR	9/1/2021	9.193	4.401	0.0024	0.0049	0.0027	0.0076	0.0164	0.756	41.7354	1.2051	97.32	45.4789	23.7673
CA29J	Segamat, JOHOR	10/1/2021	9.426	4.359	0.0024	0.0057	0.0031	0.0088	0.02	0.82	45.8155	1.6975	94.2695	41.0435	23.846
Standard Query (DAILY)															
STATION ID	LOCATIO N	DATE TIME	PM ₁₀ AVG (µg/m ³)	PM _{2.5} AVG (µg/m ³)	SO ₂ AVG (ppm)	NO ₂ AVG (ppm)	NO AVG (ppm)	NOX AVG (ppm)	O ₃ AVG (ppm)	CO AVG (ppm)	WIND DIRECTI ON AVG(°)	WIND SPEED AVG(m/s)	RELATIVE HUMIDIT Y AVG(%)	SOLAR RADIATI ON AVG(W/m ²)	Ambient Tempera ture AVG(°c)
CA66J	Tangkak, JOHOR	1/1/2021	16.369	8.69	0.0012	0.0026	0.0008	0.0035	0.0212	0.764	227.693	0.5183	86.7938	39.5353	25.5413
CA66J	Tangkak, JOHOR	2/1/2021	14.496	8.28	0.0012	0.0029	0.001	0.0039	0.0215	0.71	280.923	1.1033	88.2935	10.1744	24.9352
CA66J	Tangkak, JOHOR	3/1/2021	14.147	8.065	0.0012	0.0025	0.0011	0.0036	0.0085	0.654	171.168	0.3419	94.2874	19.765	23.5737
CA66J	Tangkak, JOHOR	4/1/2021	20.603	11.134	0.0012	0.0045	0.0025	0.007	0.0059	0.678	175.904	0.3104	88.2664	91.675	24.9666
CA66J	Tangkak, JOHOR	5/1/2021	16.877	8.255	0.0012	0.0038	0.0018	0.0056	0.0123	0.678	179.035	0.6293	80.6825	171.625	27.2465
CA66J	Tangkak, JOHOR	6/1/2021	13.949	6.391	0.0012	0.0031	0.0016	0.0047	0.0095	0.645	269.235	1.0413	85.1545	135.09	25.8956
CA66J	Tangkak, JOHOR	7/1/2021	14.779	8.654	0.0012	0.0027	0.0017	0.0044	0.0099	0.744	178.06	0.6821	85.8495	39.516	25.4754
CA66J	Tangkak, JOHOR	8/1/2021	20.207	10.216	0.0013	0.0028	0.0014	0.0042	0.0126	0.655	144.278	0.9827	86.3019	160.42	26.0454
CA66J	Tangkak, JOHOR	9/1/2021	13.943	6.61	0.0013	0.0022	0.0009	0.0032	0.0179	0.678	127.116	0.5668	85.9925	63.6724	25.7466
CA66J	Tangkak, JOHOR	10/1/2021	15.3	7.837	0.0013	0.0017	0.0007	0.0024	0.023	0.708	196.675	0.9454	83.7475	32.198	25.516
CA66J	Tangkak, JOHOR	11/1/2021	11.102	5.874	0.0014	0.0014	0.0007	0.0021	0.0286	0.702	309.092	1.7062	78.1584	71.5345	26.1354

APPENDIX B: R CODE FOR MODEL DEVELOPMENT

#RCODE FOR BATU PAHAT

```
library(forecast)
library(ggplot2)
library(tseries)
library(psych)
library(FinTS)
library(lmtest)

analyze_station_ARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (ARMA) \n")
  cat("===== \n")

  # Convert DATE column
  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Check missing values
  cat("\nMissing Values: \n")
  print(colSums(is.na(df)))

  # Descriptive stats
  cat("\nDescriptive Statistics: \n")
  print(psych::describe(df$`PM2.5 AVG (µg/m3)`))

  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`)
  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")

  acf(ts_pm25, lag.max = 120, main = paste("Original ACF -",
station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("Original PACF -",
station_name))
  model_ARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = FALSE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test: \n")
  print(tseries::adf.test(ts_pm25))

  cat("\nARMA Model Summary: \n")
  print(summary(model_ARMA))
  cat("\nARMA Coefficient Significance: \n")
  print(coeftest(model_ARMA))

  forecast_ARMA <- forecast::forecast(model_ARMA, h = 14)
  print(autoplot(forecast_ARMA) + ggtitle(paste("ARMA Forecast -",
station_name)))

  cat("\nARMA Accuracy: \n")
  print(accuracy(model_ARMA))
  cat("AIC:", AIC(model_ARMA), "| BIC:", BIC(model_ARMA), "\n")

  checkresiduals(model_ARMA, lag = 12, test = "LB")
}
```

```

    acf(model_ARMA$residuals, lag.max = 40, main = paste("ACF of
ARMA Residuals -", station_name))
    pacf(model_ARMA$residuals, lag.max = 40, main = paste("PACF of
ARMA Residuals -", station_name))
    qqnorm(model_ARMA$residuals); qqline(model_ARMA$residuals, col =
"red")

    forecast_df <- data.frame(
      Day = 1:14,
      Forecast = as.numeric(forecast_ARMA$mean),
      Lower_95 = forecast_ARMA$lower[, 2],
      Upper_95 = forecast_ARMA$upper[, 2]
    )
    write.csv(forecast_df, paste0("forecast ARMA ", station_name,
".csv"), row.names = FALSE)
  }

analyze_station_sARMA <- function(station_name, df) {
  cat("\n=====\n")
  cat(" Station:", station_name, " (SARMA)\n")
  cat("=====\n")

  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Set frequency = 7 for weekly seasonality in daily data
  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m³)`, frequency = 7)

  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")
  # ACF & PACF of Original Series
  acf(ts_pm25, lag.max = 120, main = paste("ACF -", station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("PACF -",
station_name))

  model_sARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = TRUE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test:\n")
  print(tseries::adf.test(ts_pm25))

  cat("\nSARMA Model Summary:\n")
  print(summary(model_sARMA))

  cat("\nSARMA Coefficient Significance:\n")
  print(coeftest(model_sARMA))

  forecast_sARMA <- forecast::forecast(model_sARMA, h = 14)
  print(autoplot(forecast_sARMA) + ggtitle(paste("SARMA Forecast -
", station_name)))

  cat("\nSARMA Accuracy:\n")
  print(accuracy(model_sARMA))
  cat("AIC:", AIC(model_sARMA), "| BIC:", BIC(model_sARMA), "\n")

  checkresiduals(model_sARMA, lag = 12, test = "LB")

```

```

    acf(model_sARMA$residuals, lag.max = 40, main = paste("ACF of
SARMA Residuals -", station_name))
    pacf(model_sARMA$residuals, lag.max = 40, main = paste("PACF of
SARMA Residuals -", station_name))
    qqnorm(model_sARMA$residuals); qqline(model_sARMA$residuals, col
= "red")

    forecast_df <- data.frame(
      Day = 1:14,
      Forecast = as.numeric(forecast_sARMA$mean),
      Lower_95 = forecast_sARMA$lower[, 2],
      Upper_95 = forecast_sARMA$upper[, 2]
    )
    write.csv(forecast_df, paste0("forecast SARMA ", station_name,
".csv"), row.names = FALSE)
  }

analyze_station_ARMA("Batu Pahat", BATU_PAHAT_JOHOR_2021)
analyze_station_sARMA("Batu Pahat", BATU_PAHAT_JOHOR_2021)

```

#RCODE FOR KLUANG

```

library(forecast)
library(ggplot2)
library(tseries)
library(psych)
library(FinTS)
library(lmtest)

analyze_station_ARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (ARMA) \n")
  cat("===== \n")

  # Convert DATE column
  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Check missing values
  cat("\nMissing Values: \n")
  print(colSums(is.na(df)))

  # Descriptive stats
  cat("\nDescriptive Statistics: \n")
  print(psych::describe(df$`PM2.5 AVG (µg/m3)`))

  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`)
  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")

  acf(ts_pm25, lag.max = 120, main = paste("Original ACF -",
station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("Original PACF -",
station_name))
  model_ARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = FALSE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")
}

```

```

# Stationarity Test (ADF)
cat("\nADF Test:\n")
print(tseries::adf.test(ts_pm25))

cat("\nARMA Model Summary:\n")
print(summary(model_ARMA))
cat("\nARMA Coefficient Significance:\n")
print(coeftest(model_ARMA))

forecast_ARMA <- forecast::forecast(model_ARMA, h = 14)
print(autoplot(forecast_ARMA) + ggtitle(paste("ARMA Forecast -",
station_name)))

cat("\nARMA Accuracy:\n")
print(accuracy(model_ARMA))
cat("AIC:", AIC(model_ARMA), "| BIC:", BIC(model_ARMA), "\n")

checkresiduals(model_ARMA, lag = 12, test = "LB")
acf(model_ARMA$residuals, lag.max = 40, main = paste("ACF of
ARMA Residuals -", station_name))
pacf(model_ARMA$residuals, lag.max = 40, main = paste("PACF of
ARMA Residuals -", station_name))
qqnorm(model_ARMA$residuals); qqline(model_ARMA$residuals, col =
"red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_ARMA$mean),
  Lower_95 = forecast_ARMA$lower[, 2],
  Upper_95 = forecast_ARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast ARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_sARMA <- function(station_name, df) {
  cat("\n=====\n")
  cat(" Station:", station_name, " (SARMA)\n")
  cat("=====\n")

  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Set frequency = 7 for weekly seasonality in daily data
  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m³)`, frequency = 7)

  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")
  # ACF & PACF of Original Series
  acf(ts_pm25, lag.max = 120, main = paste("ACF -", station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("PACF -",
station_name))

  model_sARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = TRUE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

```



```

# Stationarity Test (ADF)
cat("\nADF Test:\n")
print(tseries::adf.test(ts_pm25))

cat("\nSARMA Model Summary:\n")
print(summary(model_sARMA))

cat("\nSARMA Coefficient Significance:\n")
print(coeftest(model_sARMA))

forecast_sARMA <- forecast::forecast(model_sARMA, h = 14)
print(autoplot(forecast_sARMA) + ggtitle(paste("SARMA Forecast -
", station_name)))

cat("\nSARMA Accuracy:\n")
print(accuracy(model_sARMA))
cat("AIC:", AIC(model_sARMA), "| BIC:", BIC(model_sARMA), "\n")

checkresiduals(model_sARMA, lag = 12, test = "LB")

acf(model_sARMA$residuals, lag.max = 40, main = paste("ACF of
SARMA Residuals -", station_name))
pacf(model_sARMA$residuals, lag.max = 40, main = paste("PACF of
SARMA Residuals -", station_name))
qqnorm(model_sARMA$residuals); qqline(model_sARMA$residuals, col
= "red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_sARMA$mean),
  Lower_95 = forecast_sARMA$lower[, 2],
  Upper_95 = forecast_sARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast SARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_ARMA("Kluang", KLUANG_JOHOR_2021)
analyze_station_sARMA("Kluang", KLUANG_JOHOR_2021
)

```

#RCODE FOR KOTA TINGGI

```

library(forecast)
library(ggplot2)
library(tseries)
library(psych)
library(FinTS)
library(lmtest)

analyze_station_ARMA <- function(station_name, df) {
  cat("\n=====\n")
  cat(" Station:", station_name, " (ARMA)\n")
  cat("=====\n")

  # Convert DATE column
  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Check missing values
  cat("\nMissing Values:\n")
  print(colSums(is.na(df)))
}

```

```

# Descriptive stats
cat("\nDescriptive Statistics:\n")
print(psych::describe(df$`PM2.5 AVG (µg/m3)`))

ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`)
plot(ts_pm25, main = paste("Time Series Plot -", station_name),
     ylab = "PM2.5 (µg/m³)", col = "steelblue")

acf(ts_pm25, lag.max = 120, main = paste("Original ACF -",
station_name))
pacf(ts_pm25, lag.max = 120, main = paste("Original PACF -",
station_name))
model_ARMA <- forecast::auto.arima(ts_pm25,
                                   seasonal = FALSE,
                                   stepwise = TRUE,
                                   trace = TRUE,
                                   approximation = FALSE,
                                   test = "adf")

# Stationarity Test (ADF)
cat("\nADF Test:\n")
print(tseries::adf.test(ts_pm25))

cat("\nARMA Model Summary:\n")
print(summary(model_ARMA))
cat("\nARMA Coefficient Significance:\n")
print(coeftest(model_ARMA))

forecast_ARMA <- forecast::forecast(model_ARMA, h = 14)
print(autoplot(forecast_ARMA) + ggtitle(paste("ARMA Forecast -",
station_name)))

cat("\nARMA Accuracy:\n")
print(accuracy(model_ARMA))
cat("AIC:", AIC(model_ARMA), "| BIC:", BIC(model_ARMA), "\n")

checkresiduals(model_ARMA, lag = 12, test = "LB")
acf(model_ARMA$residuals, lag.max = 40, main = paste("ACF of
ARMA Residuals -", station_name))
pacf(model_ARMA$residuals, lag.max = 40, main = paste("PACF of
ARMA Residuals -", station_name))
qqnorm(model_ARMA$residuals); qqline(model_ARMA$residuals, col =
"red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_ARMA$mean),
  Lower_95 = forecast_ARMA$lower[, 2],
  Upper_95 = forecast_ARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast ARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_sARMA <- function(station_name, df) {
  cat("\n=====\n")
  cat(" Station:", station_name, " (SARMA)\n")
  cat("=====\n")

  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

```

```

# Set frequency = 7 for weekly seasonality in daily data
ts_pm25 <- ts(df$`PM2.5 AVG (µg/m³)`, frequency = 7)

plot(ts_pm25, main = paste("Time Series Plot -", station_name),
     ylab = "PM2.5 (µg/m³)", col = "steelblue")
# ACF & PACF of Original Series
acf(ts_pm25, lag.max = 120, main = paste("ACF -", station_name))
pacf(ts_pm25, lag.max = 120, main = paste("PACF -",
station_name))

model_sARMA <- forecast::auto.arima(ts_pm25,
                                   seasonal = TRUE,
                                   stepwise = TRUE,
                                   trace = TRUE,
                                   approximation = FALSE,
                                   test = "adf")

# Stationarity Test (ADF)
cat("\nADF Test:\n")
print(tseries::adf.test(ts_pm25))

cat("\nSARMA Model Summary:\n")
print(summary(model_sARMA))

cat("\nSARMA Coefficient Significance:\n")
print(coeftest(model_sARMA))

forecast_sARMA <- forecast::forecast(model_sARMA, h = 14)
print(autoplot(forecast_sARMA) + ggtitle(paste("SARMA Forecast -",
station_name)))

cat("\nSARMA Accuracy:\n")
print(accuracy(model_sARMA))
cat("AIC:", AIC(model_sARMA), "| BIC:", BIC(model_sARMA), "\n")

checkresiduals(model_sARMA, lag = 12, test = "LB")

acf(model_sARMA$residuals, lag.max = 40, main = paste("ACF of
SARMA Residuals -", station_name))
pacf(model_sARMA$residuals, lag.max = 40, main = paste("PACF of
SARMA Residuals -", station_name))
qqnorm(model_sARMA$residuals); qqline(model_sARMA$residuals, col
= "red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_sARMA$mean),
  Lower_95 = forecast_sARMA$lower[, 2],
  Upper_95 = forecast_sARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast SARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_ARMA("Kota Tinggi", KOTA_TINGGI_JOHOR_2021)
analyze_station_sARMA("Kota Tinggi", KOTA_TINGGI_JOHOR_2021
)

```

#RCODE FOR LARKIN

```

library(forecast)
library(ggplot2)
library(tseries)
library(psych)
library(FinTS)
library(lmtest)

analyze_station_ARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (ARMA) \n")
  cat("===== \n")

  # Convert DATE column
  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Check missing values
  cat("\nMissing Values: \n")
  print(colSums(is.na(df)))

  # Descriptive stats
  cat("\nDescriptive Statistics: \n")
  print(psych::describe(df$`PM2.5 AVG (µg/m3)`)

  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`)
  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")

  acf(ts_pm25, lag.max = 120, main = paste("Original ACF -",
station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("Original PACF -",
station_name))
  model_ARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = FALSE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test: \n")
  print(tseries::adf.test(ts_pm25))

  cat("\nARMA Model Summary: \n")
  print(summary(model_ARMA))
  cat("\nARMA Coefficient Significance: \n")
  print(coeftest(model_ARMA))

  forecast_ARMA <- forecast::forecast(model_ARMA, h = 14)
  print(autoplot(forecast_ARMA) + ggtitle(paste("ARMA Forecast -",
station_name)))

  cat("\nARMA Accuracy: \n")
  print(accuracy(model_ARMA))
  cat("AIC:", AIC(model_ARMA), "| BIC:", BIC(model_ARMA), "\n")

  checkresiduals(model_ARMA, lag = 12, test = "LB")
  acf(model_ARMA$residuals, lag.max = 40, main = paste("ACF of
ARMA Residuals -", station_name))
  pacf(model_ARMA$residuals, lag.max = 40, main = paste("PACF of
ARMA Residuals -", station_name))

```

```

    qqnorm(model_ARMA$residuals); qqline(model_ARMA$residuals, col =
"red")

    forecast_df <- data.frame(
      Day = 1:14,
      Forecast = as.numeric(forecast_ARMA$mean),
      Lower_95 = forecast_ARMA$lower[, 2],
      Upper_95 = forecast_ARMA$upper[, 2]
    )
    write.csv(forecast_df, paste0("forecast ARMA ", station_name,
".csv"), row.names = FALSE)
  }

analyze_station_sARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (SARMA) \n")
  cat("===== \n")

  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Set frequency = 7 for weekly seasonality in daily data
  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m³)`, frequency = 7)

  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")

  # ACF & PACF of Original Series
  acf(ts_pm25, lag.max = 120, main = paste("ACF -", station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("PACF -",
station_name))

  model_sARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = TRUE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test: \n")
  print(tseries::adf.test(ts_pm25))

  cat("\nSARMA Model Summary: \n")
  print(summary(model_sARMA))

  cat("\nSARMA Coefficient Significance: \n")
  print(coeftest(model_sARMA))

  forecast_sARMA <- forecast::forecast(model_sARMA, h = 14)
  print(autoplot(forecast_sARMA) + ggtitle(paste("SARMA Forecast -",
station_name)))

  cat("\nSARMA Accuracy: \n")
  print(accuracy(model_sARMA))
  cat("AIC:", AIC(model_sARMA), "| BIC:", BIC(model_sARMA), "\n")

  checkresiduals(model_sARMA, lag = 12, test = "LB")

  acf(model_sARMA$residuals, lag.max = 40, main = paste("ACF of
SARMA Residuals -", station_name))

```

```

pacf(model_sARMA$residuals, lag.max = 40, main = paste("PACF of
SARMA Residuals -", station_name))
qqnorm(model_sARMA$residuals); qqline(model_sARMA$residuals, col
= "red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_sARMA$mean),
  Lower_95 = forecast_sARMA$lower[, 2],
  Upper_95 = forecast_sARMA$upper[, 2]
)

write.csv(forecast_df, paste0("forecast SARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_ARMA("Larkin", LARKIN_JOHOR_2021)
analyze_station_sARMA("Larkin", LARKIN_JOHOR_2021)

```

#RCODE FOR PASIR GUDANG

```

library(forecast)
library(ggplot2)
library(tseries)
library(psych)
library(FinTS)
library(lmtest)

analyze_station_ARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (ARMA) \n")
  cat("===== \n")

  # Convert DATE column
  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Check missing values
  cat("\nMissing Values: \n")
  print(colSums(is.na(df)))

  # Descriptive stats
  cat("\nDescriptive Statistics: \n")
  print(psych::describe(df$`PM2.5 AVG (µg/m3)`))

  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`)
  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")

  acf(ts_pm25, lag.max = 120, main = paste("Original ACF -",
station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("Original PACF -",
station_name))
  model_ARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = FALSE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test: \n")
  print(tseries::adf.test(ts_pm25))
}

```

```

cat("\nARMA Model Summary:\n")
print(summary(model_ARMA))
cat("\nARMA Coefficient Significance:\n")
print(coefest(model_ARMA))

forecast_ARMA <- forecast::forecast(model_ARMA, h = 14)
print(autoplot(forecast_ARMA) + ggtitle(paste("ARMA Forecast -",
station_name)))

cat("\nARMA Accuracy:\n")
print(accuracy(model_ARMA))
cat("AIC:", AIC(model_ARMA), "| BIC:", BIC(model_ARMA), "\n")

checkresiduals(model_ARMA, lag = 12, test = "LB")
acf(model_ARMA$residuals, lag.max = 40, main = paste("ACF of
ARMA Residuals -", station_name))
pacf(model_ARMA$residuals, lag.max = 40, main = paste("PACF of
ARMA Residuals -", station_name))
qqnorm(model_ARMA$residuals); qqline(model_ARMA$residuals, col =
"red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_ARMA$mean),
  Lower_95 = forecast_ARMA$lower[, 2],
  Upper_95 = forecast_ARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast ARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_sARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (SARMA)\n")
  cat("===== \n")

  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Set frequency = 7 for weekly seasonality in daily data
  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m³)`, frequency = 7)

  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")
  # ACF & PACF of Original Series
  acf(ts_pm25, lag.max = 120, main = paste("ACF -", station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("PACF -",
station_name))

  model_sARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = TRUE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test:\n")
  print(tseries::adf.test(ts_pm25))

```

```

cat("\nSARMA Model Summary:\n")
print(summary(model_sARMA))

cat("\nSARMA Coefficient Significance:\n")
print(coeftest(model_sARMA))

forecast_sARMA <- forecast::forecast(model_sARMA, h = 14)
print(autoplot(forecast_sARMA) + ggtitle(paste("SARMA Forecast -",
station_name)))

cat("\nSARMA Accuracy:\n")
print(accuracy(model_sARMA))
cat("AIC:", AIC(model_sARMA), "| BIC:", BIC(model_sARMA), "\n")

checkresiduals(model_sARMA, lag = 12, test = "LB")

acf(model_sARMA$residuals, lag.max = 40, main = paste("ACF of
SARMA Residuals -", station_name))
pacf(model_sARMA$residuals, lag.max = 40, main = paste("PACF of
SARMA Residuals -", station_name))
qqnorm(model_sARMA$residuals); qqline(model_sARMA$residuals, col
= "red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_sARMA$mean),
  Lower_95 = forecast_sARMA$lower[, 2],
  Upper_95 = forecast_sARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast SARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_ARMA("Pasir Gudang",PASIR_GUDANG_JOHOR_2021)
analyze_station_sARMA("Pasir Gudang",PASIR_GUDANG_JOHOR_2021
)

```

#RCODE FOR PENDERANG

```

library(forecast)
library(ggplot2)
library(tseries)
library(psych)
library(FinTS)
library(lmtest)

analyze_station_ARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (ARMA) \n")
  cat("===== \n")

  # Convert DATE column
  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Check missing values
  cat("\nMissing Values: \n")
  print(colSums(is.na(df)))

  # Descriptive stats
  cat("\nDescriptive Statistics: \n")
  print(psych::describe(df$`PM2.5 AVG (µg/m3)`))
}

```



```

ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)` )
plot(ts_pm25, main = paste("Time Series Plot -", station_name),
     ylab = "PM2.5 (µg/m³)", col = "steelblue")

acf(ts_pm25, lag.max = 120, main = paste("Original ACF -",
station_name))
pacf(ts_pm25, lag.max = 120, main = paste("Original PACF -",
station_name))
model_ARMA <- forecast::auto.arima(ts_pm25,
                                   seasonal = FALSE,
                                   stepwise = TRUE,
                                   trace = TRUE,
                                   approximation = FALSE,
                                   test = "adf")

# Stationarity Test (ADF)
cat("\nADF Test:\n")
print(tseries::adf.test(ts_pm25))

cat("\nARMA Model Summary:\n")
print(summary(model_ARMA))
cat("\nARMA Coefficient Significance:\n")
print(coeftest(model_ARMA))

forecast_ARMA <- forecast::forecast(model_ARMA, h = 14)
print(autoplot(forecast_ARMA) + ggtitle(paste("ARMA Forecast -",
station_name)))

cat("\nARMA Accuracy:\n")
print(accuracy(model_ARMA))
cat("AIC:", AIC(model_ARMA), "| BIC:", BIC(model_ARMA), "\n")

checkresiduals(model_ARMA, lag = 12, test = "LB")
acf(model_ARMA$residuals, lag.max = 40, main = paste("ACF of
ARMA Residuals -", station_name))
pacf(model_ARMA$residuals, lag.max = 40, main = paste("PACF of
ARMA Residuals -", station_name))
qqnorm(model_ARMA$residuals); qqline(model_ARMA$residuals, col =
"red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_ARMA$mean),
  Lower_95 = forecast_ARMA$lower[, 2],
  Upper_95 = forecast_ARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast ARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_sARMA <- function(station_name, df) {
  cat("\n=====\n")
  cat(" Station:", station_name, " (SARMA)\n")
  cat("=====\n")

  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Set frequency = 7 for weekly seasonality in daily data
  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`, frequency = 7)

```

```

plot(ts_pm25, main = paste("Time Series Plot -", station_name),
     ylab = "PM2.5 ( $\mu\text{g}/\text{m}^3$ )", col = "steelblue")
# ACF & PACF of Original Series
acf(ts_pm25, lag.max = 120, main = paste("ACF -", station_name))
pacf(ts_pm25, lag.max = 120, main = paste("PACF -",
station_name))

model_sARMA <- forecast::auto.arima(ts_pm25,
                                   seasonal = TRUE,
                                   stepwise = TRUE,
                                   trace = TRUE,
                                   approximation = FALSE,
                                   test = "adf")

# Stationarity Test (ADF)
cat("\nADF Test:\n")
print(tseries::adf.test(ts_pm25))

cat("\nSARMA Model Summary:\n")
print(summary(model_sARMA))

cat("\nSARMA Coefficient Significance:\n")
print(coeftest(model_sARMA))

forecast_sARMA <- forecast::forecast(model_sARMA, h = 14)
print(autoplot(forecast_sARMA) + ggtitle(paste("SARMA Forecast -
", station_name)))

cat("\nSARMA Accuracy:\n")
print(accuracy(model_sARMA))
cat("AIC:", AIC(model_sARMA), "| BIC:", BIC(model_sARMA), "\n")

checkresiduals(model_sARMA, lag = 12, test = "LB")

acf(model_sARMA$residuals, lag.max = 40, main = paste("ACF of
SARMA Residuals -", station_name))
pacf(model_sARMA$residuals, lag.max = 40, main = paste("PACF of
SARMA Residuals -", station_name))
qqnorm(model_sARMA$residuals); qqline(model_sARMA$residuals, col
= "red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_sARMA$mean),
  Lower_95 = forecast_sARMA$lower[, 2],
  Upper_95 = forecast_sARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast SARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_ARMA("Pengerang", Pengerang_Johor_2021)
analyze_station_sARMA("Pengerang", Pengerang_Johor_2021)

```

#RCODE FOR SEGAMAT

```

library(forecast)
library(ggplot2)
library(tseries)
library(psych)

```

```

library(FinTS)
library(lmtest)

analyze_station_ARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (ARMA) \n")
  cat("===== \n")

  # Convert DATE column
  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Check missing values
  cat("\nMissing Values: \n")
  print(colSums(is.na(df)))

  # Descriptive stats
  cat("\nDescriptive Statistics: \n")
  print(psych::describe(df$`PM2.5 AVG (µg/m3)`))

  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`)
  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")

  acf(ts_pm25, lag.max = 120, main = paste("Original ACF -",
station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("Original PACF -",
station_name))
  model_ARMA <- forecast::auto.arima(ts_pm25,
                                   seasonal = FALSE,
                                   stepwise = TRUE,
                                   trace = TRUE,
                                   approximation = FALSE,
                                   test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test: \n")
  print(tseries::adf.test(ts_pm25))

  cat("\nARMA Model Summary: \n")
  print(summary(model_ARMA))
  cat("\nARMA Coefficient Significance: \n")
  print(coeftest(model_ARMA))

  forecast_ARMA <- forecast::forecast(model_ARMA, h = 14)
  print(autoplot(forecast_ARMA) + ggtitle(paste("ARMA Forecast -",
station_name)))

  cat("\nARMA Accuracy: \n")
  print(accuracy(model_ARMA))
  cat("AIC:", AIC(model_ARMA), "| BIC:", BIC(model_ARMA), "\n")

  checkresiduals(model_ARMA, lag = 12, test = "LB")
  acf(model_ARMA$residuals, lag.max = 40, main = paste("ACF of
ARMA Residuals -", station_name))
  pacf(model_ARMA$residuals, lag.max = 40, main = paste("PACF of
ARMA Residuals -", station_name))
  qqnorm(model_ARMA$residuals); qqline(model_ARMA$residuals, col =
"red")

  forecast_df <- data.frame(

```

```

    Day = 1:14,
    Forecast = as.numeric(forecast_ARMA$mean),
    Lower_95 = forecast_ARMA$lower[, 2],
    Upper_95 = forecast_ARMA$upper[, 2]
  )
  write.csv(forecast_df, paste0("forecast ARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_sARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, "(SARMA) \n")
  cat("===== \n")

  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Set frequency = 7 for weekly seasonality in daily data
  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`, frequency = 7)

  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")
  # ACF & PACF of Original Series

  acf(ts_pm25, lag.max = 120, main = paste("ACF -", station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("PACF -",
station_name))

  model_sARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = TRUE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test: \n")
  print(tseries::adf.test(ts_pm25))

  cat("\nSARMA Model Summary: \n")
  print(summary(model_sARMA))

  cat("\nSARMA Coefficient Significance: \n")
  print(coeftest(model_sARMA))

  forecast_sARMA <- forecast::forecast(model_sARMA, h = 14)
  print(autoplot(forecast_sARMA) + ggtitle(paste("SARMA Forecast -",
station_name)))

  cat("\nSARMA Accuracy: \n")
  print(accuracy(model_sARMA))
  cat("AIC:", AIC(model_sARMA), "| BIC:", BIC(model_sARMA), "\n")

  checkresiduals(model_sARMA, lag = 12, test = "LB")

  acf(model_sARMA$residuals, lag.max = 40, main = paste("ACF of
SARMA Residuals -", station_name))
  pacf(model_sARMA$residuals, lag.max = 40, main = paste("PACF of
SARMA Residuals -", station_name))

```

```
qqnorm(model_sARMA$residuals); qqline(model_sARMA$residuals, col =
"red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_sARMA$mean),
  Lower_95 = forecast_sARMA$lower[, 2],
  Upper_95 = forecast_sARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast SARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_ARMA("SEGAMAT", SEGAMAT_JOHOR_2021)
analyze_station_sARMA("SEGAMAT", SEGAMAT_JOHOR_2021)
```

#RCODE FOR TANGKAK

```
library(forecast)
library(ggplot2)
library(tseries)
library(psych)
library(FinTS)
library(lmtest)

analyze_station_ARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, " (ARMA) \n")
  cat("===== \n")

  # Convert DATE column
  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Check missing values
  cat("\nMissing Values: \n")
  print(colSums(is.na(df)))

  # Descriptive stats
  cat("\nDescriptive Statistics: \n")
  print(psych::describe(df$`PM2.5 AVG (µg/m3)`))

  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`)
  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")

  acf(ts_pm25, lag.max = 120, main = paste("Original ACF -",
station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("Original PACF -",
station_name))
  model_ARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = FALSE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test: \n")
  print(tseries::adf.test(ts_pm25))

  cat("\nARMA Model Summary: \n")
```

```

print(summary(model_ARMA))
cat("\nARMA Coefficient Significance:\n")
print(coefest(model_ARMA))

forecast_ARMA <- forecast::forecast(model_ARMA, h = 14)
print(autoplot(forecast_ARMA) + ggtitle(paste("ARMA Forecast -",
station_name)))

cat("\nARMA Accuracy:\n")
print(accuracy(model_ARMA))
cat("AIC:", AIC(model_ARMA), "| BIC:", BIC(model_ARMA), "\n")

checkresiduals(model_ARMA, lag = 12, test = "LB")
acf(model_ARMA$residuals, lag.max = 40, main = paste("ACF of
ARMA Residuals -", station_name))
pacf(model_ARMA$residuals, lag.max = 40, main = paste("PACF of
ARMA Residuals -", station_name))
qqnorm(model_ARMA$residuals); qqline(model_ARMA$residuals, col =
"red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_ARMA$mean),
  Lower_95 = forecast_ARMA$lower[, 2],
  Upper_95 = forecast_ARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast ARMA ", station_name,
".csv"), row.names = FALSE)
}

analyze_station_sARMA <- function(station_name, df) {
  cat("\n===== \n")
  cat(" Station:", station_name, "(SARMA)\n")
  cat("===== \n")

  df$`DATE TIME` <- as.Date(df$`DATE TIME`)

  # Set frequency = 7 for weekly seasonality in daily data
  ts_pm25 <- ts(df$`PM2.5 AVG (µg/m3)`, frequency = 7)

  plot(ts_pm25, main = paste("Time Series Plot -", station_name),
       ylab = "PM2.5 (µg/m³)", col = "steelblue")
  # ACF & PACF of Original Series
  acf(ts_pm25, lag.max = 120, main = paste("ACF -", station_name))
  pacf(ts_pm25, lag.max = 120, main = paste("PACF -",
station_name))

  model_sARMA <- forecast::auto.arima(ts_pm25,
                                     seasonal = TRUE,
                                     stepwise = TRUE,
                                     trace = TRUE,
                                     approximation = FALSE,
                                     test = "adf")

  # Stationarity Test (ADF)
  cat("\nADF Test:\n")
  print(tseries::adf.test(ts_pm25))

  cat("\nSARMA Model Summary:\n")
  print(summary(model_sARMA))

```

```

cat("\nSARMA Coefficient Significance:\n")
print(coeftest(model_sARMA))

forecast_sARMA <- forecast::forecast(model_sARMA, h = 14)
print(autoplot(forecast_sARMA) + ggtitle(paste("SARMA Forecast -", station_name)))

cat("\nSARMA Accuracy:\n")
print(accuracy(model_sARMA))
cat("AIC:", AIC(model_sARMA), "| BIC:", BIC(model_sARMA), "\n")

checkresiduals(model_sARMA, lag = 12, test = "LB")

acf(model_sARMA$residuals, lag.max = 40, main = paste("ACF of SARMA Residuals -", station_name))
pacf(model_sARMA$residuals, lag.max = 40, main = paste("PACF of SARMA Residuals -", station_name))
qqnorm(model_sARMA$residuals); qqline(model_sARMA$residuals, col = "red")

forecast_df <- data.frame(
  Day = 1:14,
  Forecast = as.numeric(forecast_sARMA$mean),
  Lower_95 = forecast_sARMA$lower[, 2],
  Upper_95 = forecast_sARMA$upper[, 2]
)
write.csv(forecast_df, paste0("forecast SARMA ", station_name, ".csv"), row.names = FALSE)
}

analyze_station_ARMA("Tangkak", TANGKAK_JOHOR_2021)
analyze_station_sARMA("Tangkak", TANGKAK_JOHOR_2021)

```

APPENDIX C: FULL ARMA AND SARMA OUTPUT

```

=====
Station: Batu Pahat (ARMA)
=====

Missing Values:

          STATION ID          LOCATION
          0              0
          DATE TIME      PM10 AVG (µg/m3)
          0              0
          PM2.5 AVG (µg/m3)      SO2 AVG (ppm)
          0              0
          NO2 AVG (ppm)          NO AVG (ppm)
          0              0
          NOX AVG (ppm)          O3 AVG (ppm)
          0              0
          CO AVG (ppm)      WIND DIRECTION AVG (°)
          0              0
          WIND SPEED AVG (m/s)  RELATIVE HUMIDITY AVG (%)
          0              0
          SOLAR RADIATION AVG (W/m2)  Ambient Temperature AVG (°C)
          0              0
          Station
          0

Descriptive Statistics:
      vars   n  mean   sd median trimmed  mad  min   max range
skew
X1      1 365 12.69 5.85   11.4   12.08 4.66 3.01 40.35 37.34
1.47
      kurtosis   se
X1           3.54 0.31

ARMA(2,0,2) with non-zero mean : 2045.35
ARMA(0,0,0) with non-zero mean : 2328.781
ARMA(1,0,0) with non-zero mean : 2048.663
ARMA(0,0,1) with non-zero mean : 2140.936
ARMA(0,0,0) with zero mean      : 2962.798
ARMA(1,0,2) with non-zero mean : 2043.328
ARMA(0,0,2) with non-zero mean : 2099.379
ARMA(1,0,1) with non-zero mean : 2050.014
ARMA(1,0,3) with non-zero mean : 2045.367
ARMA(0,0,3) with non-zero mean : 2081.147
ARMA(2,0,1) with non-zero mean : 2051.923
ARMA(2,0,3) with non-zero mean : 2047.102
ARMA(1,0,2) with zero mean      : Inf

Best model: ARMA(1,0,2) with non-zero mean

ADF Test: Augmented Dickey-Fuller Test

data:  ts_pm25

```



```

Dickey-Fuller = -4.4233, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary

ARMA Model Summary:
Series: ts_pm25
ARMA(1,0,2) with non-zero mean

Coefficients:
          ar1          ma1          ma2          mean
          0.8844    -0.1843    -0.2104    12.4392
s.e.    0.0438     0.0725     0.0687     1.0589

sigma^2 = 15.5:  log likelihood = -1016.58
AIC=2043.16  AICc=2043.33  BIC=2062.66

Training set error measures:
              ME          RMSE          MAE          MPE          MAPE
MASE
Training set 0.03822098 3.915769 2.957619 -9.372716 26.33912
0.9311564
              ACF1
Training set 0.0008145776

ARMA Coefficient Significance:

z test of coefficients:

          Estimate Std. Error z value  Pr(>|z|)
ar1         0.884414   0.043787 20.1983 < 2.2e-16 ***
ma1        -0.184329   0.072512 -2.5420  0.011021 *
ma2        -0.210373   0.068689 -3.0627  0.002194 **
intercept 12.439194   1.058857 11.7478 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ARMA Accuracy:
              ME          RMSE          MAE          MPE          MAPE
MASE
Training set 0.03822098 3.915769 2.957619 -9.372716 26.33912
0.9311564
              ACF1
Training set 0.0008145776
AIC: 2043.161 | BIC: 2062.66

Ljung-Box test

data:  Residuals from ARMA(1,0,2) with non-zero mean
Q* = 4.9098, df = 9, p-value = 0.8421

Model df: 3.    Total lags used: 12

=====

```

```

Station: Batu Pahat (SARMA)
=====

ARMA(2,0,2) (1,0,1) [7] with non-zero mean : 2048.528
ARMA(0,0,0) with non-zero mean : 2328.781
ARMA(1,0,0) (1,0,0) [7] with non-zero mean : 2047.5
ARMA(0,0,1) (0,0,1) [7] with non-zero mean : 2130.629
ARMA(0,0,0) with zero mean : 2962.798
ARMA(1,0,0) with non-zero mean : 2048.663
ARMA(1,0,0) (2,0,0) [7] with non-zero mean : 2048.324
ARMA(1,0,0) (1,0,1) [7] with non-zero mean : 2048.98
ARMA(1,0,0) (0,0,1) [7] with non-zero mean : 2047.878
ARMA(1,0,0) (2,0,1) [7] with non-zero mean : 2050.277
ARMA(0,0,0) (1,0,0) [7] with non-zero mean : 2294.033
ARMA(2,0,0) (1,0,0) [7] with non-zero mean : 2049.346
ARMA(1,0,1) (1,0,0) [7] with non-zero mean : 2049.229
ARMA(0,0,1) (1,0,0) [7] with non-zero mean : 2127.923
ARMA(2,0,1) (1,0,0) [7] with non-zero mean : 2051.024
ARMA(1,0,0) (1,0,0) [7] with zero mean : 2089.917

Best model: ARMA(1,0,0) (1,0,0) [7] with non-zero mean

ADF Test:Augmented Dickey-Fuller Test
data: ts_pm25
Dickey-Fuller = -4.4233, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary

SARMA Model Summary:
Series: ts_pm25
ARMA(1,0,0) (1,0,0) [7] with non-zero mean

Coefficients:
          ar1      sar1      mean
      0.7224  0.0957  12.5344
s.e.  0.0371  0.0534   0.8166

sigma^2 = 15.73: log likelihood = -1019.69
AIC=2047.39 AICc=2047.5 BIC=2062.99

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE
MASE
Training set 0.02802052 3.949619 2.986938 -9.767566 26.79508
0.5945725
              ACF1
Training set -0.0177696

SARMA Coefficient Significance:

z test of coefficients:

          Estimate Std. Error z value Pr(>|z|)
ar1      0.722395   0.037145 19.4482 < 2e-16 ***
sar1      0.095696   0.053371  1.7931  0.00296 .

```

```

intercept 12.534360    0.816557 15.3503 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

SARMA Accuracy:
                                ME      RMSE      MAE      MPE      MAPE
MASE
Training set 0.02802052 3.949619 2.986938 -9.767566 26.79508
0.5945725
                                ACF1
Training set -0.0177696
AIC: 2047.389 | BIC: 2062.989

Ljung-Box test

data:  Residuals from ARMA(1,0,0)(1,0,0)[7] with non-zero
mean
Q* = 9.6945, df = 10, p-value = 0.4677

Model df: 2.    Total lags used: 12
=====
Station: Kluang (ARMA)
=====

Missing Values:

      STATION ID      LOCATION
      0            0
      DATE TIME      PM10 AVG (µg/m3)
      0            0
      PM2.5 AVG (µg/m3)      SO2 AVG (ppm)
      0            365
      NO2 AVG (ppm)      NO AVG (ppm)
      365            365
      NOX AVG (ppm)      O3 AVG (ppm)
      365            365
      CO AVG (ppm)      WIND DIRECTION AVG(°)
      365            0
      WIND SPEED AVG(m/s)      RELATIVE HUMIDITY AVG(%)
      0            0
      SOLAR RADIATION AVG(W/m2)      Ambient Temperature AVG(°c)
      2            0
      Station
      0

Descriptive Statistics:
vars  n mean  sd median trimmed  mad  min  max range
skew
X1    1 365 10.6 4.3   9.71   10.26 3.94 2.14 24.35 22.21
0.74
      kurtosis  se
X1    0.25 0.22

ARMA(2,0,2) with non-zero mean : 1834.74
ARMA(0,0,0) with non-zero mean : 2103.111

```

```

ARMA(1,0,0) with non-zero mean : 1835.069
ARMA(0,0,1) with non-zero mean : 1911.236
ARMA(0,0,0) with zero mean      : 2816.57
ARMA(1,0,2) with non-zero mean : 1832.984
ARMA(0,0,2) with non-zero mean : 1876.597
ARMA(1,0,1) with non-zero mean : 1837.02
ARMA(1,0,3) with non-zero mean : 1834.964
ARMA(0,0,3) with non-zero mean : 1855.141
ARMA(2,0,1) with non-zero mean : 1836.856
ARMA(2,0,3) with non-zero mean : 1836.551
ARMA(1,0,2) with zero mean      : Inf

Best model: ARMA(1,0,2) with non-zero mean

ADF Test:

      Augmented Dickey-Fuller Test

data:  ts_pm25
Dickey-Fuller = -4.408, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary

ARMA Model Summary:
Series: ts_pm25
ARMA(1,0,2) with non-zero mean

Coefficients:
      ar1      ma1      ma2      mean
      0.8364 -0.0965 -0.1954 10.5033
s.e.  0.0585  0.0842  0.0800  0.6588

sigma^2 = 8.714:  log likelihood = -911.41
AIC=1832.82  AICc=1832.98  BIC=1852.32

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE
MASE
Training set 0.00566888 2.935768 2.172389 -8.594562 23.45322
0.9268773
              ACF1
Training set 0.0007327751

ARMA Coefficient Significance:

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1      0.836438   0.058500 14.2981 < 2e-16 ***
ma1     -0.096472   0.084214 -1.1456  0.25198
ma2     -0.195377   0.079988 -2.4426  0.01458 *
intercept 10.503345   0.658808 15.9429 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
1

```

```

ARMA Accuracy:
                                ME      RMSE      MAE      MPE      MAPE
MASE
Training set 0.00566888 2.935768 2.172389 -8.594562 23.45322
0.9268773
                                ACF1
Training set 0.0007327751
AIC: 1832.817 | BIC: 1852.317

Ljung-Box test

data: Residuals from ARMA(1,0,2) with non-zero mean
Q* = 6.9319, df = 9, p-value = 0.6442

Model df: 3. Total lags used: 12

=====
Station: Kluang (SARMA)
=====

ARMA(2,0,2) (1,0,1) [7] with non-zero mean : 1837.362
ARMA(0,0,0) with non-zero mean : 2103.111
ARMA(1,0,0) (1,0,0) [7] with non-zero mean : 1836.046
ARMA(0,0,1) (0,0,1) [7] with non-zero mean : 1902.616
ARMA(0,0,0) with zero mean : 2816.57
ARMA(1,0,0) with non-zero mean : 1835.069
ARMA(1,0,0) (0,0,1) [7] with non-zero mean : 1835.973
ARMA(1,0,0) (1,0,1) [7] with non-zero mean : 1837.319
ARMA(2,0,0) with non-zero mean : 1837.052
ARMA(1,0,1) with non-zero mean : 1837.02
ARMA(0,0,1) with non-zero mean : 1911.236
ARMA(2,0,1) with non-zero mean : 1836.856
ARMA(1,0,0) with zero mean : 1881.082

Best model: ARMA(1,0,0) with non-zero mean

ADF Test:Augmented Dickey-Fuller Test
data: ts_pm25
Dickey-Fuller = -4.408, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary

SARMA Model Summary:
Series: ts_pm25
ARMA(1,0,0) with non-zero mean

Coefficients:
      ar1      mean
      0.7244 10.5498
s.e. 0.0361 0.5585

sigma^2 = 8.816: log likelihood = -914.5
AIC=1835 AICc=1835.07 BIC=1846.7

```

Training set error measures:

	ME	RMSE	MAE	MPE
MAPE				
MASE				
ACF1				
Training set	-0.0002027516	2.960996	2.195027	-8.819198
	23.69122	0.5346002	0.01142109	

SARMA Coefficient Significance:

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
arl	0.724444	0.036067	20.086	< 2.2e-16 ***
intercept	10.549811	0.558503	18.890	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

SARMA Accuracy:

	ME	RMSE	MAE	MPE
MAPE				
MASE				
ACF1				
Training set	-0.0002027516	2.960996	2.195027	-8.819198
	23.69122	0.5346002	0.01142109	
AIC: 1835.003 BIC: 1846.703				

Ljung-Box test

data: Residuals from ARMA(1,0,0) with non-zero mean
Q* = 11.421, df = 11, p-value = 0.4087

Model df: 1. Total lags used: 12

=====
Station: Kota Tinggi (ARMA)
=====

Missing Values:

STATION ID	LOCATION
0	0
DATE TIME	PM10 AVG (µg/m3)
0	0
PM2.5 AVG (µg/m3)	SO2 AVG (ppm)
0	0
NO2 AVG (ppm)	NO AVG (ppm)
0	0
NOX AVG (ppm)	O3 AVG (ppm)
0	0
CO AVG (ppm)	WIND DIRECTION AVG (°)
1	0
WIND SPEED AVG(m/s)	RELATIVE HUMIDITY AVG(%)
0	0
SOLAR RADIATION AVG(W/m2)	Ambient Temperature AVG(°c)
8	2
Station	
0	

```

Descriptive Statistics:
      vars   n  mean    sd median trimmed  mad   min   max range
X1      1 365 10.14  3.96   9.56    9.86  3.96  2.38 24.81 22.43
      skew kurtosis    se
X1 0.71      0.3 0.21

ARMA(2,0,2) with non-zero mean : 1846.338
ARMA(0,0,0) with non-zero mean : 2043.316
ARMA(1,0,0) with non-zero mean : 1843.112
ARMA(0,0,1) with non-zero mean : 1897.137
ARMA(0,0,0) with zero mean      : 2780.756
ARMA(2,0,0) with non-zero mean : 1844.7
ARMA(1,0,1) with non-zero mean : 1844.668
ARMA(2,0,1) with non-zero mean : 1846.045
ARMA(1,0,0) with zero mean      : 1902.496

Best model: ARMA(1,0,0) with non-zero mean

ADF Test:

      Augmented Dickey-Fuller Test

data:  ts_pm25
Dickey-Fuller = -5.2971, Lag order = 7, p-value =
0.01
alternative hypothesis: stationary

ARMA Model Summary:
Series: ts_pm25
ARMA(1,0,0) with non-zero mean

Coefficients:
      ar1      mean
      0.6544 10.0898
s.e.  0.0397  0.4513

sigma^2 = 9.017:  log likelihood = -918.52
AIC=1843.05  AICc=1843.11  BIC=1854.75

Training set error measures:
              ME      RMSE      MAE      MPE
Training set 0.01047544 2.994559 2.326345 -9.360785
              MAPE      MASE      ACF1
Training set 25.79962 0.9219076 0.02359763

ARMA Coefficient Significance:

z test of coefficients:

      Estimate Std. Error z value  Pr(>|z|)
ar1      0.654374   0.039683  16.490 < 2.2e-16 ***
intercept 10.089762   0.451266  22.359 < 2.2e-16 ***
---
Signif. codes:

```

0 '***' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1

ARMA Accuracy:

	ME	RMSE	MAE	MPE
Training set	0.01047544	2.994559	2.326345	-9.360785
	MAPE	MASE	ACF1	
Training set	25.79962	0.9219076	0.02359763	

AIC: 1843.046 | BIC: 1854.745

Ljung-Box test

data: Residuals from ARMA(1,0,0) with non-zero mean
Q* = 11.253, df = 11, p-value = 0.4223

Model df: 1. Total lags used: 12

=====
Station: Kota Tinggi (SARMA)
=====

ARMA(2,0,2) (1,0,1) [7]	with non-zero mean	: 1852.235
ARMA(0,0,0)	with non-zero mean	: 2043.316
ARMA(1,0,0) (1,0,0) [7]	with non-zero mean	: 1845.147
ARMA(0,0,1) (0,0,1) [7]	with non-zero mean	: 1897.311
ARMA(0,0,0)	with zero mean	: 2780.756
ARMA(1,0,0)	with non-zero mean	: 1843.112
ARMA(1,0,0) (0,0,1) [7]	with non-zero mean	: 1845.147
ARMA(1,0,0) (1,0,1) [7]	with non-zero mean	: Inf
ARMA(2,0,0)	with non-zero mean	: 1844.7
ARMA(1,0,1)	with non-zero mean	: 1844.668
ARMA(0,0,1)	with non-zero mean	: 1897.137
ARMA(2,0,1)	with non-zero mean	: 1846.045
ARMA(1,0,0)	with zero mean	: 1902.496

Best model: ARMA(1,0,0) with non-zero mean

ADF Test: Augmented Dickey-Fuller Test

data: ts_pm25

Dickey-Fuller = -5.2971, Lag order = 7, p-value = 0.01

alternative hypothesis: stationary

SARMA Model Summary:

Series: ts_pm25

ARMA(1,0,0) with non-zero mean

Coefficients:

	ar1	mean
	0.6544	10.0898
s.e.	0.0397	0.4513

sigma^2 = 9.017: log likelihood = -918.52

AIC=1843.05 AICc=1843.11 BIC=1854.75

Training set error measures:

	ME	RMSE	MAE	MPE
Training set	0.01047544	2.994559	2.326345	-9.360785

	MAPE	MASE	ACF1
Training set	25.79962	0.5579195	0.02359763

SARMA Coefficient Significance:

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
arl	0.654374	0.039683	16.490	< 2.2e-16 ***
intercept	10.089762	0.451266	22.359	< 2.2e-16 ***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

SARMA Accuracy:

	ME	RMSE	MAE	MPE
Training set	0.01047544	2.994559	2.326345	-9.360785

	MAPE	MASE	ACF1
Training set	25.79962	0.5579195	0.02359763

AIC: 1843.046 | BIC: 1854.745

Ljung-Box test

data: Residuals from ARMA(1,0,0) with non-zero mean
Q* = 11.253, df = 11, p-value = 0.4223

Model df: 1. Total lags used: 12

=====
Station: Larkin (ARMA)
=====

Missing Values:

STATION ID	LOCATION
0	0
DATE TIME	PM10 AVG (µg/m3)
0	0
PM2.5 AVG (µg/m3)	SO2 AVG (ppm)
0	0
NO2 AVG (ppm)	NO AVG (ppm)
0	0
NOX AVG (ppm)	O3 AVG (ppm)
0	0
CO AVG (ppm)	WIND DIRECTION AVG (°)
0	0
WIND SPEED AVG(m/s)	RELATIVE HUMIDITY AVG(%)
0	0
SOLAR RADIATION AVG(W/m2)	Ambient Temperature AVG(°c)
0	0
Station	
0	

```

Descriptive Statistics:
      vars   n  mean   sd median trimmed mad   min    max range
X1      1 365 16.57 5.47  15.82   16.24 5.3 5.63 35.88 30.25
      skew kurtosis   se
X1   0.6      0.18 0.29

ARMA(2,0,2) with non-zero mean : 2094.138
ARMA(0,0,0) with non-zero mean : 2279.636
ARMA(1,0,0) with non-zero mean : 2092.235
ARMA(0,0,1) with non-zero mean : 2147.239
ARMA(0,0,0) with zero mean      : 3125.19
ARMA(2,0,0) with non-zero mean : 2094.275
ARMA(1,0,1) with non-zero mean : 2094.276
ARMA(2,0,1) with non-zero mean : 2092.621
ARMA(1,0,0) with zero mean      : 2156.379

Best model: ARMA(1,0,0) with non-zero mean

ADF Test: Augmented Dickey-Fuller Test
data: ts_pm25
Dickey-Fuller = -4.9281, Lag order = 7, p-value =
0.01
alternative hypothesis: stationary

ARMA Model Summary:
Series: ts_pm25
ARMA(1,0,0) with non-zero mean

Coefficients:
      ar1      mean
      0.6401 16.4858
s.e.  0.0405  0.6099

sigma^2 = 17.84: log likelihood = -1043.08
AIC=2092.17 AICc=2092.23 BIC=2103.87

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE
Training set 0.0216095 4.212702 3.310456 -6.927653 22.13493
      MASE      ACF1
Training set 0.9093555 0.0002023648

ARMA Coefficient Significance:

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1      0.640079  0.040498  15.805 < 2.2e-16 ***
intercept 16.485846  0.609868  27.032 < 2.2e-16 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

ARMA Accuracy:
              ME      RMSE      MAE      MPE      MAPE
Training set 0.0216095 4.212702 3.310456 -6.927653 22.13493
              MASE      ACF1
Training set 0.9093555 0.0002023648
AIC: 2092.168 | BIC: 2103.868

      Ljung-Box test

data:  Residuals from ARMA(1,0,0) with non-zero mean
Q* = 13.683, df = 11, p-value = 0.251

Model df: 1.    Total lags used: 12

=====
Station: Larkin (SARMA)
=====

ARMA(2,0,2) (1,0,1) [7] with non-zero mean : 2102.349
ARMA(0,0,0)              with non-zero mean : 2279.636
ARMA(1,0,0) (1,0,0) [7] with non-zero mean : 2094.228
ARMA(0,0,1) (0,0,1) [7] with non-zero mean : 2147.775
ARMA(0,0,0)              with zero mean      : 3125.19
ARMA(1,0,0)              with non-zero mean : 2092.235
ARMA(1,0,0) (0,0,1) [7] with non-zero mean : 2094.229
ARMA(1,0,0) (1,0,1) [7] with non-zero mean : Inf
ARMA(2,0,0)              with non-zero mean : 2094.275
ARMA(1,0,1)              with non-zero mean : 2094.276
ARMA(0,0,1)              with non-zero mean : 2147.239
ARMA(2,0,1)              with non-zero mean : 2092.621
ARMA(1,0,0)              with zero mean      : 2156.379

Best model: ARMA(1,0,0)              with non-zero mean

ADF Test:Augmented Dickey-Fuller Test
data:  ts_pm25
Dickey-Fuller = -4.9281, Lag order = 7, p-value =
0.01
alternative hypothesis: stationary

SARMA Model Summary:
Series: ts_pm25
ARMA(1,0,0) with non-zero mean

Coefficients:
      ar1      mean
      0.6401 16.4858
s.e.  0.0405  0.6099

sigma^2 = 17.84:  log likelihood = -1043.08
AIC=2092.17  AICc=2092.23  BIC=2103.87

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE

```

```

Training set 0.0216095 4.212702 3.310456 -6.927653 22.13493
              MASE          ACF1
Training set 0.5921077 0.0002023648

```

SARMA Coefficient Significance:

z test of coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
ar1          0.640079    0.040498  15.805 < 2.2e-16 ***
intercept 16.485846    0.609868  27.032 < 2.2e-16 ***
---

```

Signif. codes:

```

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

SARMA Accuracy:

```

              ME      RMSE      MAE      MPE      MAPE
Training set 0.0216095 4.212702 3.310456 -6.927653 22.13493

```

```

              MASE          ACF1
Training set 0.5921077 0.0002023648

```

AIC: 2092.168 | BIC: 2103.868

Ljung-Box test

data: Residuals from ARMA(1,0,0) with non-zero mean
Q* = 13.683, df = 11, p-value = 0.251

Model df: 1. Total lags used: 12

```

=====
Station: PASIR GUDANG (ARMA)
=====

```

```

ARMA(2,0,2) with non-zero mean : 1997.774
ARMA(0,0,0) with non-zero mean : 2186.646
ARMA(1,0,0) with non-zero mean : 1996.656
ARMA(0,0,1) with non-zero mean : 2057.316
ARMA(0,0,0) with zero mean      : 2991.914
ARMA(2,0,0) with non-zero mean : 1996.079
ARMA(3,0,0) with non-zero mean : 1997.31
ARMA(2,0,1) with non-zero mean : 1993.213
ARMA(1,0,1) with non-zero mean : 1995.75
ARMA(3,0,1) with non-zero mean : 1997.605
ARMA(1,0,2) with non-zero mean : 1997.558
ARMA(3,0,2) with non-zero mean : Inf
ARMA(2,0,1) with zero mean     : Inf

```

Best model: ARMA(2,0,1) with non-zero mean

ARMA Model Summary:

Series: ts_pm25

ARMA(2,0,1) with non-zero mean

Coefficients:

```

              ar1      ar2      ma1      mean

```

```

      1.5101  -0.5206  -0.9242  12.9621
s.e.  0.0732   0.0650   0.0493   1.4833

sigma^2 = 13.52:  log likelihood = -991.52
AIC=1993.05   AICc=1993.21   BIC=2012.55

Training set error measures:
              ME      RMSE      MAE      MPE      MAPE
MASE
Training set 0.1058313 3.656298 2.881328 -6.523337 22.80744
0.9209968
              ACF1
Training set -0.01546432

ARMA Coefficient Significance:

z test of coefficients:

      Estimate Std. Error  z value  Pr(>|z|)
ar1      1.510075    0.073232  20.6204 < 2.2e-16 ***
ar2     -0.520590    0.065010  -8.0078 1.168e-15 ***
ma1     -0.924219    0.049342 -18.7310 < 2.2e-16 ***
intercept 12.962094    1.483314   8.7386 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ARMA Accuracy:
              ME      RMSE      MAE      MPE      MAPE
MASE
Training set 0.1058313 3.656298 2.881328 -6.523337 22.80744
0.9209968
              ACF1
Training set -0.01546432
AIC: 1993.046 | BIC: 2012.546

Ljung-Box test

data:  Residuals from ARMA(2,0,1) with non-zero mean
Q* = 8.7849, df = 9, p-value = 0.4574

Model df: 3.    Total lags used: 12

=====
Station: PASIR GUDANG (SARMA)
=====

ARMA(2,0,2) (1,0,1) [7] with non-zero mean : 2001.668
ARMA(0,0,0)              with non-zero mean : 2186.646
ARMA(1,0,0) (1,0,0) [7] with non-zero mean : 1998.111
ARMA(0,0,1) (0,0,1) [7] with non-zero mean : 2054.835
ARMA(0,0,0)              with zero mean      : 2991.914
ARMA(1,0,0)              with non-zero mean : 1996.656
ARMA(1,0,0) (0,0,1) [7] with non-zero mean : 1998.168
ARMA(1,0,0) (1,0,1) [7] with non-zero mean : 1999.555

```

```

ARMA(2,0,0) with non-zero mean : 1996.079
ARMA(2,0,0) (1,0,0) [7] with non-zero mean : 1997.927
ARMA(2,0,0) (0,0,1) [7] with non-zero mean : 1997.947
ARMA(2,0,0) (1,0,1) [7] with non-zero mean : 1999.513
ARMA(3,0,0) with non-zero mean : 1997.31
ARMA(2,0,1) with non-zero mean : 1993.213
ARMA(2,0,1) (1,0,0) [7] with non-zero mean : 1995.265
ARMA(2,0,1) (0,0,1) [7] with non-zero mean : 1995.267
ARMA(2,0,1) (1,0,1) [7] with non-zero mean : 2001.3
ARMA(1,0,1) with non-zero mean : 1995.75
ARMA(3,0,1) with non-zero mean : 1997.605
ARMA(2,0,2) with non-zero mean : 1997.774
ARMA(1,0,2) with non-zero mean : 1997.558
ARMA(3,0,2) with non-zero mean : Inf
ARMA(2,0,1) with zero mean : Inf

Best model: ARMA(2,0,1) with non-zero mean

SARMA Model Summary:
Series: ts_pm25
ARMA(2,0,1) with non-zero mean

Coefficients:
          ar1          ar2          ma1          mean
      1.5101   -0.5206   -0.9242   12.9621
s.e.   0.0732    0.0650    0.0493    1.4833

sigma^2 = 13.52: log likelihood = -991.52
AIC=1993.05 AICc=1993.21 BIC=2012.55

Training set error measures:
              ME          RMSE          MAE          MPE          MAPE
MASE
Training set 0.1058313 3.656298 2.881328 -6.523337 22.80744
0.5995925
              ACF1
Training set -0.01546432

SARMA Coefficient Significance:

z test of coefficients:

          Estimate Std. Error z value Pr(>|z|)
ar1         1.510075   0.073232  20.6204 < 2.2e-16 ***
ar2        -0.520590   0.065010  -8.0078 1.168e-15 ***
ma1        -0.924219   0.049342 -18.7310 < 2.2e-16 ***
intercept 12.962094   1.483314   8.7386 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

SARMA Accuracy:
              ME          RMSE          MAE          MPE          MAPE
MASE

```

Training set 0.1058313 3.656298 2.881328 -6.523337 22.80744
0.5995925

ACF1

Training set -0.01546432

AIC: 1993.046 | BIC: 2012.546

Ljung-Box test

data: Residuals from ARMA(2,0,1) with non-zero mean
Q* = 8.7849, df = 9, p-value = 0.4574

Model df: 3. Total lags used: 12

=====
Station: Pengerang (ARMA)
=====

Missing Values:

STATION ID	LOCATION
0	0
DATE TIME	PM10 AVG (µg/m3)
0	0
PM2.5 AVG (µg/m3)	SO2 AVG (ppm)
0	0
NO2 AVG (ppm)	NO AVG (ppm)
0	0
NOX AVG (ppm)	O3 AVG (ppm)
0	365
CO AVG (ppm)	WIND DIRECTION AVG (°)
365	0
WIND SPEED AVG (m/s)	RELATIVE HUMIDITY AVG (%)
0	0
SOLAR RADIATION AVG (W/m2)	Ambient Temperature AVG (°C)
0	0
Station	
0	

Descriptive Statistics:

vars	n	mean	sd	median	trimmed	mad	min	max	range	
X1	1	365	11.48	5.17	10.64	10.87	3.64	2.12	39.52	37.4
skew	kurtosis	se								
X1	2.13	7.33	0.27							

ARMA(2,0,2) with non-zero mean : 1986.784
ARMA(0,0,0) with non-zero mean : 2237.635
ARMA(1,0,0) with non-zero mean : 1996.728
ARMA(0,0,1) with non-zero mean : 2088.465
ARMA(0,0,0) with zero mean : 2886.812
ARMA(1,0,2) with non-zero mean : 1987.985
ARMA(2,0,1) with non-zero mean : 1987.886
ARMA(3,0,2) with non-zero mean : 1989.75
ARMA(2,0,3) with non-zero mean : 1988.83
ARMA(1,0,1) with non-zero mean : 1987
ARMA(1,0,3) with non-zero mean : 1989.706
ARMA(3,0,1) with non-zero mean : 1988.852
ARMA(3,0,3) with non-zero mean : 1987.738
ARMA(2,0,2) with zero mean : 2006.189

```

Best model: ARMA(2,0,2) with non-zero mean

ADF Test:Augmented Dickey-Fuller Test
data: ts_pm25
Dickey-Fuller = -4.399, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary

ARMA Model Summary:
Series: ts_pm25
ARMA(2,0,2) with non-zero mean

Coefficients:
            ar1      ar2      ma1      ma2      mean
      -0.0343  0.743  0.6143 -0.2989  11.3088
s.e.    0.0986  0.074  0.1108  0.0712  0.8442

sigma^2 = 13.24:  log likelihood = -987.27
AIC=1986.55  AICc=1986.78  BIC=2009.95

Training set error measures:
            ME      RMSE      MAE      MPE
Training set 0.03192686 3.613776 2.660831 -8.520339
            MAPE      MASE      ACF1
Training set 25.19411 0.9443527 0.0006838395

ARMA Coefficient Significance:

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1    -0.034321  0.098633 -0.3480  0.7279
ar2     0.742952  0.074030 10.0359 < 2.2e-16 ***
ma1     0.614319  0.110833  5.5428 2.977e-08 ***
ma2    -0.298869  0.071235 -4.1955 2.722e-05 ***
intercept 11.308819  0.844159 13.3966 < 2.2e-16 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ARMA Accuracy:
            ME      RMSE      MAE      MPE
Training set 0.03192686 3.613776 2.660831 -8.520339
            MAPE      MASE      ACF1
Training set 25.19411 0.9443527 0.0006838395
AIC: 1986.549 | BIC: 2009.948

Ljung-Box test

data: Residuals from ARMA(2,0,2) with non-zero mean
Q* = 11.92, df = 8, p-value = 0.1548

Model df: 4.  Total lags used: 12

```



```

=====
Station: Pengerang (SARMA)
=====

ARMA(2,0,2) (1,0,1) [7] with non-zero mean : 1989.917
ARMA(0,0,0) with non-zero mean : 2237.635
ARMA(1,0,0) (1,0,0) [7] with non-zero mean : 1994.079
ARMA(0,0,1) (0,0,1) [7] with non-zero mean : 2069.57
ARMA(0,0,0) with zero mean : 2886.812
ARMA(2,0,2) (0,0,1) [7] with non-zero mean : 1988.173
ARMA(2,0,2) with non-zero mean : 1986.784
ARMA(2,0,2) (1,0,0) [7] with non-zero mean : 1988.223
ARMA(1,0,2) with non-zero mean : 1987.985
ARMA(2,0,1) with non-zero mean : 1987.886
ARMA(3,0,2) with non-zero mean : 1989.75
ARMA(2,0,3) with non-zero mean : 1988.83
ARMA(1,0,1) with non-zero mean : 1987
ARMA(1,0,3) with non-zero mean : 1989.706
ARMA(3,0,1) with non-zero mean : 1988.852
ARMA(3,0,3) with non-zero mean : 1987.738
ARMA(2,0,2) with zero mean : 2006.189

Best model: ARMA(2,0,2) with non-zero mean

ADF Test:Augmented Dickey-Fuller Test
data: ts_pm25
Dickey-Fuller = -4.399, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary

SARMA Model Summary:
Series: ts_pm25
ARMA(2,0,2) with non-zero mean

Coefficients:
      ar1      ar2      ma1      ma2      mean
    -0.0343  0.743  0.6143 -0.2989  11.3088
s.e.   0.0986  0.074  0.1108  0.0712  0.8442

sigma^2 = 13.24: log likelihood = -987.27
AIC=1986.55 AICc=1986.78 BIC=2009.95

Training set error measures:
              ME      RMSE      MAE      MPE
Training set 0.03192686 3.613776 2.660831 -8.520339
              MAPE      MASE      ACF1
Training set 25.19411 0.6178642 0.0006838395

SARMA Coefficient Significance:

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1      -0.034321   0.098633 -0.3480   0.7279

```

```

ar2      0.742952    0.074030 10.0359 < 2.2e-16 ***
ma1      0.614319    0.110833  5.5428 2.977e-08 ***
ma2     -0.298869    0.071235 -4.1955 2.722e-05 ***
intercept 11.308819    0.844159 13.3966 < 2.2e-16 ***

```

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

SARMA Accuracy:

	ME	RMSE	MAE	MPE
Training set	0.03192686	3.613776	2.660831	-8.520339

	MAPE	MASE	ACF1
Training set	25.19411	0.6178642	0.0006838395

AIC: 1986.549 | BIC: 2009.948

Ljung-Box test

data: Residuals from ARMA(2,0,2) with non-zero mean
Q* = 11.92, df = 8, p-value = 0.1548

Model df: 4. Total lags used: 12

```

=====
Station: Segamat (ARMA)
=====

```

Missing Values:

STATION ID	LOCATION
0	0
DATE TIME	
0	
PM10 AVG (µg/m3)	
0	
PM2.5 AVG (µg/m3)	
0	
SO2 AVG (ppm)	
0	
NO2 AVG (ppm)	
1	
NO AVG (ppm)	
1	
NOX AVG (ppm)	
1	
O3 AVG (ppm)	
0	
CO AVG (ppm)	
0	
WIND DIRECTION AVG (°)	
0	
WIND SPEED AVG (m/s)	
0	
RELATIVE HUMIDITY AVG (%)	
0	
SOLAR RADIATION AVG (W/m2)	
0	
Ambient Temperature AVG (°C)	

Descriptive Statistics:

	vars	n	mean	sd	median	trimmed	mad	min	max
X1	1	365	13.32	5.63	12.35	12.87	4.78	3.38	47.63
			range	skew	kurtosis	se			
X1	44.26	1.59		6.04	0.29				

ARMA(2,0,2) with non-zero mean : 2047.655
 ARMA(0,0,0) with non-zero mean : 2300.846
 ARMA(1,0,0) with non-zero mean : 2052.705
 ARMA(0,0,1) with non-zero mean : 2106.063
 ARMA(0,0,0) with zero mean : 2987.994
 ARMA(1,0,2) with non-zero mean : 2047.96
 ARMA(2,0,1) with non-zero mean : 2046.471
 ARMA(1,0,1) with non-zero mean : 2048.444
 ARMA(2,0,0) with non-zero mean : 2049.944
 ARMA(3,0,1) with non-zero mean : 2044.681
 ARMA(3,0,0) with non-zero mean : 2047.911
 ARMA(4,0,1) with non-zero mean : 2046.698
 ARMA(3,0,2) with non-zero mean : 2046.659
 ARMA(4,0,0) with non-zero mean : 2049.334
 ARMA(4,0,2) with non-zero mean : 2044.455
 ARMA(5,0,2) with non-zero mean : 2046.146
 ARMA(4,0,3) with non-zero mean : 2037.207
 ARMA(3,0,3) with non-zero mean : 2045.427
 ARMA(5,0,3) with non-zero mean : 2047.816
 ARMA(4,0,4) with non-zero mean : 2042.679
 ARMA(3,0,4) with non-zero mean : 2041.542
 ARMA(5,0,4) with non-zero mean : 2043.99
 ARMA(4,0,3) with zero mean : Inf

Best model: ARMA(4,0,3) with non-zero mean

ARMA Model Summary:

Series: ts_pm25

ARMA(4,0,3) with non-zero mean

Coefficients:

	ar1	ar2	ar3	ar4	ma1
	1.6190	-0.1460	-1.0967	0.5572	-0.8518
s.e.	0.0925	0.1862	0.1722	0.0696	0.0818
	ma2	ma3	mean		
	-0.6943	0.8313	13.1100		
s.e.	0.0955	0.0705	0.8534		

sigma^2 = 15.04: log likelihood = -1009.35

AIC=2036.7 AICc=2037.21 BIC=2071.8

Training set error measures:

	ME	RMSE	MAE	MPE
Training set	0.03564268	3.83574	2.826224	-7.69678
	MAPE	MASE	ACF1	
Training set	23.40763	0.9328901	-0.001542915	

ARMA Coefficient Significance:

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	1.619026	0.092529	17.4975	< 2.2e-16	***
ar2	-0.146015	0.186161	-0.7843	0.4328	
ar3	-1.096701	0.172201	-6.3687	1.906e-10	***
ar4	0.557247	0.069560	8.0111	1.137e-15	***
ma1	-0.851772	0.081844	-10.4072	< 2.2e-16	***
ma2	-0.694339	0.095487	-7.2716	3.553e-13	***
ma3	0.831322	0.070535	11.7859	< 2.2e-16	***
intercept	13.109951	0.853379	15.3624	< 2.2e-16	***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ARMA Accuracy:

	ME	RMSE	MAE	MPE
Training set	0.03564268	3.83574	2.826224	-7.69678

	MAPE	MASE	ACF1
Training set	23.40763	0.9328901	-0.001542915

AIC: 2036.7 | BIC: 2071.799

Ljung-Box test

data: Residuals from ARMA(4,0,3) with non-zero mean
 $Q^* = 11.102$, $df = 5$, $p\text{-value} = 0.05939$

Model df: 7. Total lags used: 12

=====

Station: Segamat (SARMA)

=====

ARMA(2,0,2) (1,0,1) [7] with non-zero mean : 2048.197
 ARMA(0,0,0) with non-zero mean : 2300.846
 ARMA(1,0,0) (1,0,0) [7] with non-zero mean : 2052.49
 ARMA(0,0,1) (0,0,1) [7] with non-zero mean : 2096.118
 ARMA(0,0,0) with zero mean : 2987.994
 ARMA(2,0,2) (0,0,1) [7] with non-zero mean : 2044.496
 ARMA(2,0,2) with non-zero mean : 2047.655
 ARMA(2,0,2) (0,0,2) [7] with non-zero mean : 2046.534
 ARMA(2,0,2) (1,0,0) [7] with non-zero mean : 2044.565
 ARMA(2,0,2) (1,0,2) [7] with non-zero mean : 2050.049
 ARMA(1,0,2) (0,0,1) [7] with non-zero mean : 2046.34
 ARMA(2,0,1) (0,0,1) [7] with non-zero mean : 2046.576
 ARMA(3,0,2) (0,0,1) [7] with non-zero mean : 2046.381
 ARMA(2,0,3) (0,0,1) [7] with non-zero mean : 2046.459
 ARMA(1,0,1) (0,0,1) [7] with non-zero mean : 2045.049
 ARMA(1,0,3) (0,0,1) [7] with non-zero mean : 2045.852
 ARMA(3,0,1) (0,0,1) [7] with non-zero mean : 2044.573
 ARMA(3,0,3) (0,0,1) [7] with non-zero mean : Inf
 ARMA(2,0,2) (0,0,1) [7] with zero mean : Inf

```

Best model: ARMA(2,0,2) (0,0,1) [7] with non-zero mean

SARMA Model Summary:
Series: ts_pm25
ARMA(2,0,2) (0,0,1) [7] with non-zero mean

Coefficients:
      ar1      ar2      ma1      ma2      sma1
      1.3577 -0.3910 -0.5611 -0.2744  0.0902
s.e.   0.1439  0.1179  0.1416  0.0705  0.0576
      mean
      12.9152
s.e.    1.1201

sigma^2 = 15.46:  log likelihood = -1015.09
AIC=2044.18  AICc=2044.5  BIC=2071.48

Training set error measures:
      ME      RMSE      MAE      MPE
Training set 0.05964365 3.89953 2.885423 -7.703672
      MAPE      MASE      ACF1
Training set 23.7244 0.5661715 0.0007035409

SARMA Coefficient Significance:

z test of coefficients:

      Estimate Std. Error z value  Pr(>|z|)
ar1      1.357724    0.143871  9.4371 < 2.2e-16 ***
ar2     -0.390971    0.117939 -3.3150 0.0009164 ***
ma1     -0.561099    0.141581 -3.9631 7.398e-05 ***
ma2     -0.274365    0.070461 -3.8939 9.866e-05 ***
sma1      0.090159    0.057603  1.5652 0.0175449 **
intercept 12.915162    1.120102 11.5303 < 2.2e-16 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

SARMA Accuracy:
      ME      RMSE      MAE      MPE
Training set 0.05964365 3.89953 2.885423 -7.703672
      MAPE      MASE      ACF1
Training set 23.7244 0.5661715 0.0007035409
AIC: 2044.182 | BIC: 2071.481

Ljung-Box test

data:  Residuals from ARMA(2,0,2) (0,0,1) [7] with non-zero
mean
Q* = 18.985, df = 7, p-value = 0.078236

Model df: 5.  Total lags used: 12
=====
Station: Tangkak (ARMA)

```

```

=====
Missing Values:
      STATION ID              LOCATION
      1              1
      DATE TIME      PM10 AVG (µg/m3)
      1              0
      PM2.5 AVG (µg/m3)      SO2 AVG (ppm)
      0              2
      NO2 AVG (ppm)      NO AVG (ppm)
      3              3
      NOX AVG (ppm)      O3 AVG (ppm)
      3              2
      CO AVG (ppm)      WIND DIRECTION AVG(°)
      2              2
      WIND SPEED AVG(m/s)      RELATIVE HUMIDITY AVG(%)
      2              2
      SOLAR RADIATION AVG(W/m2)      Ambient Temperature AVG(°c)
      10              2
      Station
      0

Descriptive Statistics:
      vars   n  mean   sd median trimmed  mad  min   max range
X1      1 366 13.87 5.82  12.6  13.07 3.98 5.76 50.49 44.73
      skew kurtosis  se
X1 2.26      8.73 0.3

ARMA(2,0,2) with non-zero mean : 2083.353
ARMA(0,0,0) with non-zero mean : 2331.477
ARMA(1,0,0) with non-zero mean : 2098.014
ARMA(0,0,1) with non-zero mean : 2137.096
ARMA(0,0,0) with zero mean      : 3025.01
ARMA(1,0,2) with non-zero mean : 2087.056
ARMA(2,0,1) with non-zero mean : 2089.222
ARMA(3,0,2) with non-zero mean : 2093.101
ARMA(2,0,3) with non-zero mean : 2085.434
ARMA(1,0,1) with non-zero mean : 2091.592
ARMA(1,0,3) with non-zero mean : 2084.444
ARMA(3,0,1) with non-zero mean : 2091.186
ARMA(3,0,3) with non-zero mean : 2085.216
ARMA(2,0,2) with zero mean      : Inf

Best model: ARMA(2,0,2) with non-zero mean

ADF Test:

      Augmented Dickey-Fuller Test

data:  ts_pm25
Dickey-Fuller = -5.0376, Lag order = 7, p-value =
0.01
alternative hypothesis: stationary

```

```

ARMA Model Summary:
Series: ts_pm25
ARMA(2,0,2) with non-zero mean

Coefficients:
          ar1          ar2          ma1          ma2          mean
      1.2506   -0.2830   -0.4836   -0.3328   13.6155
s.e.   0.1184    0.1064    0.1165    0.0747    1.1768

sigma^2 = 16.98:  log likelihood = -1035.56
AIC=2083.12   AICc=2083.35   BIC=2106.53

Training set error measures:
              ME          RMSE          MAE          MPE
Training set 0.04451554  4.092934  2.847743  -6.571309
              MAPE          MASE          ACF1
Training set 21.36266  0.9318479  0.0009312748

ARMA Coefficient Significance:

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1      1.250567    0.118408 10.5615 < 2.2e-16 ***
ar2     -0.282998    0.106381  -2.6602  0.007809 **
ma1     -0.483630    0.116543  -4.1498  3.328e-05 ***
ma2     -0.332798    0.074654  -4.4578  8.279e-06 ***
intercept 13.615456    1.176815 11.5698 < 2.2e-16 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ARMA Accuracy:
              ME          RMSE          MAE          MPE
Training set 0.04451554  4.092934  2.847743  -6.571309
              MAPE          MASE          ACF1
Training set 21.36266  0.9318479  0.0009312748
AIC: 2083.119 | BIC: 2106.535

Ljung-Box test

data:  Residuals from ARMA(2,0,2) with non-zero mean
Q* = 15.192, df = 8, p-value = 0.05552

Model df: 4.    Total lags used: 12

=====
Station: Tangkak (SARMA)
=====

ARMA(2,0,2) (1,0,1) [7] with non-zero mean : 2090.516
ARMA(0,0,0)              with non-zero mean : 2331.477
ARMA(1,0,0) (1,0,0) [7] with non-zero mean : 2098.336
ARMA(0,0,1) (0,0,1) [7] with non-zero mean : 2128.991

```

```

ARMA(0,0,0)           with zero mean      : 3025.01
ARMA(2,0,2) (0,0,1) [7] with non-zero mean : 2084.563
ARMA(2,0,2)           with non-zero mean : 2083.353
ARMA(2,0,2) (1,0,0) [7] with non-zero mean : 2090.793
ARMA(1,0,2)           with non-zero mean : 2087.056
ARMA(2,0,1)           with non-zero mean : 2089.222
ARMA(3,0,2)           with non-zero mean : 2093.101
ARMA(2,0,3)           with non-zero mean : 2085.434
ARMA(1,0,1)           with non-zero mean : 2091.592
ARMA(1,0,3)           with non-zero mean : 2084.444
ARMA(3,0,1)           with non-zero mean : 2091.186
ARMA(3,0,3)           with non-zero mean : 2085.216
ARMA(2,0,2)           with zero mean      : Inf

```

Best model: ARMA(2,0,2) with non-zero mean

ADF Test:

Augmented Dickey-Fuller Test

```

data: ts_pm25
Dickey-Fuller = -5.0376, Lag order = 7, p-value =
0.01
alternative hypothesis: stationary

```

SARMA Model Summary:

```

Series: ts_pm25
ARMA(2,0,2) with non-zero mean

```

Coefficients:

	ar1	ar2	ma1	ma2	mean
	1.2506	-0.2830	-0.4836	-0.3328	13.6155
s.e.	0.1184	0.1064	0.1165	0.0747	1.1768

```

sigma^2 = 16.98: log likelihood = -1035.56
AIC=2083.12 AICc=2083.35 BIC=2106.53

```

Training set error measures:

	ME	RMSE	MAE	MPE
Training set	0.04451554	4.092934	2.847743	-6.571309

	MAPE	MASE	ACF1
Training set	21.36266	0.5580331	0.0009312748

SARMA Coefficient Significance:

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	1.250567	0.118408	10.5615	< 2.2e-16 ***
ar2	-0.282998	0.106381	-2.6602	0.007809 **
ma1	-0.483630	0.116543	-4.1498	3.328e-05 ***
ma2	-0.332798	0.074654	-4.4578	8.279e-06 ***
intercept	13.615456	1.176815	11.5698	< 2.2e-16 ***

```

Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

SARMA Accuracy:
              ME      RMSE      MAE      MPE
Training set 0.04451554 4.092934 2.847743 -6.571309
              MAPE      MASE      ACF1
Training set 21.36266 0.5580331 0.0009312748
AIC: 2083.119 | BIC: 2106.535

      Ljung-Box test

data:  Residuals from ARMA(2,0,2) with non-zero mean
Q* = 15.192, df = 8, p-value = 0.05552

Model df: 4.    Total lags used: 12

```